

GE Healthcare

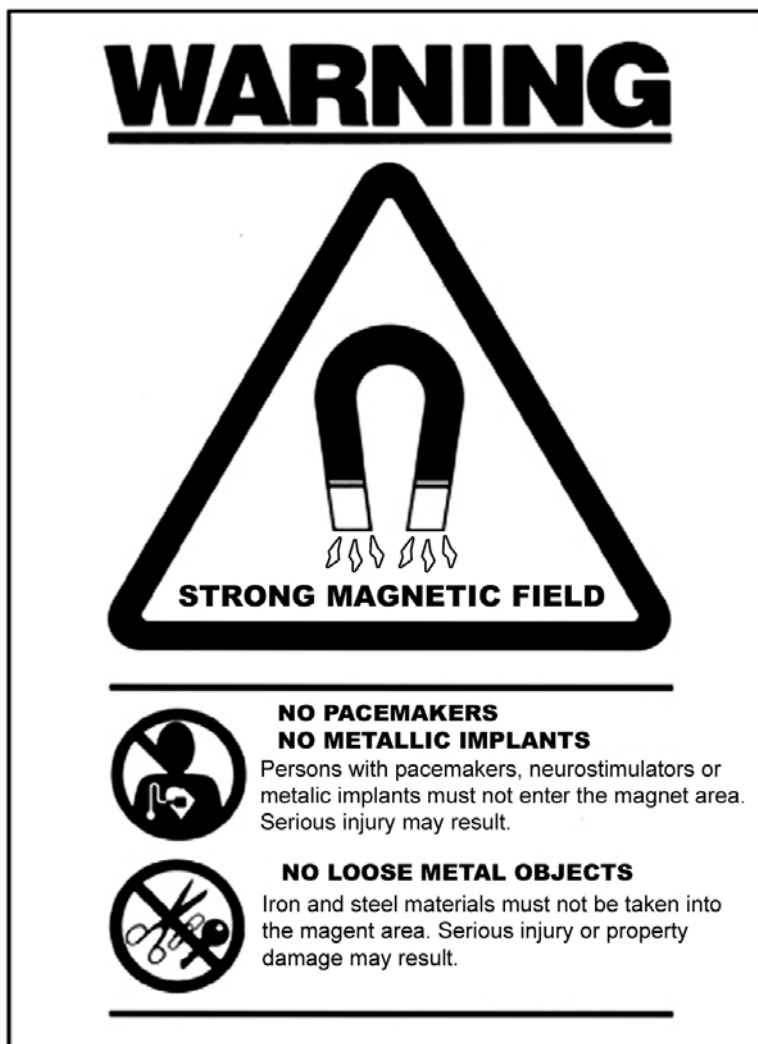
Signa EXCITE HD 3.0T Upgrades

Pre-Installation



OPERATING DOCUMENTATION

5133305
Revision 5



Important Information

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
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- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A' CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

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Revision History

Revision	Date	Description
1	23 March 2005	initial upgrade product release
2	28 April 2005	Pre-Installation Introduction Chapter: Added Critical To Image Quality section Chapter 1, Section 2 , Critical To Image Quality
		Pre-Installation Checklist Chapter: Chapter 10, Section 2 , 3.0T Upgrade Pre-Installation Checklist added CTIQ numbers and updated for wording symmetry with MR Engineering wording for Upgrade Critical To Image Quality Checklist
		Tools and Test Equipment Chapter: Chapter 11, Section 7 , Tool Kit Added Non-Magnetic Tool large & small kits entries
3	30 March 2006	Topics modified throughout direction
		- Added NEC 2005 references: Chapter 6, Section 1, System Power Introduction, Chapter 6, Section 2, Critical Power Requirements, Chapter 6, Section 4, Grounding, Chapter 6, Section 5, Ground Fault Protection, - Added Remote MRU Option: Remote MRU Option, Chapter 6, Section 1, System Power Introduction, Chapter 7, Section 11, Remote MRU Option Interconnects - Add IPCM Option information: Chapter 2, Section 4, Additional System Options, Chapter 3, Section 1, Proximity Limits, Chapter 3, Section 6.3, IPCM Option For LCC300 Magnets ONLY, Chapter 3, Section 8.2, IPCM Option, Chapter 5, Section 4, Air Cooling, Chapter 6, Section 1, System Power Introduction, Chapter 6, Section 9, IPCM Option Power Requirements, Chapter 7, Section 5, Upgrade Catalogs And Interconnects Information Index, and Chapter 9, Section 1, Shipment.
		System Configuration Chapter: Pre-EXCITE Signa With 3.0T94 To Signa EXCITE HD Upgrades revised Pre-EXCITE Signa system with 3T94 Magnet upgrade to include M3000PT for system with ACGD Gradient Cabinet and M3000PF for system with SGD Gradient Cabinet.
		Room Layouts Chapter: Chapter 3, Section 4 , MR System Interconnects Routing MR System Interconnects Routing Requirements/Recommendations table: Row titled Entire MR System: recommendation for use of access flooring deleted Magnet Room (changed to "recommended for use in ONLY the Equipment Room "); Row titled Magnet Room: Row titled Magnet Room: deleted "e.g. aluminum" from 1st bullet and Note. Chapter 3, Section 8.4 , Operator Workspace "seismic anchor holes" changed to BRACKETS MOUNTING HOLES and brackets callout to "BRACKETS SUPPLIED WITH CABINET FOR MOUNTING, IF REQUIRED."
		Site Environment Chapter: Chapter 5, Section 4 , Air Cooling added DC Lighting Controller option equipment values. Chapter 5, Section 9 , Room Ventilation Magnet Room exhaust fan and air inlet requirements change to minimum CFM (m ³ /minute) and minimum of room air exchanges per hour. Also revised pressure equalizing waveguide location to include wall requirements.

Revision	Date	Description
3 (cont'd)		RF Shielded Room Chapter: Chapter 8, Section 1, RF Shielded Room Requirements RF Shielded Room Requirements table, row titled Construction, Note text deleted 'e.g. aluminum'. Physical Considerations subsection changes: • Chapter 8, Section 6.1, Introduction added missing 1st paragraph • Chapter 8, Section 6.2, Doors and Other Openings added bullets text. • Chapter 8, Section 6.3, Walls added missing paragraph text. Chapter 8, Section 5, RF Penetration Panel requirements reordered for clarity and completeness. Also added Notice pertaining to Penetration Panel mounting hardware must not loosen over time to maintain RF attenuation requirement. Chapter 8, Section 6.4, Magnet Room Floors "Magnet Room floor Requirements and Recommendations" table changes: • Table row titled Floor Design: Bullet stating 42.125 dimension finished floor to center of Magnet opening added missing minus (-) for tolerance +/- value. Access flooring Notice deleted "e.g. aluminum". • Through out table deleted "e.g. aluminum" for Notice! pertaining to removable panels known source of RF Noise. • Entire Floor Levelness row contents revised, Magnet Mounting recess area requirements including tolerance (+/-) values changed and recommendation added. • Majority of Floor Construction row revised: Magnet Mounting Recessed area required with and without VibroAcoustic Damping Option and clarified requirements for both configurations. Added RF Shield material seams MUST NOT be under projected areas for VibroAcoustic Damping Option mats. Added Access flooring Notice. • Added tables with illustrations & references of revised Magnet Load Pattern with & without VibroAcoustic Damping Option. Shipping And Delivery Chapter: Chapter 9, Section 1, Shipment incorporated Storage Requirements section info into this section. Tools and Test Equipment Chapter: Chapter 11, Section 3, Cryogenic Equipment Supplied By Rigger row contents changed to "Refer to <i>Direction 2340869 GE LCC300 Active Shield Magnet Delivery and Installation</i> for equipment and requirements." Chapter 11, Section 4, Installation Equipment Added 3T shim related tools: -Shim case / cable kit 2135558, LCC300 Shim Camera Kit 2386028, LCC300 Passive Shim Kit 2386029. Updated several part numbers: -Field Plotting Kit 46-251865G5 -Magnet Ramping Equipment Kit part number updated to 46-260703G5 -Non-Magnetic Tool Kit replaced by new Titanium Non-Magnetic Tool Kits.
4	22 August 2006	Room Layouts Chapter: Room Sizes added exit route wording to heading of Room Dimensions table. Chapter 3, Section 4, MR System Interconnects Routing MR System Interconnects Routing Requirements/Recommendations table reformatted to number requirements and updated requirements to be consistent with revised Magnet Room floor requirement of metal access floor tiles not allowed anywhere: Row titled Entire MR System; recommendation for use of access flooring deleted.; Row titled Magnet Room: deleted access floor note and added metal floor access tiles not allowed with reference to Magnet Room Floors requirement in RF Shielded Room chapter. Chapter 3, Section 6.2, Customer Site Storage Requirements added section with reference to <i>Direction 5182674 GE Healthcare Signa MRI Scanners Customer Site Storage Requirements</i> for size and weight of surface coils and phantoms that require customer provided storage locations. SPT Phantom Set Shipping/Storage Cart information included in Customer Site Storage Requirements subsection. Chapter 3, Section 8.8, RFS Cabinet (MR2) added side Service Access Requirement. Site Environment Chapter: Chapter 5, Section 6, Altitude added Notice with Magnet Systems approval/direction contact info for Magnet located above 8100 ft (2469 m). Chapter 5, Section 9, Room Ventilation Ventilation Requirements/Recommendations table reformatted for 1 column with numbered list of requirements. Incorporated Magnet Room ventilation switch placement requirement (removed from RF Shielded Room Chapter). Interconnect Data Chapter: Reformatted Introduction section so subsections are now included in direction table of contents.

Revision	Date	Description
4 (continued)		RF Shielded Room Chapter: Chapter 8, Section 6.4, Magnet Room Floors major changes incorporated: • Rewrote entire section to simplify requirements, used feedback from Project Managers - Installations and Zone Support Engineers. • Same levelness dimension requirements for all Magnet Room floor areas. • Metal access floor time are NOT allowed in Magnet Room. • Illustration revised to show recessed area with Vibroacoustic mats projected locations.
5	5 April 2007	Interconnect routing requirements moved to Chapter 3, Section 4, MR System Interconnects Routing Table titled "MR System Interconnects Routing Requirements/Recommendations", row titled "Magnet Room" from Chapter 8, Section 1, RF Shielded Room Requirements RF Shielded Room Requirements table. Also reworded moved requirements: - added anchors to mechanical fasteners list and clarified mechanical fastener not allowed within 2 ft (0.61 m) of Gradient Cables; - added unsecured or overlapped electrically conductive materials not allowed within 2 ft (0.61m of Gradient Cables. [Resolution for iTrak 13067413, iTrak 13066436]. RF Shielded Room Requirements table, Construction row deleted note concerning access floor since that material is not allowed in Magnet Room [iTrak 13076809]. Room Layout Chapter: Chapter 3, Section 8.4, Operator Workspace dimensions revised for new OW Table configuration. Site Environment Chapter: Chapter 5, Section 7, Lighting requirements changed to DC Lighting required in Magnet Room and additional rewording for clarification of requirements. [iTrak 13068863].

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Chapter 1 Pre-Installation Introduction

1 Introduction

This document contains the physical, magnetic, cryogenic, plumbing and electrical data necessary for planning and preparing a site for a magnetic resonance system. "Preinstallation work" is done to prepare the customer's premises for the installation of the products sold. It is the responsibility of the purchaser to arrange for performance of this work. Such work includes:

- Installation of the electrical conduit, junction boxes, ducts, surface raceways, outlets and line safety switches.
- Installation of wires not supplied by GE Healthcare such as: the facility input power line to the power distribution unit, as well as emergency power lines and facility power lines to the Magnet Room. The electrical contractor shall ring out and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Insulation on all equipment ground wires must be green with a yellow stripe.
- Installation of non-electrical lines such as: water plumbing, helium venting systems and air conditioning equipment. Also, installation of recommended air, vacuum and oxygen lines into the Magnet Room. All lines must be clearly labeled.
- Renovation of RF shielding in Magnet Room.
- Acoustic properties of the MR Magnet Room including the RF Shield need to meet local regulations and customer requirements. All aspects of the room including the constructed walls and all purchased items such as windows, doors, vents, etc. need to be considered.
- Site construction or renovation.
- Installation of structural reinforcements as required.
- Installation and testing of selected magnetic shielding as required.
- Installation of water treatment equipment if necessary.

All work **MUST** be in compliance with national and local building and safety codes.

A structural steel analysis may be necessary during site evaluation for a magnetic resonance system. All site plans and preliminary concepts should be reviewed by GE Healthcare MR Siting group prior to construction.

Unless specifically mentioned, GE Healthcare does not provide or install: the facility input power lines to the Main Disconnect Panel and Power Distribution Unit or the power lines required in the Magnet Room, nor flooring, conduit, junction boxes, room lighting, warning lights, ducts, plumbing, water treatment equipment, cryogenic venting outside the Magnet Room, non standard cryogenic venting inside the Magnet Room, acoustic treatment to contain acoustic noise (airborne and structureborne) within the RF shielded room, or the RF shielded room illustrated in this document.

NOTE: GE does offer the VibroAcoustic Damping Option to suppress LCC type Magnet generated vibration (this is not an acoustic treatment). All floor preparations for this option are the responsibility of the customer.

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In

addition, electrical feeds into the Main Disconnect Panel and Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE Healthcare personnel or third-party service companies with equivalent training. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes. The purchaser of GE equipment shall only utilize qualified personnel (i.e. GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Pre-installation information is continually changing due to the evolution of the product. GE will make every reasonable effort to maintain accuracy of pre-installation information.

2 Critical To Image Quality

Certain aspects of the MR site have the potential to cause image quality issues. [Table 1-1](#) lists the site aspects that have currently been identified as Critical To Image Quality (CTIQ).

NOTE: [Table 1-1](#) includes only CTIQ numbers which apply to specifications contained in this pre-installation manual.

Table 1-1: MR Site Critical To Image Quality

Number	Topic	Specifications
CTIQ#2	Magnetic Field	Refer to Chapter 3, Section 1 , Proximity Limits
CTIQ#5	Acoustics	Refer to Chapter 5, Section 8.3 , System Acoustic Noise Levels and Chapter 5, Section 8.4 , Acoustic Design Guidelines
CTIQ#6	Power Quality	Refer to Chapter 6, Section 1 , System Power Introduction and Chapter 6, Section 2 , Critical Power Requirements
CTIQ#7	Lighting	Refer to Chapter 5, Section 7 , Lighting
CTIQ#8	Multiple MR System Site	Refer to Chapter 3, Section 2.1 , Two Magnet Site Layout and Chapter 3, Section 2.2 , Equipment Room Shared By Multiple MR Systems
CTIQ#10	Broadband Network Connection & Telephone	Refer to Chapter 3, Section 6.1 , System Monitoring and Support Connectivity
CTIQ#11	Room Ventilation & Air Conditioning	Refer to • Chapter 5, Section 9, Room Ventilation • Chapter 5, Section 3, Temperature and Humidity Specifications and Chapter 5, Section 4, Air Cooling
CTIQ#15	Main Disconnect Panel (MDP)	Refer to Chapter 6, Section 1 , System Power Introduction and Chapter 6, Section 2 , Critical Power Requirements
CTIQ#16	Dust-Free Environment	Refer to Chapter 5, Section 12 , Pollution
CTIQ#17	Environmental Controls	Refer to • Chapter 5, Section 3, Temperature and Humidity Specifications • Chapter 5, Section 4, Air Cooling • Chapter 5, Section 5, Water Cooling Requirements
CTIQ#18	RF Shielding	Refer to Chapter 8, Section 1 , RF Shielded Room Requirements

3 Glossary

BB

Abbreviation for Broadband

Cryogen

A substance for producing low temperatures. Liquid helium is the cryogen used to cool the magnet to approximately 4 Kelvin (-269°C or -452°F).

Cryostat

An apparatus maintaining a very low constant temperature. The cryostat consists of one concentric, cylindrical container housed in an outer vacuum tight vessel. The magnet and shim coils are mounted in the inner container. The container is filled with liquid helium. The shields surrounding the inner container are kept cold by a refrigeration device.

Dewar

A container with an evacuated space between two highly reflective walls used to keep low temperature substances at near-constant temperatures. Liquid helium is usually stored and shipped in dewars.

Exclusion Zone

Area where the magnetic flux density is greater than five gauss. Personnel with cardiac pacemakers, neurostimulators and other biostimulation devices must NOT enter this zone. Signs are posted outside the five gauss line alerting personnel of this requirement. Since the magnetic field is three-dimensional, signs are also posted on floors above and below the Magnet Room in which the five gauss line exists.

Ferrous Material

Any substance containing iron which is strongly attracted by a magnetic field.

Gauss (G)

A unit of magnetic flux density. The earth's magnetic field strength is approximately one half gauss to one gauss depending on location. The internationally accepted unit is the tesla (1 Tesla = 10,000G and 1 milli Tesla = 10G).

Gradient

The amount and direction of the rate of change in space of the magnetic field strength. In the magnetic resonance system, gradient amplifiers and coils are used to vary the magnetic field strength in the x, y, and z planes.

Homogeneity

Uniformity. The homogeneity of the static magnetic field is an important quality of the magnet.

Isocenter

Center of the imaging volume ideally located at the magnet center.

Isogauss Line

An imaginary line or a line on a field plot connecting identical magnetic field strength points.

Magnetic Field (B)

A condition in a region of space established by the presence of a magnet and characterized by the presence of a detectable magnetic force at every point in the region. A magnetic field exists in the space around a magnet (or current carrying conductor) and can produce a magnetizing force on a body within it.

Magnetic Resonance (MR)

The absorption or emission of electromagnetic energy by nuclei in a static magnetic field, after excitation by a suitable radio frequency field.

Magnetic Shielding

Using material (e.g. steel) to redistribute a magnetic field , usually to reduce fringe fields.

MGD Chassis

Abbreviation for Multi Generational Data Acquisition Chassis

NB

Abbreviation for Narrow Band

Quench

Condition when a superconducting magnet becomes resistive thus rapidly boiling off liquid helium. The magnetic field reduces rapidly after a quench.

Radio Frequency (RF)

Frequency intermediate between audio frequency and infrared frequencies. Used in magnetic resonance systems to excite nuclei to resonance. Typical frequency range for magnetic resonance systems is 5-130 Mhz.

Radio Frequency Shielding

Using material (e.g. copper, aluminium, or steel) to reduce interference from external radio frequencies. A radio frequency shielded room usually encloses the entire magnet room.

Resonance

A large amplitude vibration caused by a relative small periodic stimulus of the same or nearly the same period as the natural vibration period of the system. In magnetic resonance imaging, the radio frequency pulses are the periodic stimuli which are at the same vibration period as the hydrogen nuclei being imaged.

RRF Chassis

Abbreviation for Remote Radio Frequency Chassis

Security Zone

Area within the Magnet Room where the magnet is located. Signs are posted outside the Magnet Room warning personnel of the high magnetic field existing in the Magnet Room and the possibility of ferrous objects becoming dangerous projectiles within this zone.

Shield Cooler Coldhead

An external refrigeration device which maintains the shields inside the cryostat at a constant temperature.

Shim Coils

Shim coils are used to provide auxiliary magnetic fields in order to compensate for inhomogeneities in the main magnetic field due to imperfections in the manufacturing of the magnet or affects of steel in the surrounding environment.

Shimming

Correction of inhomogeneity of the main magnetic field due to imperfections in the magnet or to the presence of external ferromagnetic objects.

Superconducting Magnet

A magnet whose magnetic field originates from current flowing through a superconductor. Such a magnet is enclosed in a cryostat.

Superconductor

A substance whose electrical resistance essentially disappears at temperatures near zero Kelvin. A commonly used superconductor in magnetic resonance imaging system magnets is niobium-titanium embedded in a copper matrix.

Tesla

The internationally accepted unit of magnetic flux density. One tesla is equal to 10,000 gauss. One milli Tesla is equal to 10 gauss.

Chapter 2 System Configuration

1 Upgrades 3.0T Configurations Introduction

Magnetic Resonance (MR) 3.0T system upgrades to Signa EXCITE HD hardware which may have site impacts are documented throughout this manual. A specific upgrade hardware content is dependent on the configuration of the system being upgraded. Refer to the appropriate section listed below for an overview of the upgrade hardware content, refer to the remainder of this manual for upgrade requirements.

- [Section 2](#), Release 11.x to Release 12.x Upgrades
- [Section 3](#), Pre-EXCITE Signa With 3.0T94 To Signa EXCITE HD Upgrades

NOTE: Only catalogs which include hardware are documented in this manual.

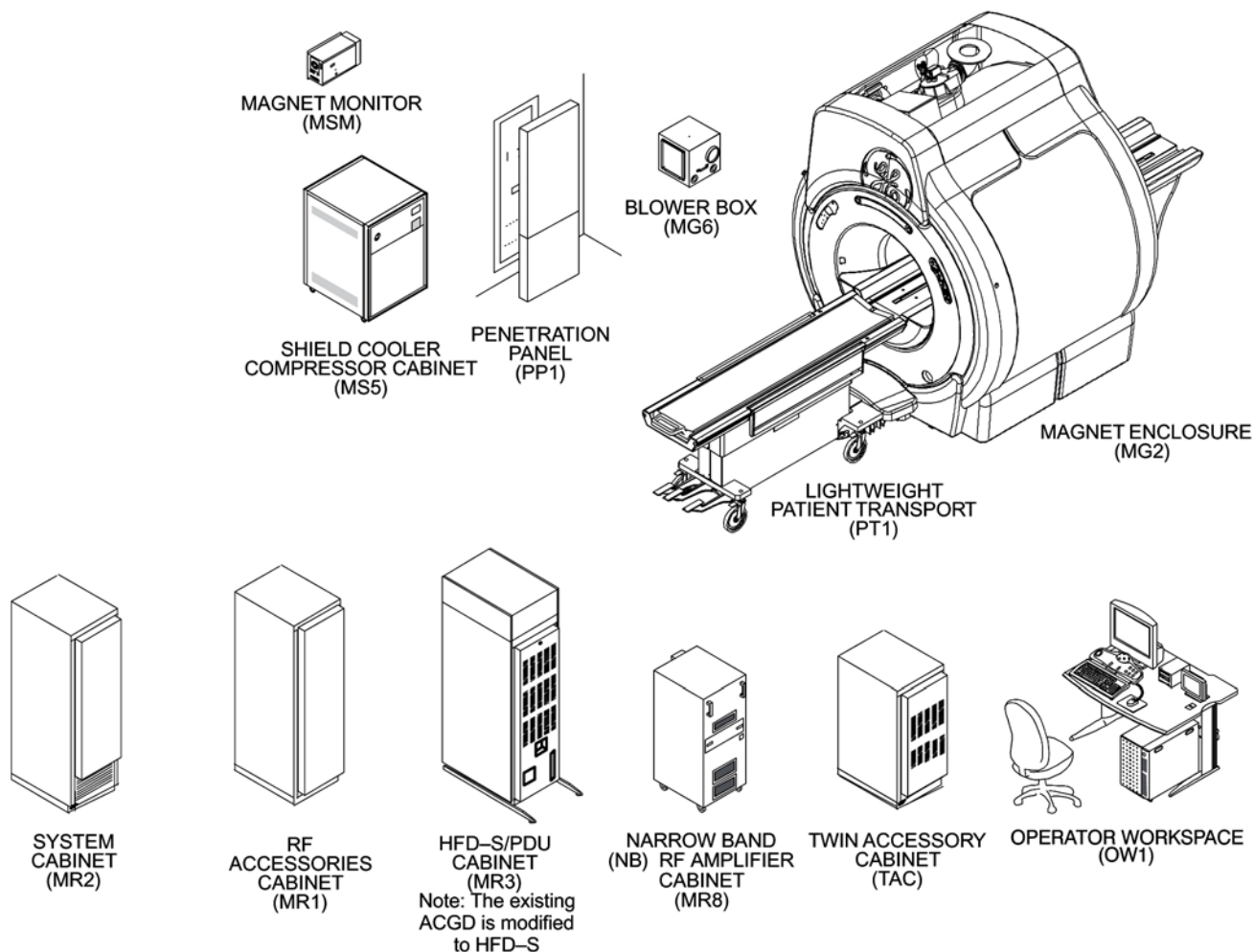
2 Release 11.x to Release 12.x Upgrades



NOTICE

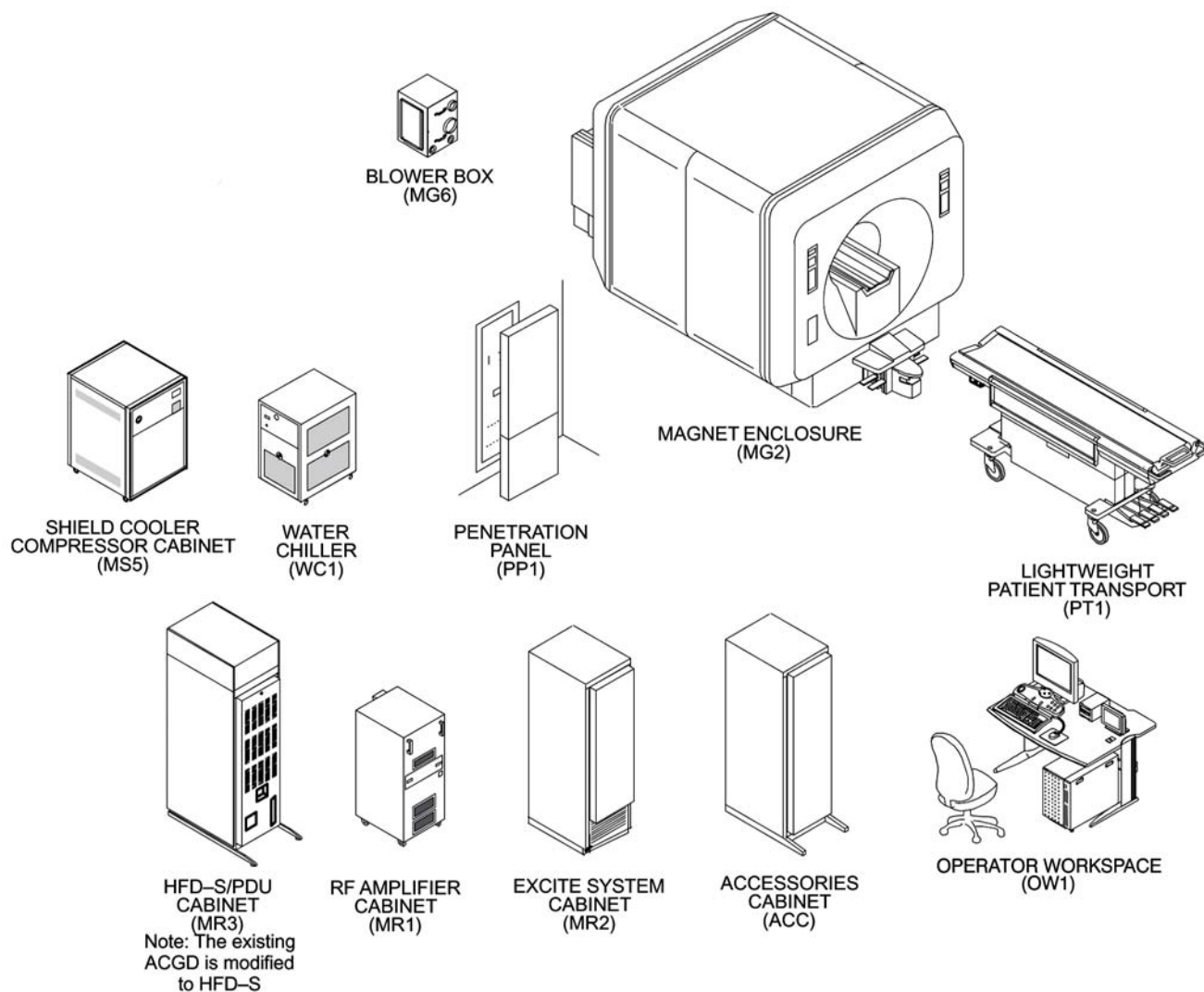
Release 11.x systems upgraded to Release 12.x final cabinets configuration is not the same as a new Signa EXCITE HD 3.0T. Many of the existing system equipment cabinets are modified by the Signa EXCITE HD upgrade.

The EXCITE HD 3.0T (Release 12.x) upgrade for an existing Signa EXCITE 3.0T (Release 11.x) system with LCC300 Magnet or an existing Signa 3.0T with EXCITE Technology (Release 11.x) with 3T/94 Magnet preforms modifications to the existing system equipment and cabinets. [Illustration 2-1](#) and [Illustration 2-2](#) the systems major equipment after a Release 11.x to Release 12.x EXCITE HD upgrade is completed.

Illustration 2-1: Release 11.x Upgrade To Release 12.x System With LCC300 Magnet Major Equipment**Existing System Support Equipment Retained (not shown)**

System retains their existing equipment for:

- Main Disconnect Panel
- Shield Cooler Compressor water cooling equipment
- Gradient Coil water cooling equipment.

Illustration 2-2: Release 11.x Upgrade To Release 12.x System With 3T94 Magnet Major Equipment**Existing System Support Equipment Retained
(not shown)**

System retains their existing equipment for:

- Main Disconnect Panel
- Shield Cooler Compressor water cooling equipment

Signa EXCITE 3.0T (Release 11.x) to Signa EXCITE HD 3.0T upgrade catalogs are listed in [Illustration 2-3](#). The equipment modified and added by the catalogs listed in the flowchart are shown in the illustrations as noted below:

- Release 11.x with LCC300 Magnet see [Illustration 2-4](#)
- Release 11.x with 3.0T/94 Magnet see [Illustration 2-5](#)

Illustration 2-3: 3.0T Release 11.x to Release 12.X Upgrade Catalogs

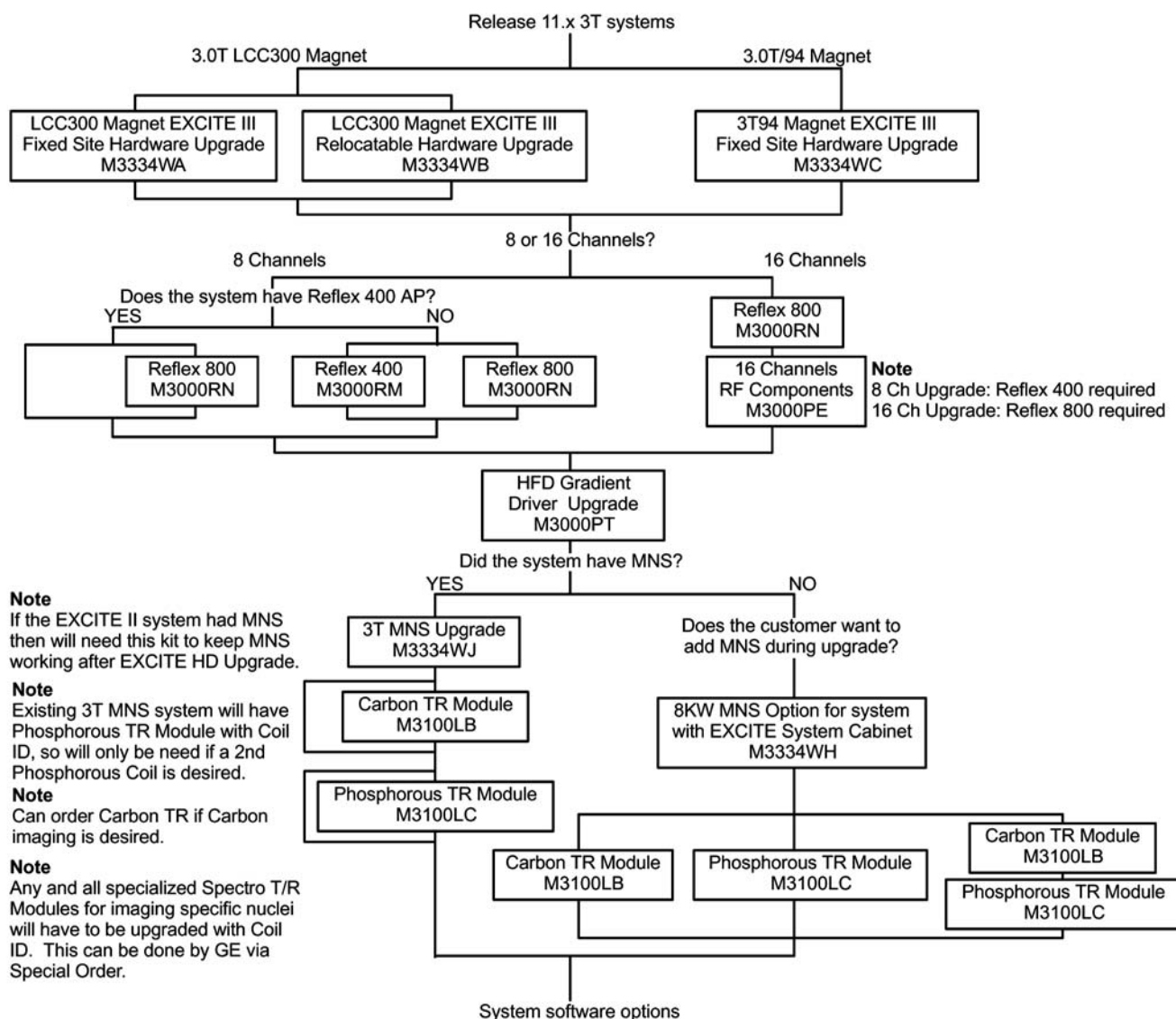


Illustration 2-4: Release 11.x To Release 12.x System With LCC300 Magnet EXCITE HD Upgrade Modified Equipment

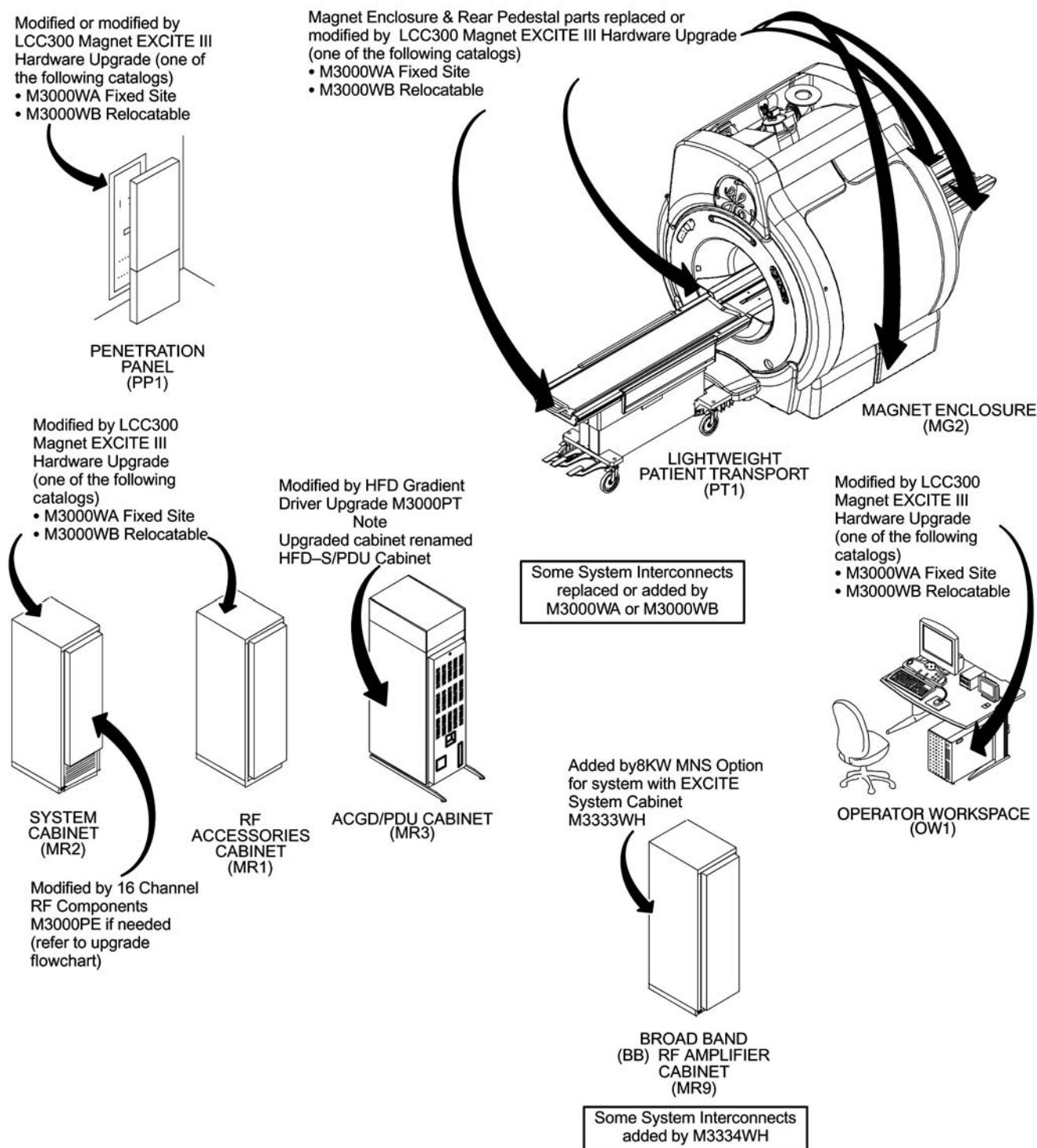
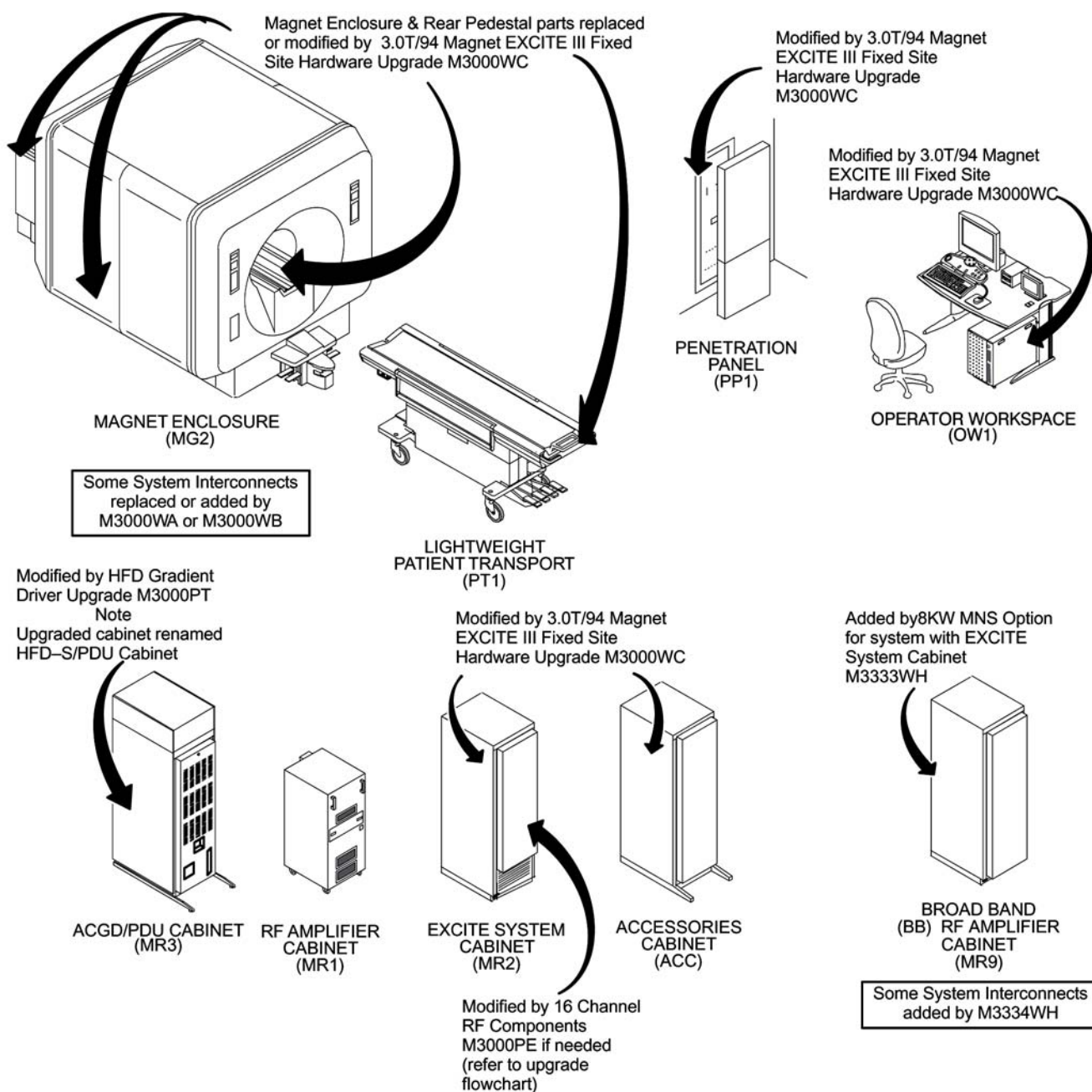


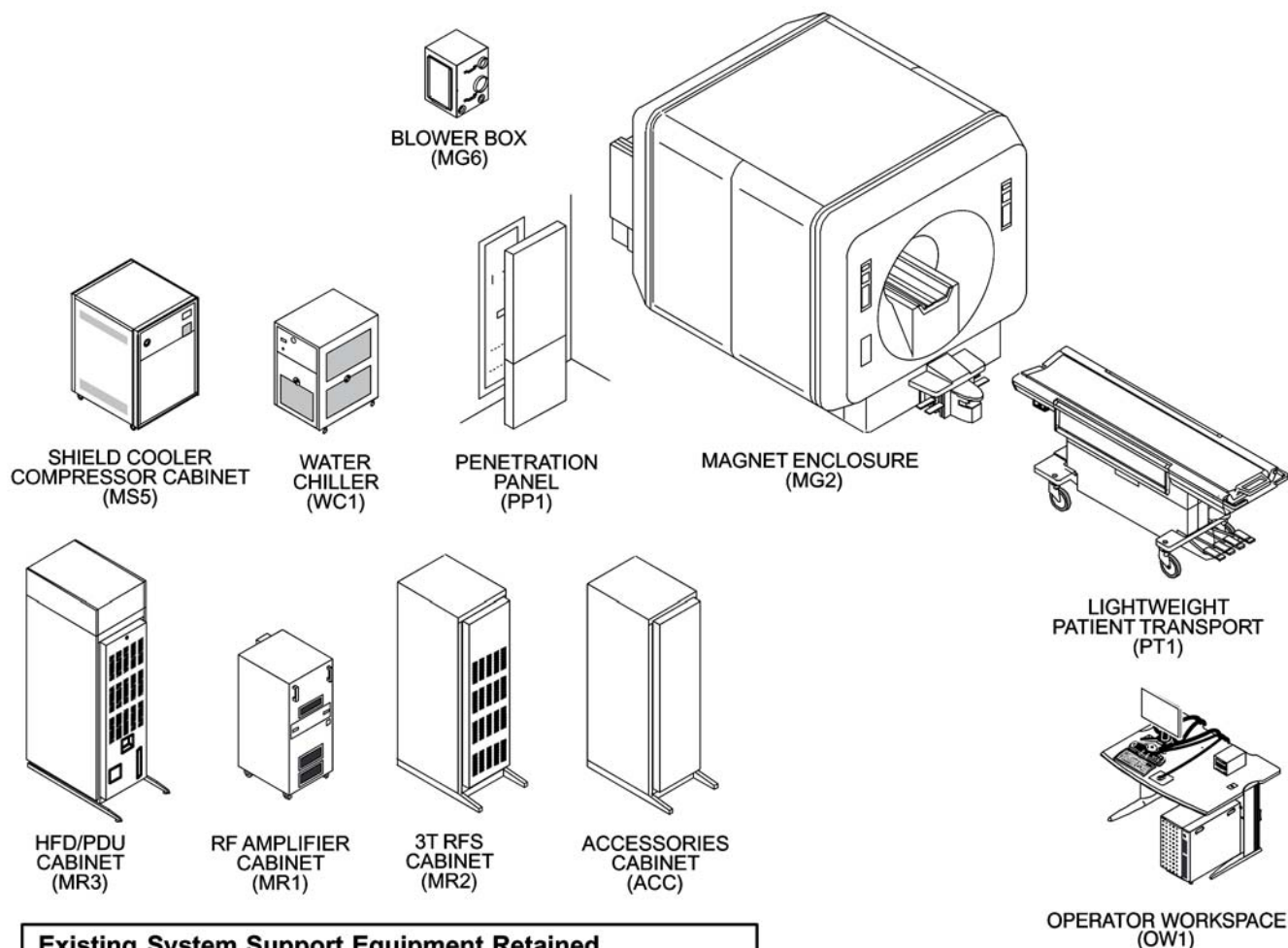
Illustration 2-5: Release 11.x To Release 12.x System With 3.0T94 Magnet EXCITE HD Upgrade Modified & Added Equipment



3 Pre-EXCITE Signa With 3.0T94 To Signa EXCITE HD Upgrades

Illustration 2-6 shows the Pre-EXCITE Signa with 3.0T94 Magnet system's major equipment after EXCITE HD (Release 12.x) upgrade.

Illustration 2-6: Pre-EXCITE SIGNA 3T94 VH3 Upgraded To Release 12.x System Major Equipment



Pre-EXCITE Signa With 3.0T/94 to Signa EXCITE HD upgrade catalogs are listed in [Illustration 2-7](#) . The equipment modified and added by the catalogs listed in the flowchart are shown in the [Illustration 2-8](#) .

Illustration 2-7: Pre-EXCITE Signa With 3.0T94 To Signa EXCITE HD Upgrade Catalogs

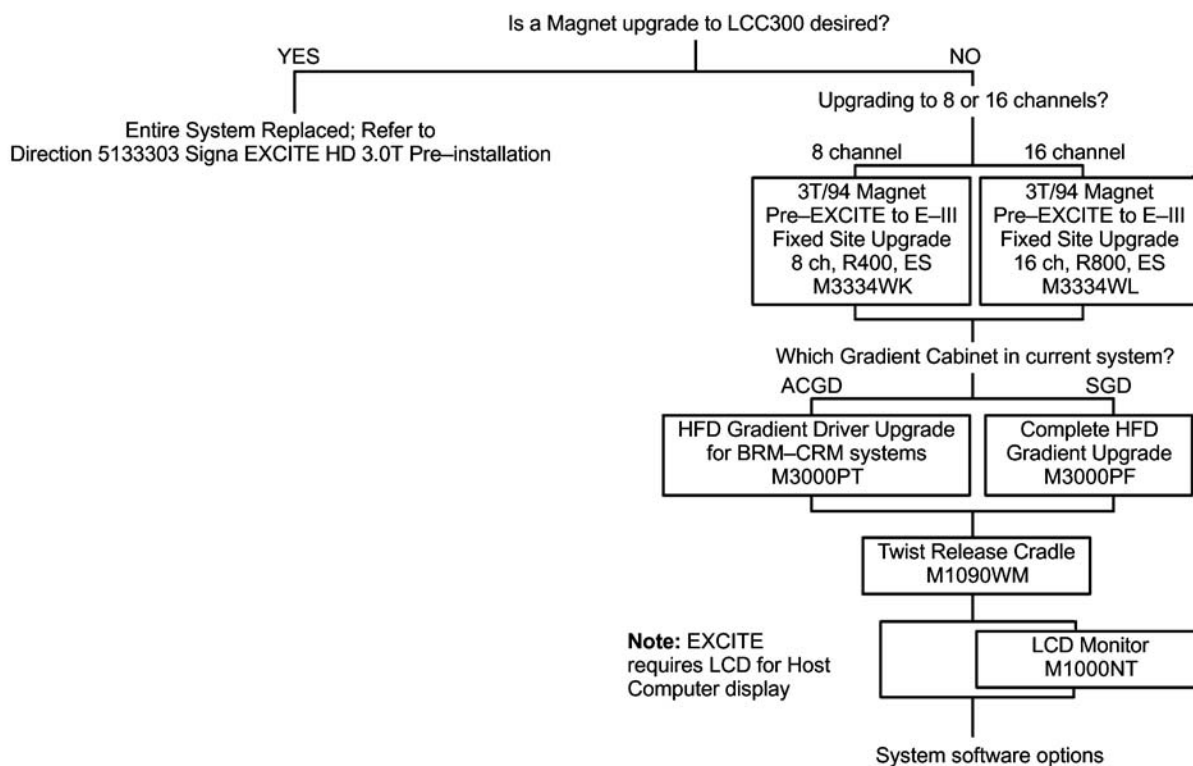
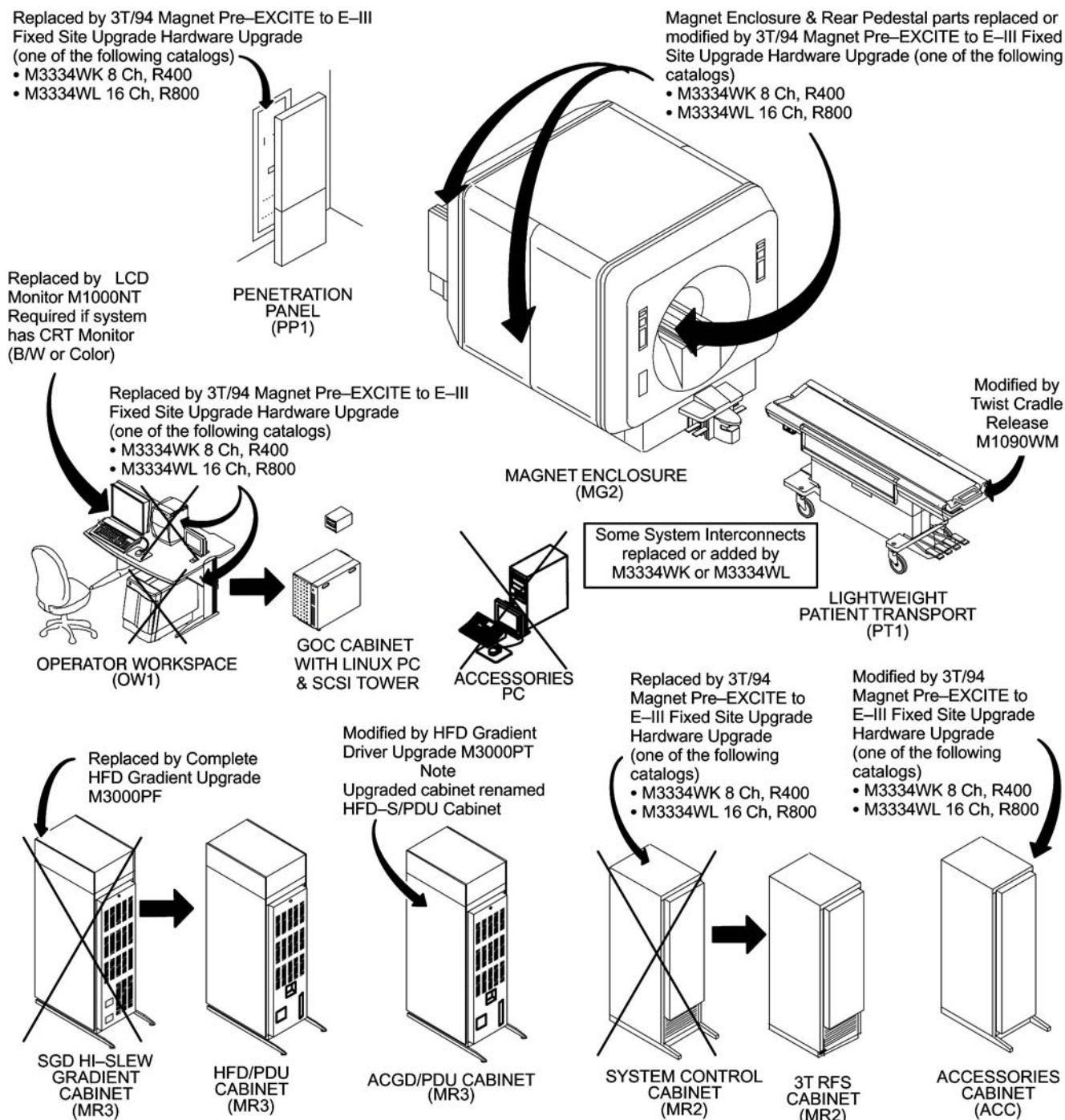


Illustration 2-8: Pre-EXCITE Signa With 3.0T94 Existing Hardware Replaced Or Modified By EXCITE HD Upgrade



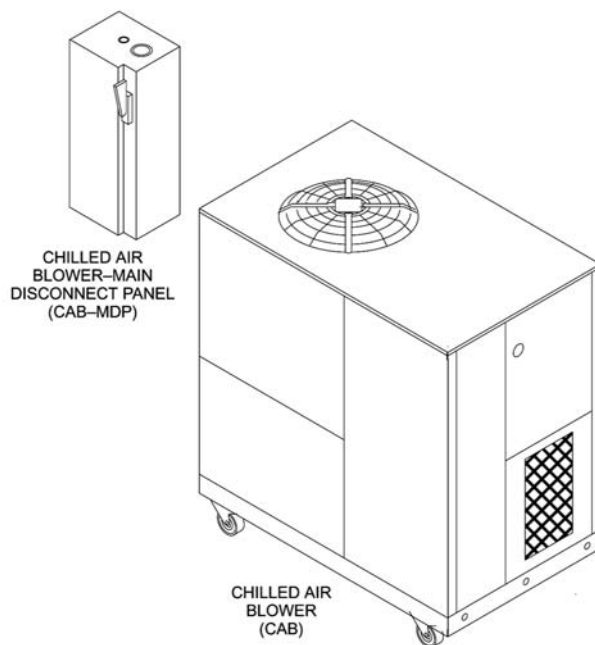
4 Additional System Options

- Integrated Patient Comfort Module (IPCM) Option (M3033LH) for LCC300 Magnet ONLY: The IPCM option adds a bore liner to the inside diameter of the RF Body Coil . The bore liner directs chilled air from the Chilled Air Blower (CAB) to the inside diameter of the RF Body Coil. The IPCM Option provides the following hardware components:
 - Chilled Air Blower (CAB) (shown in [Illustration 2-9](#))
 - Chilled Air Blower Main Disconnect Panel (CAB-MDP) (shown in [Illustration 2-9](#))

NOTE: The CAB-MDP requires an additional power feed, not powered from the MR system MDP or PDU.

- Thermal Bore Liner Assembly
- Modified Penetration Panel waveguide
- Bridge with notched sides
- Bore Liner hardware including hoses and connectors
- IPCM Option Key

Illustration 2-9: IPCM Cabinet & Panel



5 Relocatable Upgrade

Relocatable MR system upgrade catalogs are listed in the previous upgrade sections.

In addition to the system, the van may also need to be upgraded to accommodate the MR system upgrades. It is customer responsibility to arrange with the appropriate Van Manufacturer for any upgrade to be performed.

- Customer must contact Van Manufacturer to determine if van upgrade is needed, obtain quote of costs and timing, and schedule for van/system upgrade. Refer to [Section 6](#), Van Manufacturers Contact Information.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- MR system check and final calibrations will be performed by GE Field Service.

6 Van Manufacturers Contact Information

- Ellis & Watts
4400 Glen Willow Lake Lane
Batavia, OH 45103
Telephone: 513-752-9000
FAX: 513-752-4983 USA
- PDC Facilities
700 Walnut Ridge Drive, PO Box 900
Hartland, WI 53029-0900
Telephone: 262-367-7700
FAX: 262-367-7744 USA
- SMIT Mobile Equipment
Buys Ballotstraat 6
3261 La Oud-Beijerland
Holland
Telephone: (31) 186-6-14322
FAX: (31) 186-6-19367
- AK Specialty Vehicles
16745 South Lathrop Avenue
Harvey, IL 60426
Telephone: 708-596-5066
FAX: 708-596-2480 USA

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Chapter 3 Room Layouts

1 Proximity Limits

The maximum magnetic field in which the equipment can be located is listed in [Table 3-1](#).

Table 3-1: Proximity Limits

GAUSS (mT) LIMIT See Notes 1	EQUIPMENT	
0.5 GAUSS (0.05mT) OR LESS	Nuclear cameras	
1 GAUSS (0.1mT) OR LESS	- Positron Emission Tomography scanner - Linear Accelerator - Cyclotrons - Accurate Measuring scale - Image intensifiers - Color TV	- Video display (color, B/W, monochrome) - CT scanner - Ultrasound - Lithotripter - Electron microscope - Advantage Workstation with CRT Monitor
3 GAUSS (0.3mT) OR LESS	- Power transformers - Main electrical distribution transformers - Moving steel equipment such as: - Vehicular traffic - Fork lift trucks - Dumb waiters - Electric transport carts	- Loading dock (truck traffic) - Elevators - Escalators - Helicopters (See Note 2)
5 GAUSS (0.5mT) OR LESS	- Cardiac pacemakers - Neurostimulators	Biostimulation devices
10 GAUSS (1mT) OR LESS	- Magnetic tapes and floppy discs - Hard copy imagers - Line printers - Video Cassette Recorder (VCR) - Film processor - Credit cards, watches, and clocks - Telephone switching station - Water cooling equipment - HVAC equipment	- Major mechanical equipment room - Large steel equipment such as: - Emergency generators - Commercial laundry equipment - Food preparation area - Air conditioning chiller - Fuel storage tanks - Motors greater than 5 horsepower - X-ray tubes
30 GAUSS (3mT) OR LESS	- RFS Cabinet - Twin Accessory Cabinet (TAC) - System Cabinet	- MRCC equipment - GWHX equipment - Chilled Air blower (CAB) for IPCM Option
50 GAUSS (5mT) OR LESS	- GOC Computer Cabinet - LCD Color Monitor for OW (See note 3) - NB RF Amp Cabinet - HFD/PDU Cabinet - HFD-S/PDU Cabinet 3T Accessories Cabinet - Main Disconnect Panel - Magnet Monitor	- Telephones - Metal detector for screening - LCD Color Monitor for Advantage Workstation - Broadband (BB) RF Amplifier Cabinet for 8KW MNS (Option) - BrainWave HW Lite Cabinet (option) - CAB-MDP for IPCM Option
100 GAUSS (10mT) OR LESS	- Shield/Cryo Cooler Compressor Cabinet - Service Tool Magnet Power Supply Cabinet - Service Tool Shim Power Supply Cabinet	- Pneumatic Patient Alert Control Box - Oxygen Monitor (Option)
200 GAUSS (20mT) OR LESS	- Penetration Panel - Blower Box	- Magnet Rundown Unit - Remote Oxygen Sensor Module

Notes

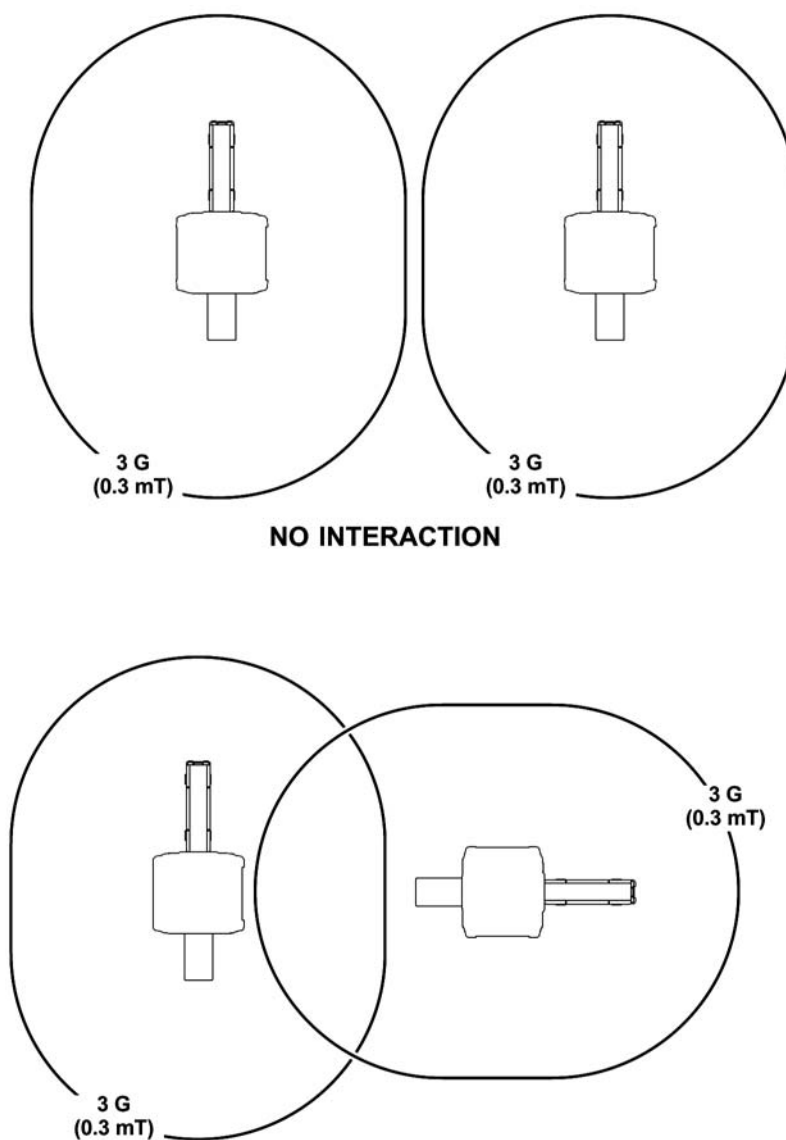
- Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable.
- Verify operating limits with provider of helicopter service
- If gauss limit is more than indicated for OW then contact GE to determine impacts on OW equipment.

2 Multiple MR Systems Site

2.1 Two Magnet Site Layout

For two magnet installations interaction can occur between the magnetic fields. For two magnets not to interact at all (including when bringing magnet to field) the 3 gauss lines of each magnet must not intersect, see [Illustration 3-1](#). If the 3 gauss lines intersect but remain outside each magnet's cryostat there will be interaction between the magnets when bringing to field, see [Illustration 3-1](#). The orientation of the magnets is irrelevant. Consult the GE Healthcare MR Siting & Shielding group for closer proximity of magnets.

Illustration 3-1: Two Magnet Installation



2.2 Equipment Room Shared By Multiple MR Systems

2.2.1 Introduction

When the Equipment Room is shared by more than one MR system of the same field strength there is a potential for cross-talk of RF energy between the MR systems. RF cross-talk may cause noise artifacts in images. Proper planning and installation of the multiple systems in the shared Equipment Room can reduce the potential for cross-talk.

The potential for cross-talk exists when the RF transmit cables and equipment of two or more MR systems are located in the same Equipment Room. For example, when one system is transmitting, the other system could be in receive mode and therefore pick up the RF energy being transmitted resulting in a cross-talk scenario.

NOTE: The potential for cross-talk exists for RF transmit cables and equipment that produces RF that are part of a non-GE MR System of the same field strength.



NOTICE

The RF Screen of the Magnet Room for each system needs to meet the RF Attenuation specifications in [Chapter 8, Section 1, RF Shielded Room Specifications](#).

The following subsections provide requirements for shared Equipment Room design, layout, and installation which reduce the potential for RF cross-talk.

2.2.2 Equipment Cabinets Relative Locations

The following are requirements for locating equipment cabinets of one MR system relative to the other MR system equipment cabinets.

- Maximize separation distance between the RF transmitter (RF Amplifier) of one MR system and the RF receiver (RRF Chassis) of the other MR system of the same field strength.
- The RF transmitter (RF Amplifier) of one MR system and the RF receiver (RRF Chassis) of the other MR system of the same field strength must be separated by a minimum of > 6.6 feet (2 meters) in all directions, see [Illustration 3-2](#) and [Illustration 3-3](#).

NOTE: The placement of the chiller, compressors and other non-RF Transmit/Receive MR System equipment relative to RF Transmit/Receive equipment should not be an RF noise issue.

Illustration 3-2: Multiple Signa EXCITE HD 3.0T (Release 12.x) Systems Electronics Cabinets Spacing

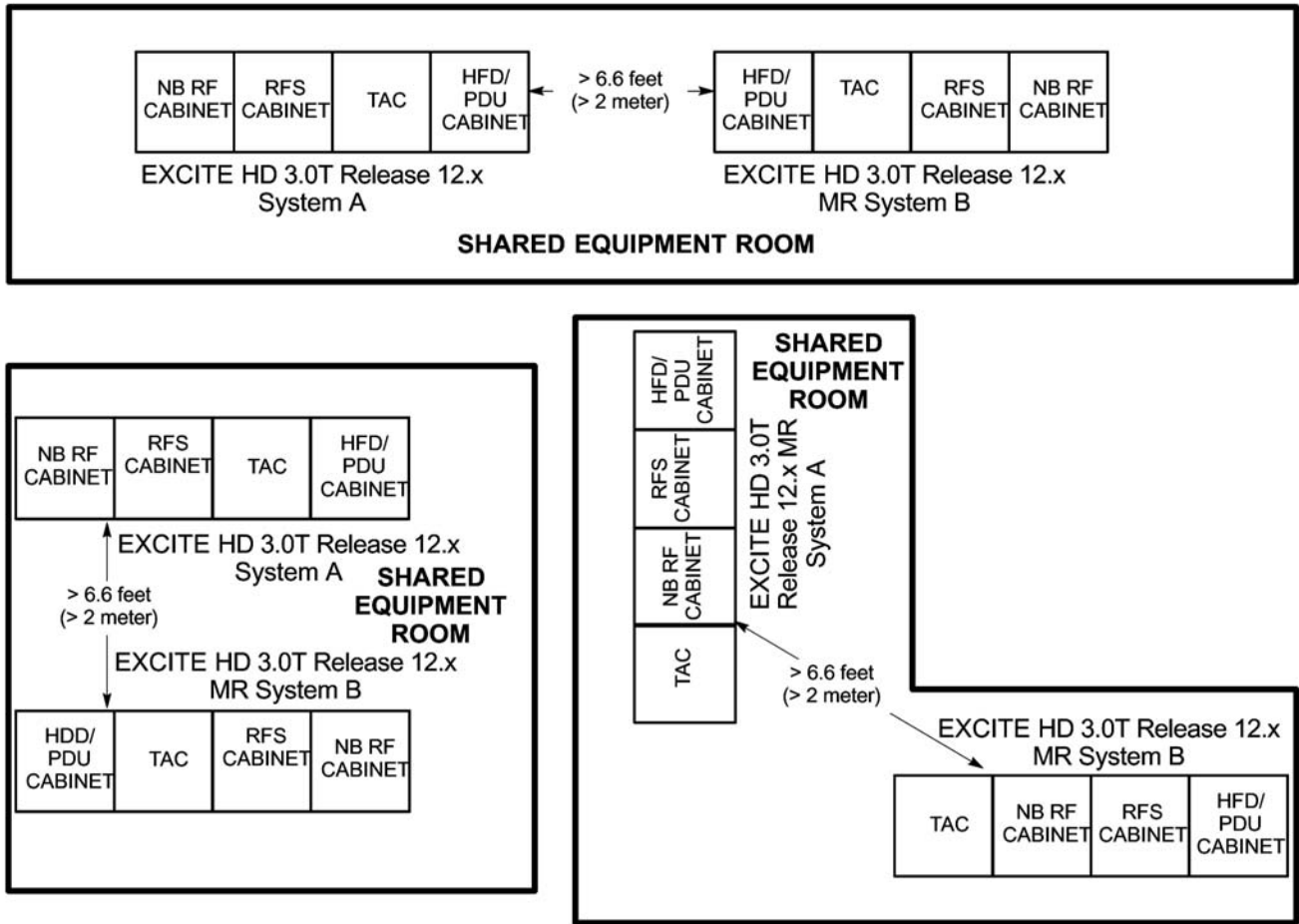
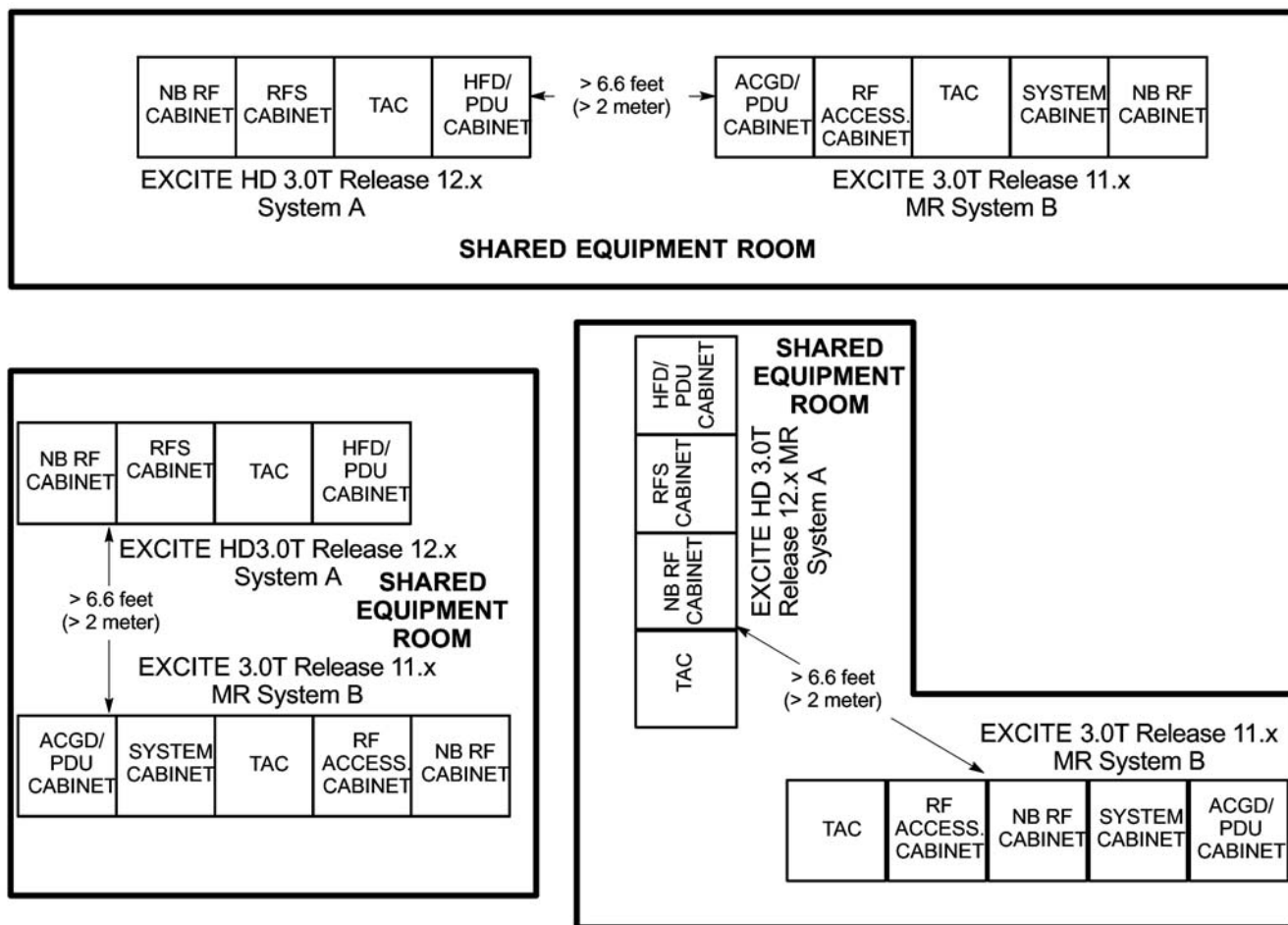


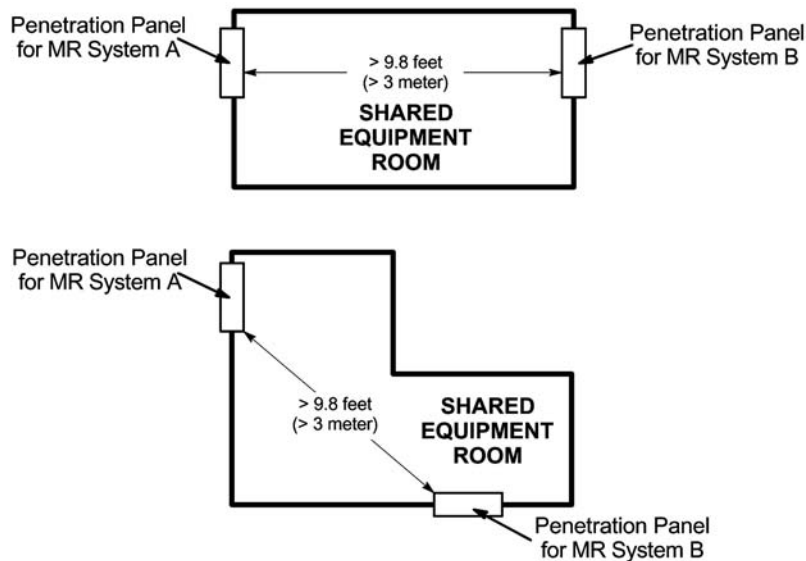
Illustration 3-3: Multiple Signa EXCITE HD 3.0T (Release 12.x) & Signa EXCITE 3.0T (Release 11.x) Systems Electronics Cabinets Spacing



2.2.3 Penetration Panels Locations

The following are requirements for locating the RF Shielded Room Penetration Panel of one MR system relative to the other MR system RF Shielded Room Penetration Panel.

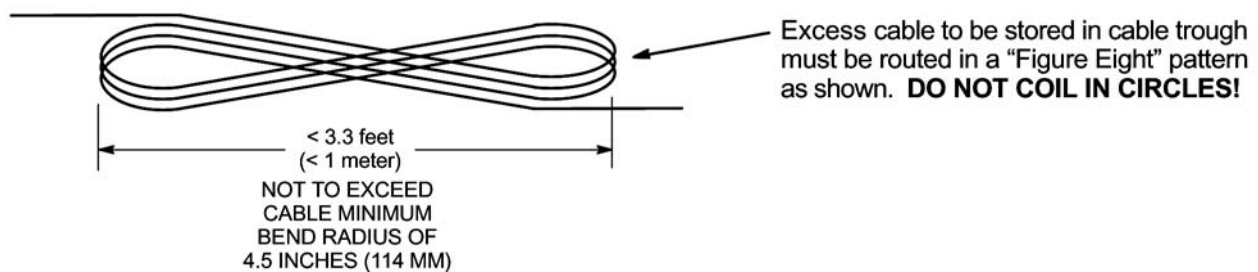
There must be > 9.8 feet (3 meters) separation between the Penetration Panels of each system sharing the Equipment Room space, see [Illustration 3-4](#).

Illustration 3-4: Multiple MR Systems Penetration Panel Spacing

2.2.4 System Cables Requirements

The following are requirements for locating and managing excess RF Receive and Transmit cables of the MR systems sharing the Equipment Room.

- In the shared Equipment Room there must be > 6.6 feet (2 meters) separation between the RF receive cables of one system and the RF transmit cable of the other system.
- There must be > 6.6 feet (2 meters) separation between the system interconnect cables of each system sharing the Equipment Room space.
- Transmit cables in the Equipment Room must be cut to length to minimize excess cable length reducing the potential for signal coupling with other cables. No excess transmit cable can be stored in the Equipment Room.
- Receive cables excess length must be stored in a "figure 8" with overall dimension of <3.3 feet (<1 meter), see [Illustration 3-5](#).

Illustration 3-5: Proper Storage Of Excess Receive Cables

3 Minimum Delivery Route Sizes and Capacity

Enter descriptive content here.

Table 3-2 lists minimum actual clearance opening dimensions for doors and hallways required by the MR system upgrade equipment. Installation or replacement of components listed in Table 3-2 must be taken into consideration when determining hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to Chapter 9, Section 1, Shipment for upgrade components shipping dimensions.

Table 3-2: Minimum Hallway/Door Dimensions

Component	Minimum Hallway/ Door Width*		Minimum Hallway/ Door Height*		Comments
	in.	mm	in.	mm	
Operator Workspace Table	32	813	80	2032	
Equipment Cabinets	36	914	80	2032	
Note * Minimum hallway and door dimensions are actual clearance openings. Width and height of rigging equipment is not included in above dimension.					

Table 3-3: Component Dimensions For Installation/Replacement

Component	Approximate Weight lbs (kg)	Overall Dimensions W x D x H in. (mm)	Comments
Split Bridge for LCC300 Magnet Enclosure	40 (18)	21.5 x 77.3 x 7 (546 x 1969 x 177.8)	Room dimensions in front of the Magnet MUST allow for bridge installation/servicing and Gradient Coil Replacement.
Bridge for 3T94 Magnet Enclosure	90 (39)	19.5 x 159 x 6 (495 x 4039 x 152.4)	Room dimensions in front of the Magnet MUST allow for bridge installation/servicing and Gradient Coil Replacement.

4 MR System Interconnects Routing

The customer, architect/consultant, and contractor must consider the design/method for routing the MR system interconnects to minimize potential adverse performance impact. Refer to [Table 3-4](#) for MR system interconnects routing requirements and recommendation.

Table 3-4: MR System Interconnects Routing Requirements/Recommendations

Area	Requirements
Entire MR System	- All MR system interconnects routing must be in accordance with local and national codes. - Consult local/national code for interconnects separation requirements (i.e. signal, power, water, air, etc.). - Must protect fiber optic interconnects, water lines, hoses and tubing from physical damage including liquids (i.e. condensation, coolants, etc.) - The majority of MR system cables require routing that accommodates the dimensions of the cable connectors. - Raceway or conduit for routing interconnects must be sized to allow for the dimension of the connectors. - Raceway or conduit always to be sized to allow the cable to pass through with all other cables already installed. - The MR system interconnect cables are FT4 or equivalent rated, not plenum rated. - If the area under any access floor is used for an air plenum then the cables may have to be in raceway depending on local and national codes. - MR system interconnects must be accessible for equipment servicing including troubleshooting and future enhancements or upgrades.
Magnet Room	- Metal access floor tiles are NOT allowed anywhere in the Magnet Room, refer to requirements in Chapter 8, Section 6.4, Magnet Room Floors. - Only non-magnetic metal material can be used when routing cables in the Magnet Room. - Any electrically conductive materials utilized for interconnects routing (i.e. raceway, access flooring) must comply with Chapter 8, Section 1, RF Shielded Room Requirements to minimize the possibilities of electrical discharge which can cause RF broadband noise. - All electrical and mechanical connections and fasteners including screws, nails, nuts, bolts, clips clamps, concrete anchors, seismic anchors, etc. must be tightened and secured to supplier specifications so as not to become a potential broadband noise source. - No mechanical fasteners are allowed within 2 feet (0.61 m) of the Gradient Cables inside the RF Shield due to the potential of screws and metal interaction becoming a source of broadband noise. - Unsecured or overlapped electrically conductive materials (i.e cable routing dividers, trough/duct z-joints) must not be located within 2 feet (0.61 m) of the Gradient Cables. - RF broadband noise can be avoided with quality design and construction that will remain solid over time and usage (i.e. no loose or micro vibrating materials). - All fasteners must be solid locking devices such as t-nuts, PEM nuts or welded nuts; no self tapping screws allowed. - Floor or wall ducts/raceways used in the Magnet Room must meet the requirements in Chapter 8, Section 1, RF Shielded Room Requirements and Chapter 8, Section 6.4, Floors. - Cable routing methods must not interfere with an unobstructed path from the Patient Table to the area directly behind the Magnet Enclosure Rear Pedestal for MR personnel usage.

Table 3-4: MR System Interconnects Routing Requirements/Recommendations (cont'd)

Area	Requirements
Recommendations	
• Raceway or conduit for routing interconnects should be oversized to allow for the possibility of additional cables being added as the system is upgraded in the future. • Conduit should not be used for running the majority of GE MR system cables in the Magnet Room due to the number and size of conduits needed.	

5 Flooring

NOTE: The following information applies to areas of the site impacted by the MR system upgrade.

Flooring consists of all materials above the structural floor support including subflooring and equipment support/mounting. Refer to [Table 3-5](#) for requirements and recommendations.

Table 3-5: MR System Flooring Requirements/Recommendations

Area	Requirements	Recommendations
Entire MR System	- Flooring materials must support the MR system equipment weight, refer to Section 7, Floor Loading. - Floors must support the equipment and any transport device needed to move the equipment. - Flooring throughout the system including Magnet Room must be in accordance with local and national codes. - Floor design must consider the MR system interconnect cables are FT4 or equivalent rated, not plenum rated.	The area under access flooring if used for an air plenum is recommended to provide a minimum 10 in. (254mm) of clear space. Cabling, plumbing (water lines), etc. routed under the access floor may affect air flow and needs to be considered if used as an air conditioning plenum.
Magnet Room	Refer to Chapter 8, Section 6.4 , Magnet Room Floors	

6 Upgrades Special Siting Considerations

6.1 System Monitoring and Support Connectivity

One of the system monitoring and support connectivity configurations listed in [Table 3-6](#) must be provided for system installation and serviceability purposes. The broadband network connection and telephone lines are to be provided and paid for by the customer.

Table 3-6: System Monitoring & Support Connectivity Requirements

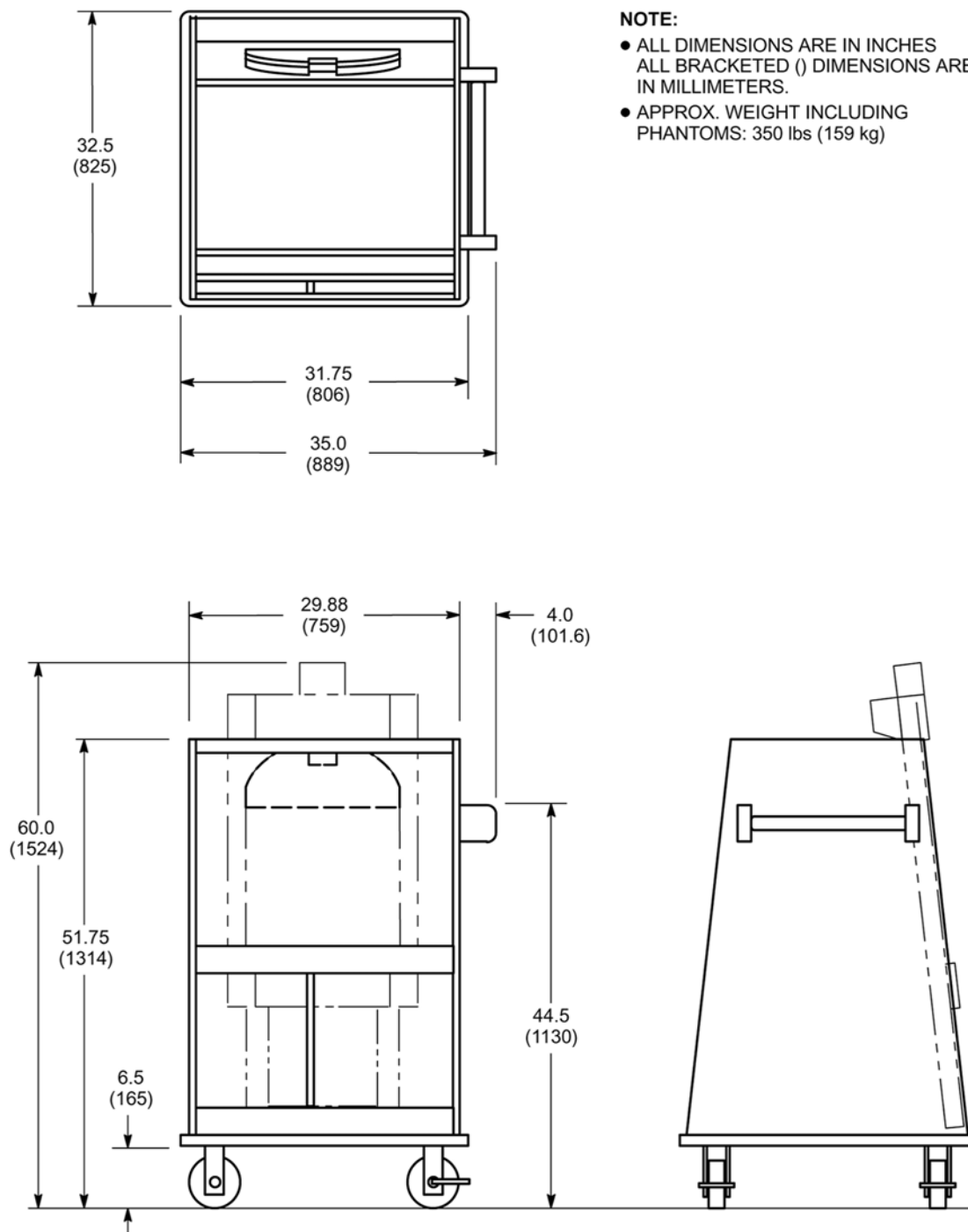
Configuration	Connection Type	Use/Location
Broadband Network Connection & Telephone Line (Recommended)	Two Broadband Internet Accessible connections with individual Static IP addresses	One access located near the Operator Workspace (OW) in the Control Room.
	See Note 1	One access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring of magnet pressure levels. This Broadband connection must not lose power when the MR system is shutdown.
	One voice-grade telephone line (voice line)	Available for Service Personnel use, located in the Control Room
Multiple telephone lines (Alternate)	One voice-grade telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	One line must be a dedicated direct-distance-dialing voice-grade line (data line)	Access located near the Operator Workspace (OW) in the Control Room. See Notes 2 & 3
	One line must be a dedicated direct-distance-dialing voice-grade line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. See Notes 2 & 3
Notes For Europe: An ISDN Connection with static IP address may be substituted for Broadband Internet Accessible connections. - A dedicated direct-distance-dialing voice-grade telephone line can be shared for Operator Workspace (OW) and Magnet Monitor (MSM) requirement through the use of a multiplexer box. The following multiplexer boxes are available for customer purchase. 46-328475P1 4 Line Phone Multiplexer box; 115 VAC input power 46-328475P3 4 Line Phone Multiplexer box; 220 VAC input power If the customer chooses not to purchase the multiplexer box then the customer must provide an additional line for each requirement as stated in this table. - If a Multiplexer Box is used then the Magnet Monitor MUST be Channel 1 to allow for call out after a power outage.		

6.2 Customer Site Storage Requirements

The MR system has several system accessories such as system phantoms and surface coils. There are many optional surface coils available. Storage space for system accessories and supplies needs to be planned for and included in room layout drawings.

Refer to *Direction 5182674 GE Healthcare Signa MRI Scanners Customer Site Storage Requirements* for size and weight of surface coils and phantoms that require customer provided storage locations. Direction 5182674 is available on-line at <http://www.gehealthcare.com/company/docs/siteplanning.html>.

SPT Phantom Set Shipping and Storage Cart is provided as part of the MR system. System Performance Test (SPT) provides the customer and GE Service with a means to quickly verify whether critical parameters affecting image quality are within specifications. The test uses a set a phantoms and a nesting plate for proper positioning of the phantoms on the Patient Table. The phantom set and nesting plate are provided on a cart which protects the pieces during shipment and storage at site. The cabinet is not magnetic therefore it can be stored inside the Magnet Room if so desired and moved to the Patient Table for ease of positioning the phantoms. See [Illustration 3-6](#) for cart dimensions information.

Illustration 3-6: SPT Phantom Set Shipping/Storage Cart

6.3 IPCM Option For LCC300 Magnets ONLY

Integrated Patient Comfort Module (IPCM) Option consists of hardware and software that can be used in combination with fMRI scanning. The IPCM Option hardware consists of the Chilled Air Blower (CAB), CAB Main Disconnect Panel (CAB-MDP), Bore Liner hardware, modified Penetration Panel waveguide, Bridge with notched sides, Bore Liner hardware including hoses and connectors, and IPCM Option Key.

The CAB provides chilled air to the 2 piece Bore Liner via an air hose routed through the Penetration Panel (provided waveguide) to the back end of the Rear Pedestal. At the Rear Pedestal additional provided hardware, hoses and connectors are used to connect the air hose to both pieces of the Bore Liner. Refer to [Chapter 7, Section 7](#), IPCM Option Interconnects for interconnect specifications.

NOTE: The CAB-MDP is not powered from the MR system MDP or PDU and therefore requires facility provided power. The CAB-MDP must also be connected to the MR system MDP Emergency Power Off control circuit via a customer supplied cable.

The CAB-MDP requires customer supplied facility power. Customer supplied Emergency Power Off control wiring is required between the CAB and CAB_MDP. Refer to [Chapter 6, Section 9](#), IPCM Option Power Requirements for power specifications.

The CAB requires airflow clearance for front, back, and top during operation. Refer to [Section 8.2](#), IPCM Option for dimensions information.

7 Floor Loading



NOTICE

It is the responsibility of the customer to obtain any and all approvals necessary for the construction of equipment support and seismic anchoring.

7.1 MR System Components

Refer to weight and dimensions shown in the illustrations of the upgrade components for the upgrade catalogs listed in [Section 8.1](#), Component Dimensions.

8 Upgrades Component Dimensions

8.1 Component Dimensions

To assist in completing your room layout, refer to [Table 3-7](#) for the list of component illustrations and the upgrade catalog numbers which provide the new components. Included are items which may impact site layout or need to be taken into consideration when determining delivery route. Modifications to existing cabinets are not included. Catalogs may list the same equipment but differ due to other modification hardware.

Table 3-7: Component Information Listed By Upgrade Catalog

Upgrade Catalog	Equipment	Dimension Information Location
M1000NT	LCD Monitor	Host LCD illustration in Section 8.4 , Operator Workspace
M3334WK M3334WL	3T94 Magnet Enclosure Bridge	Section 8.5 , 3T94 Magnet Enclosure Bridge
	GOC Cabinet, 15 inch LCD Monitor, mouse and pad, SCSI Tower	Section 8.4 , Operator Workspace
	Pneumatic Patient Alert	Section 8.6 , Pneumatic Patient Alert
	HFD/PDU Cabinet	Section 8.7 , HFD/PDU Cabinet (MR3)
	3.0T RFS Cabinet	Section 8.8 , RFS Cabinet (MR2)
	Penetration Panel	Section 8.9 , Penetration Panel
M3334WH	Broadband RF Amplifier Cabinet	Section 8.3 , Broadband RF Amplifier Cabinet For 8KW MNS Option

8.2 IPCM Option

Table 3-8: Integrated Patient Comfort Module (IPCM) Equipment List of Dimension Illustrations

Illustration Name	Illustration Number
Chilled Air Blower (CAB)	Illustration 3-7
Chilled Air Blower - Main Disconnect Panel (CAB-MDP)	Illustration 3-8

Illustration 3-7: Chilled Air Blower (CAB)

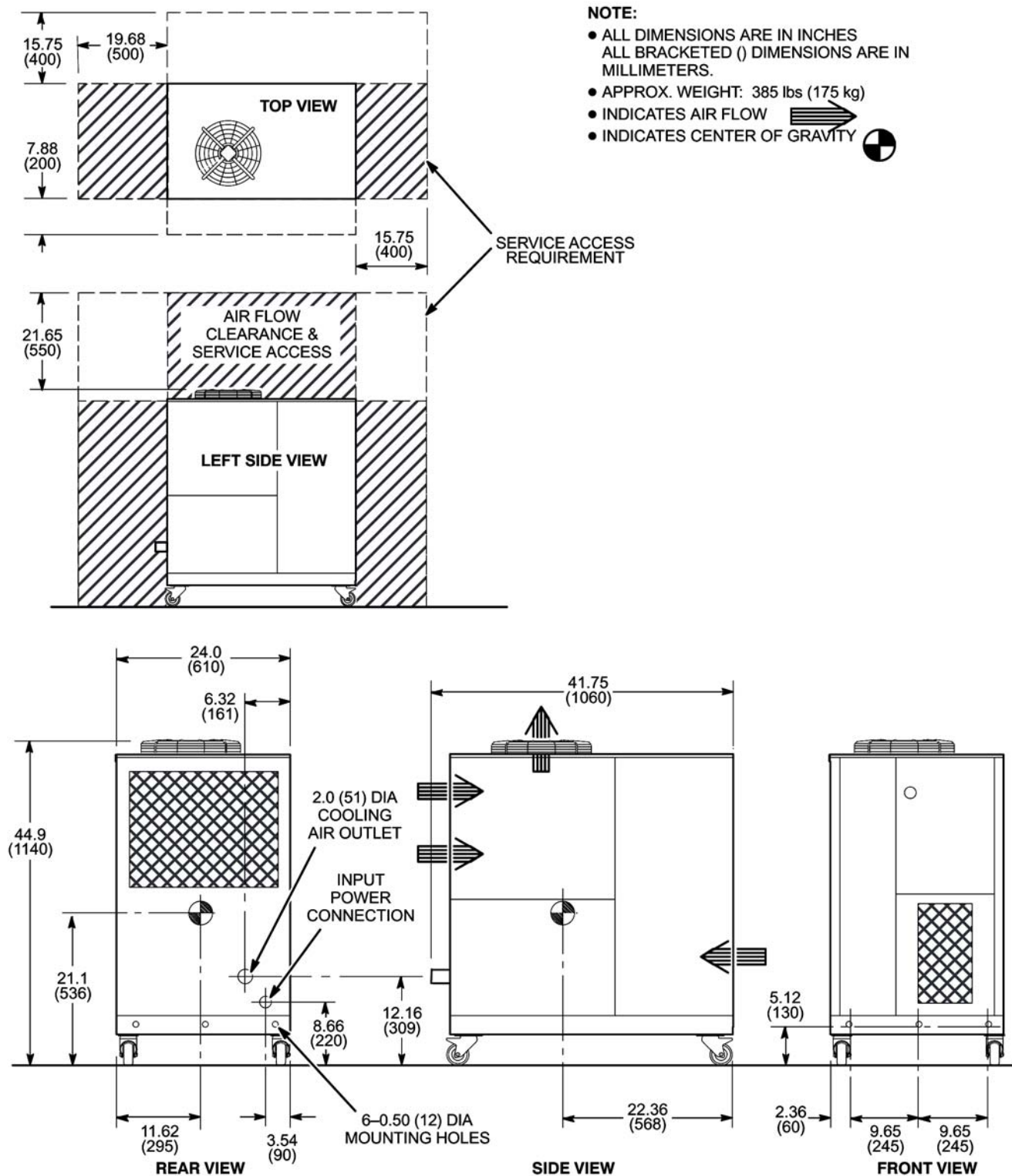
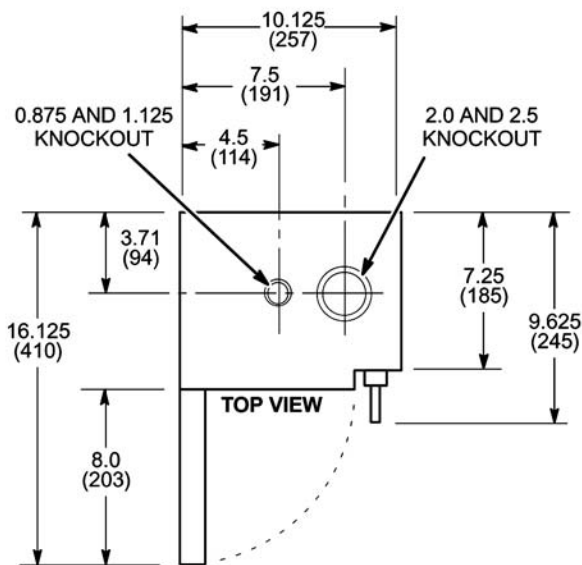
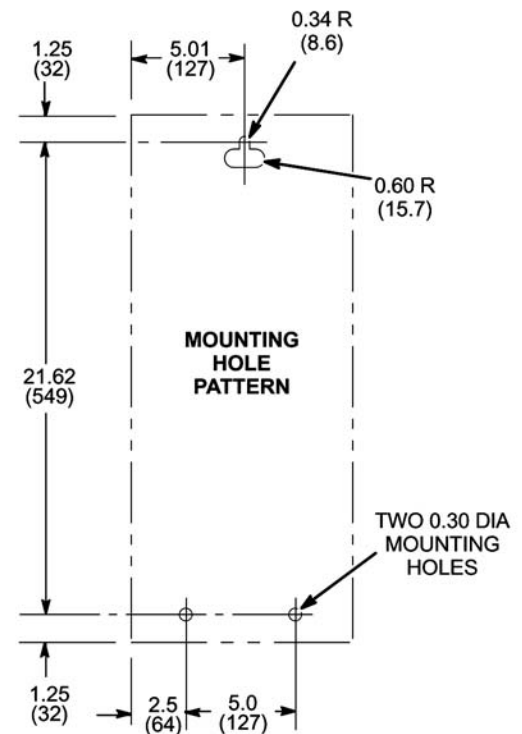
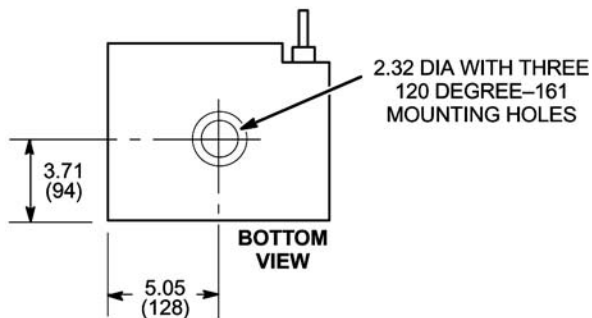
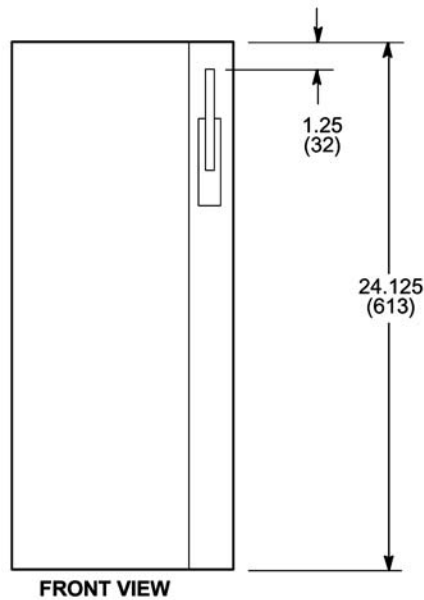


Illustration 3-8: Chilled Air Blower-Main Disconnect Panel (CAB-MDP)

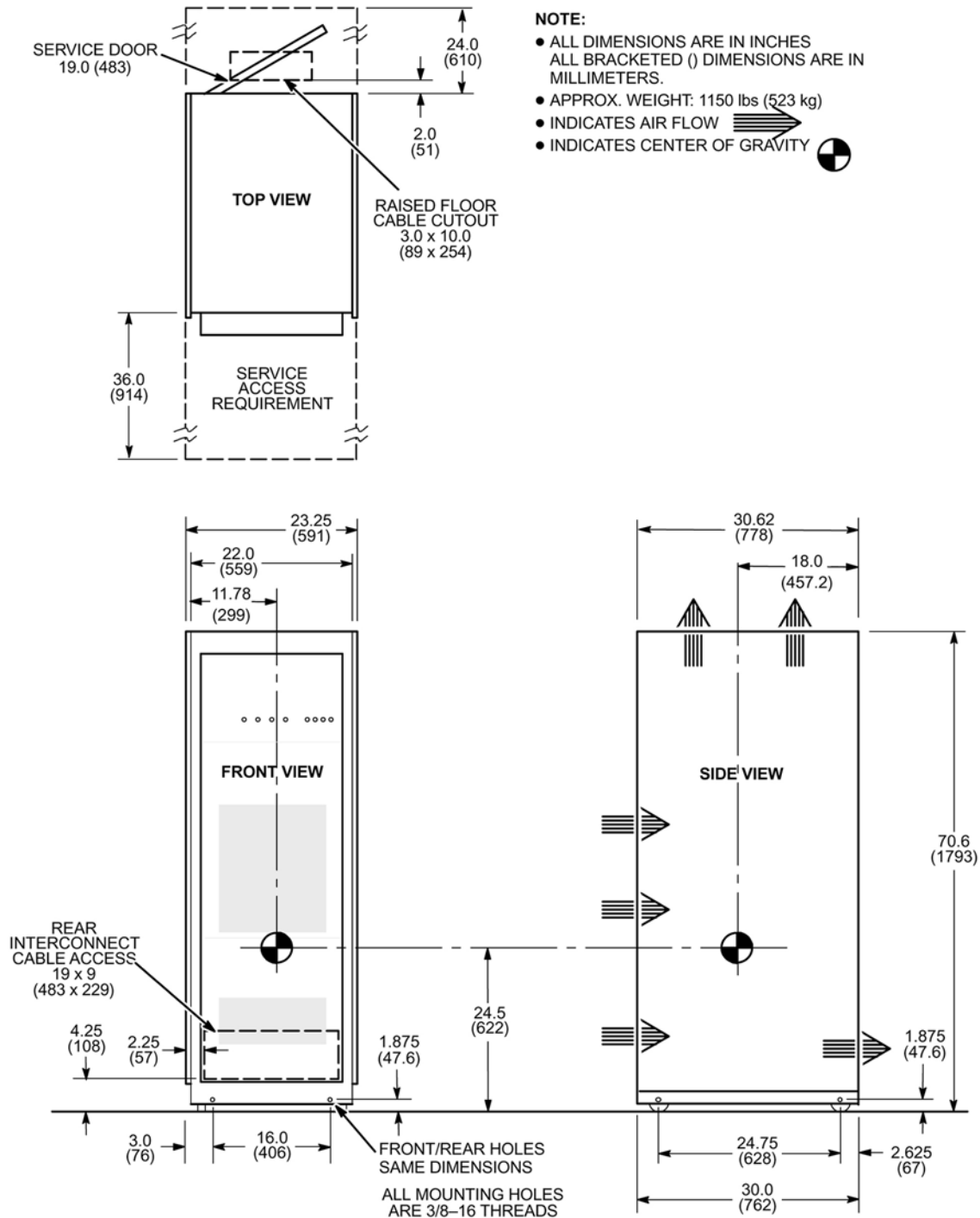
**NOTE:**

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 45 lbs (20.4 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY



8.3 Broadband RF Amplifier Cabinet For 8KW MNS Option

Illustration 3-9: Broadband (BB) RF Amplifier Cabinet (MR9) - 8KW MNS Option



8.4 Operator Workspace

Table 3-9: Operator Workspace List of Illustrations

Illustration Name	Illustration Number
Operator Workspace (OW1) Overall Dimensions	Illustration 3-10
GOC Computer Cabinet (OW1 A2)	Illustration 3-11
Operator Workspace Components Position on Table Top - Host LCD	Illustration 3-12
Operator Workspace Components Position on Table Top - SCSI Tower and 15 Inch LCD	Illustration 3-13
Operator Workspace Components Position on Table Top - Keyboard	Illustration 3-14

Illustration 3-10: Operator Workspace (OW1) Overall Dimensions

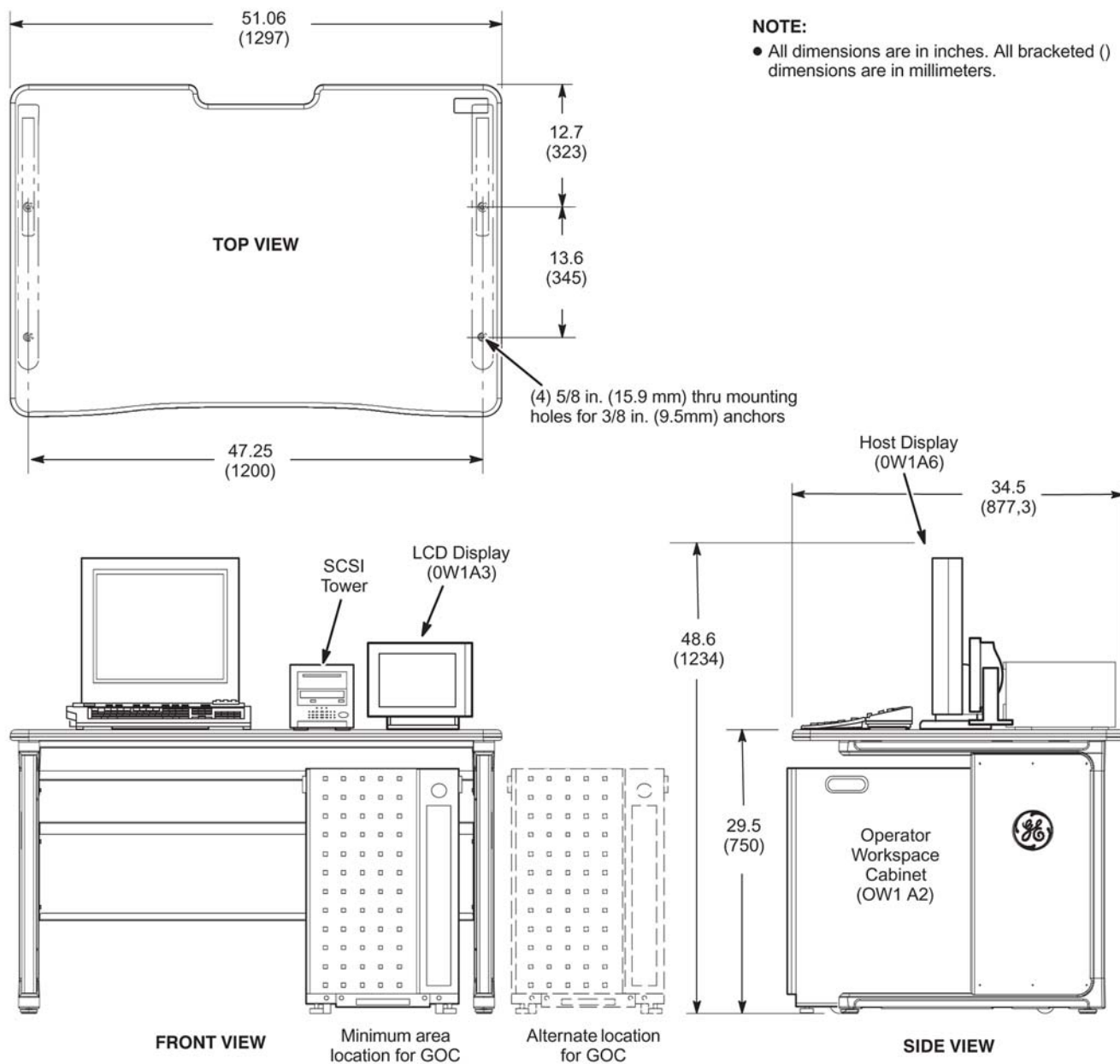


Illustration 3-11: GOC Computer Cabinet (OW1 A2)

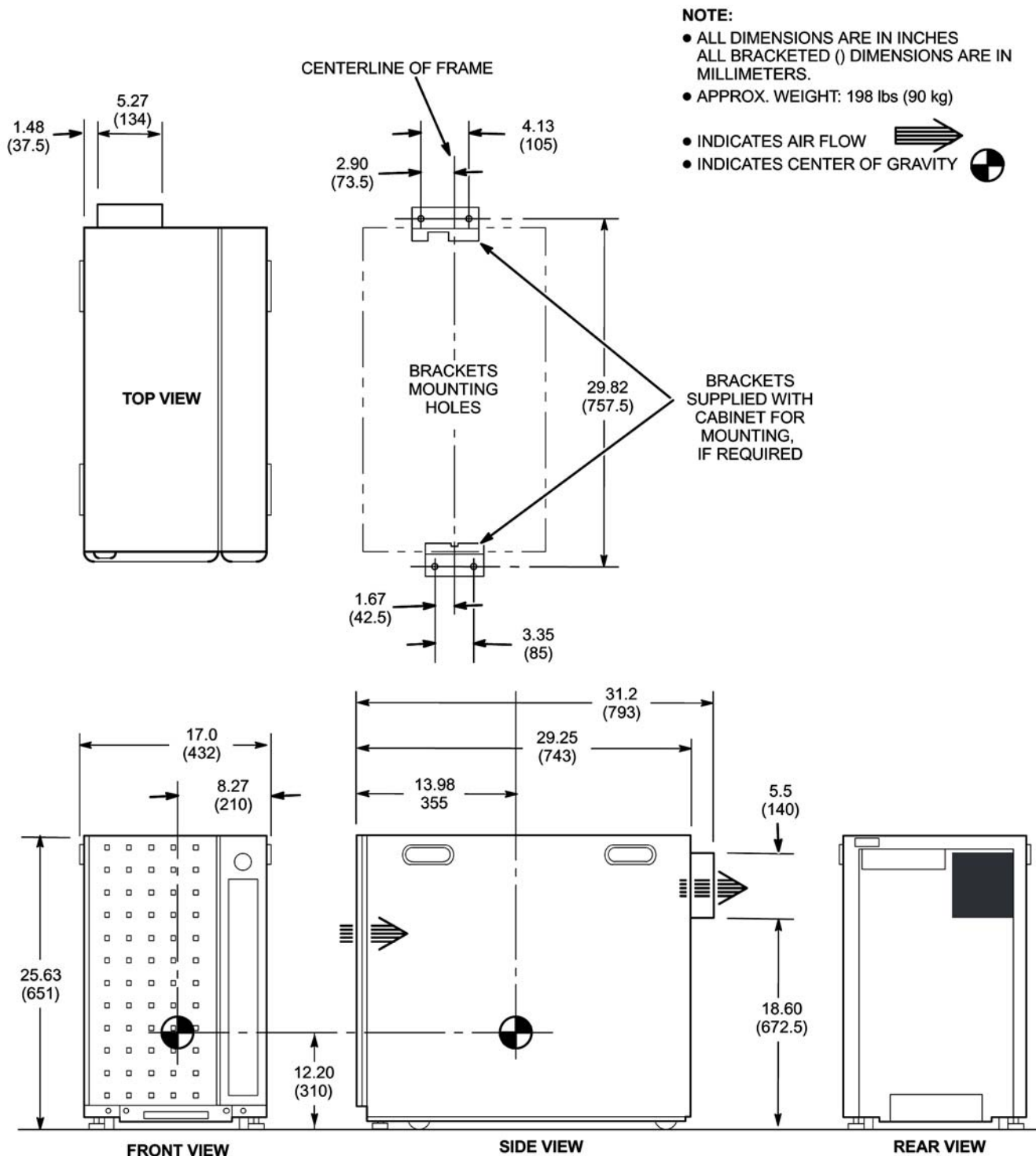
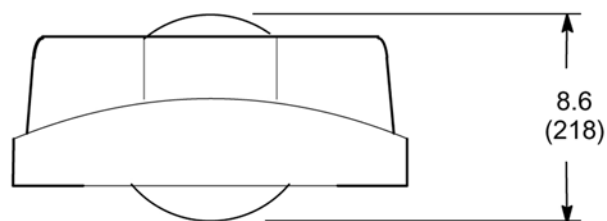


Illustration 3-12: Operator Workspace Components Position on Table Top - Host LCD**TOP VIEW****NOTE:**

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

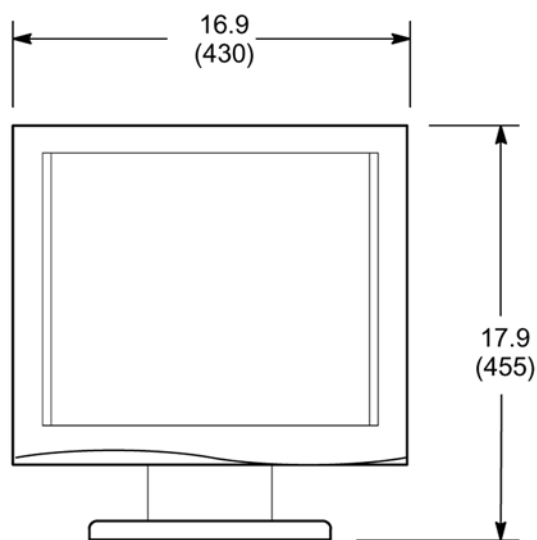
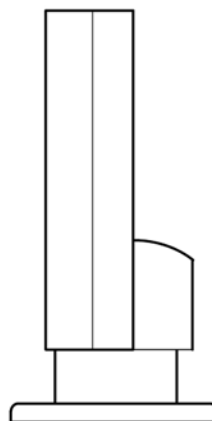
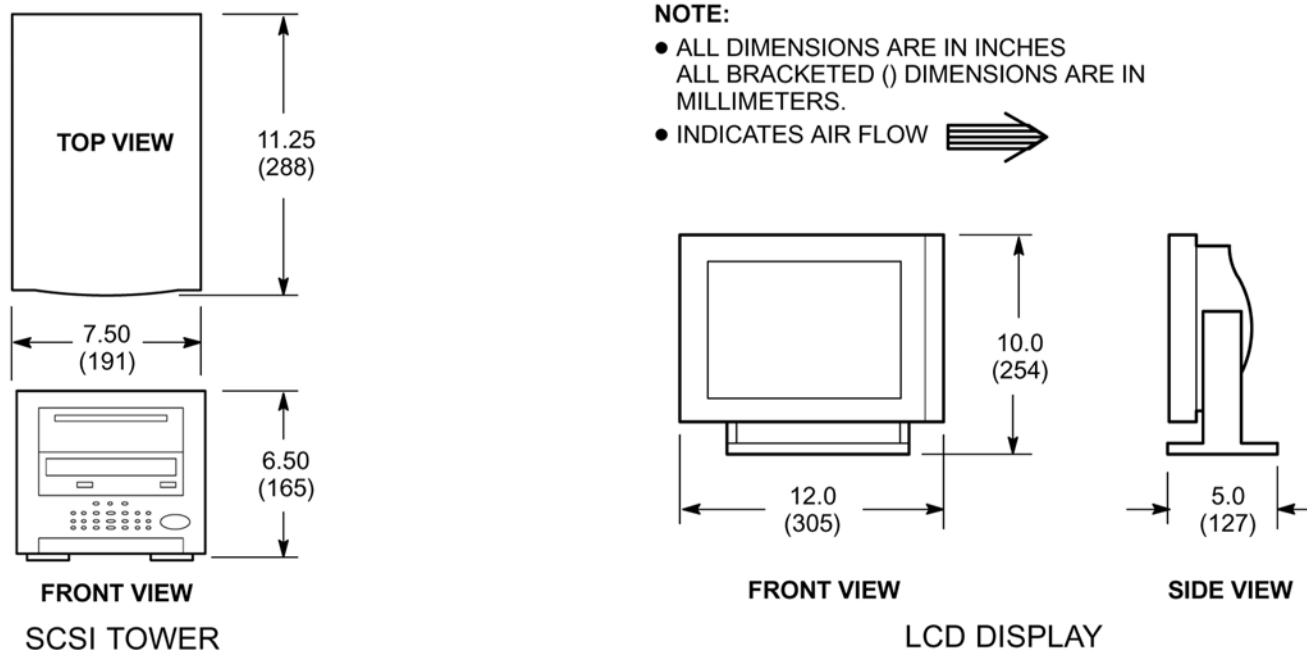
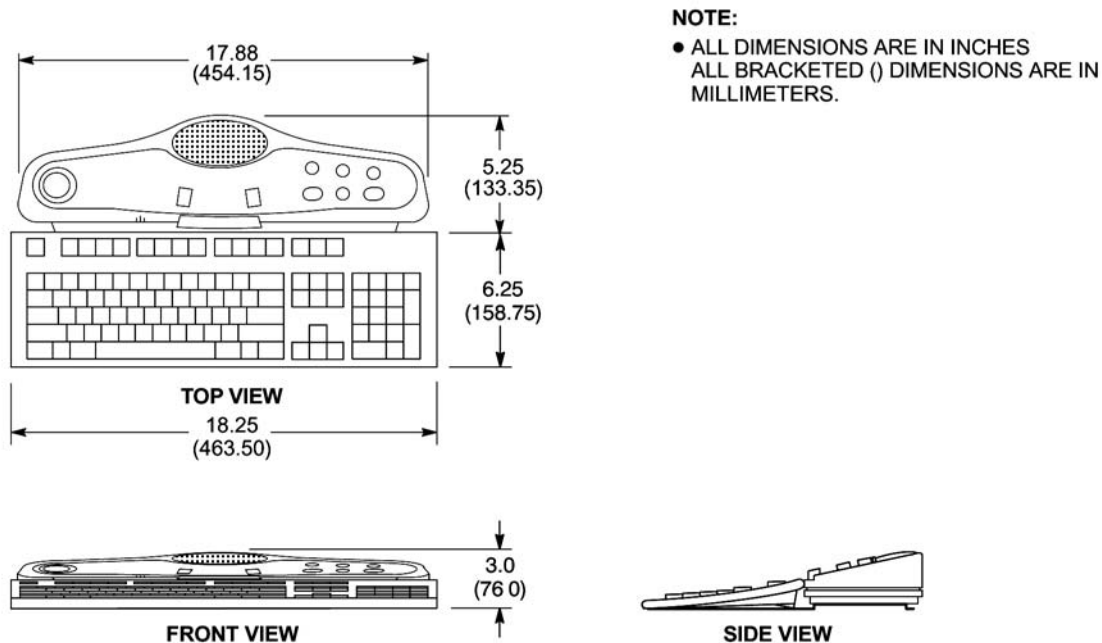
**FRONT VIEW****SIDE VIEW**

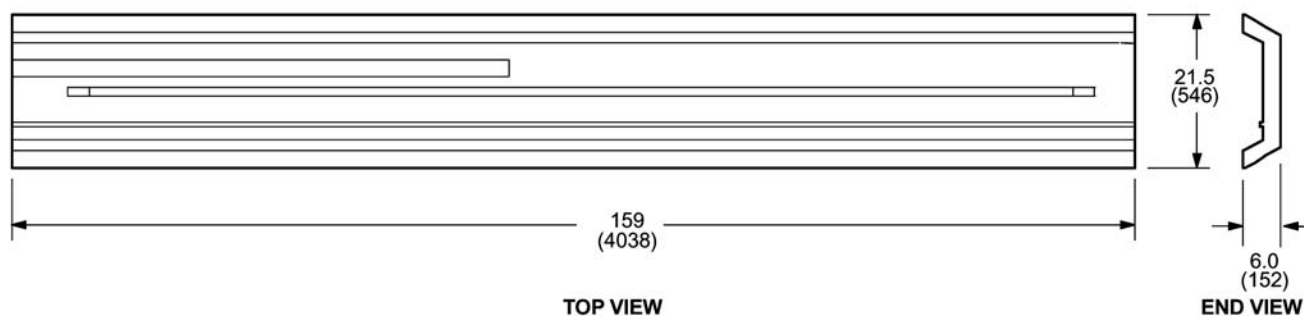
Illustration 3-13: Operator Workspace Components Position on Table Top - SCSI Tower and 15 Inch LCD**Illustration 3-14: Operator Workspace Components Position on Table Top - Keyboard**

8.5 3T94 Magnet Enclosure Bridge

Illustration 3-15: 3T94 Magnet Enclosure Bridge

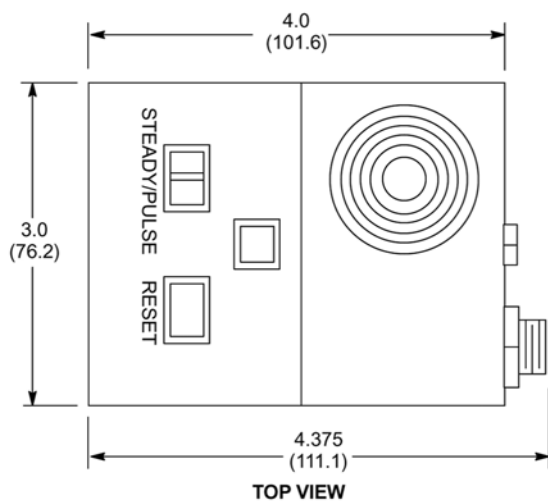
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 90 lbs (39 kg)



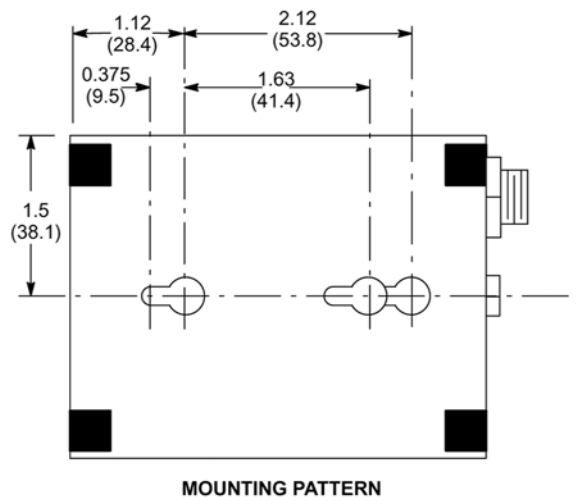
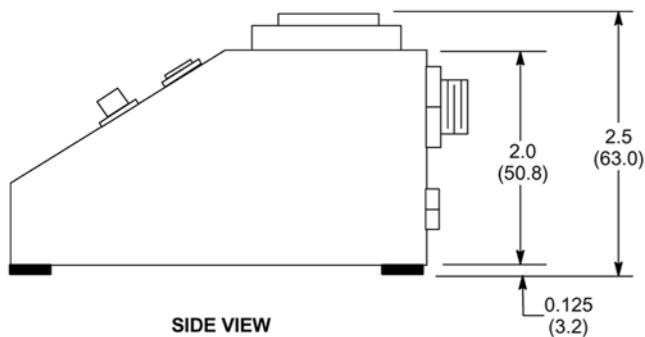
8.6 Pneumatic Patient Alert

Illustration 3-16: Pneumatic Patient Alert Control Box (PA1)



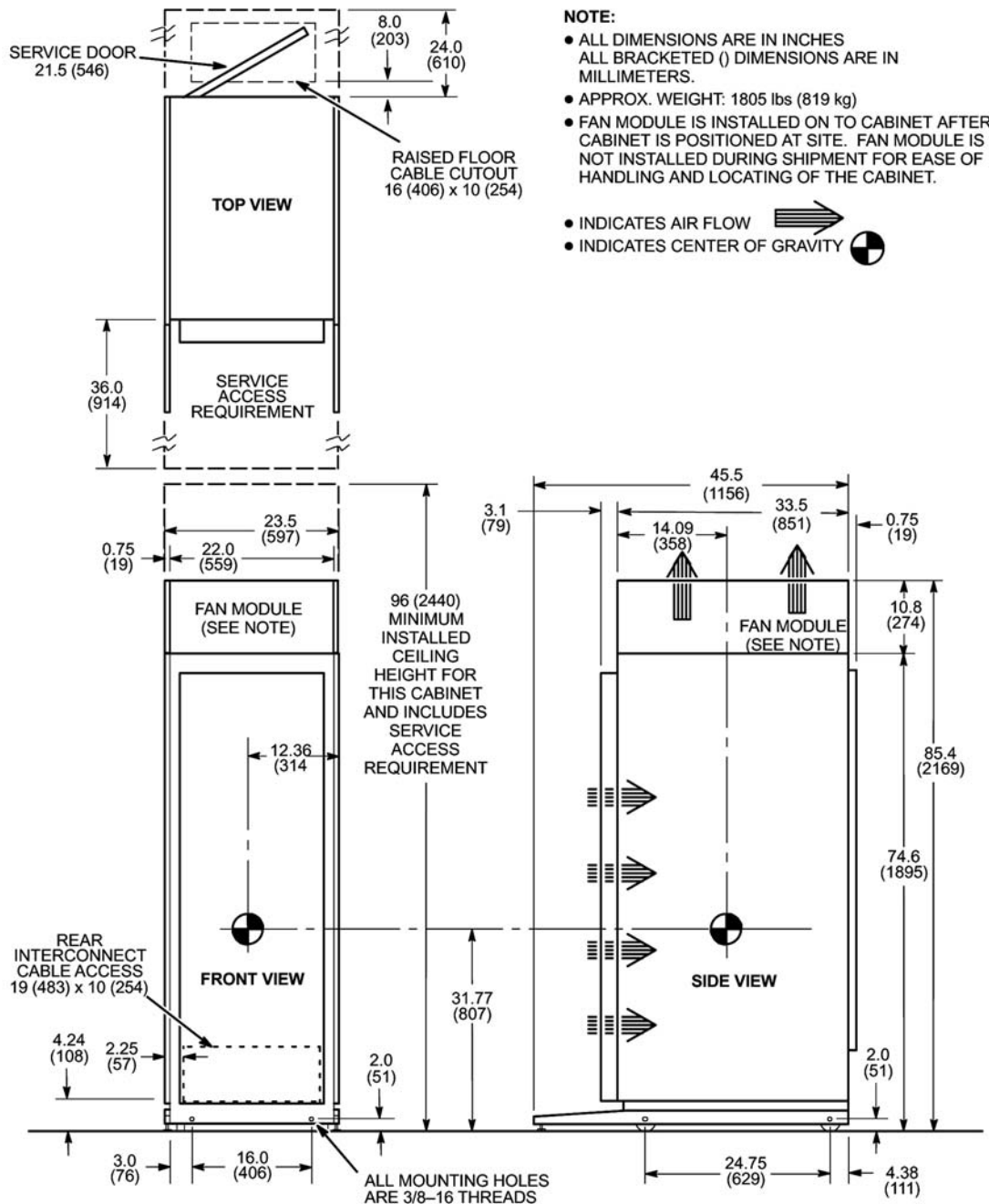
NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 0.5 lbs (0.2 kg)



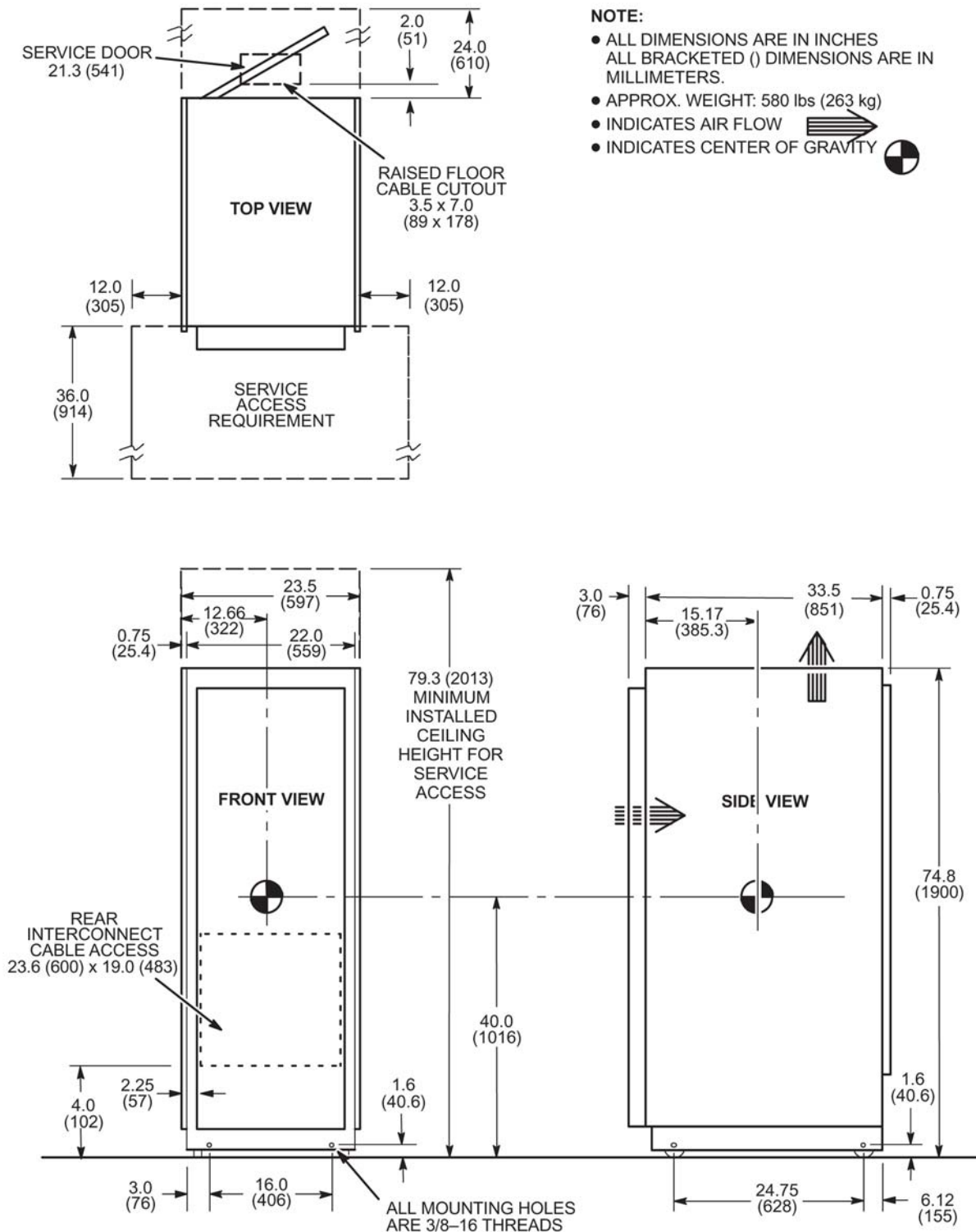
8.7 HFD/PDU Cabinet (MR3)

Illustration 3-17: HFD/PDU Cabinet (MR3)



8.8 RFS Cabinet (MR2)

Illustration 3-18: 3T RFS Cabinet



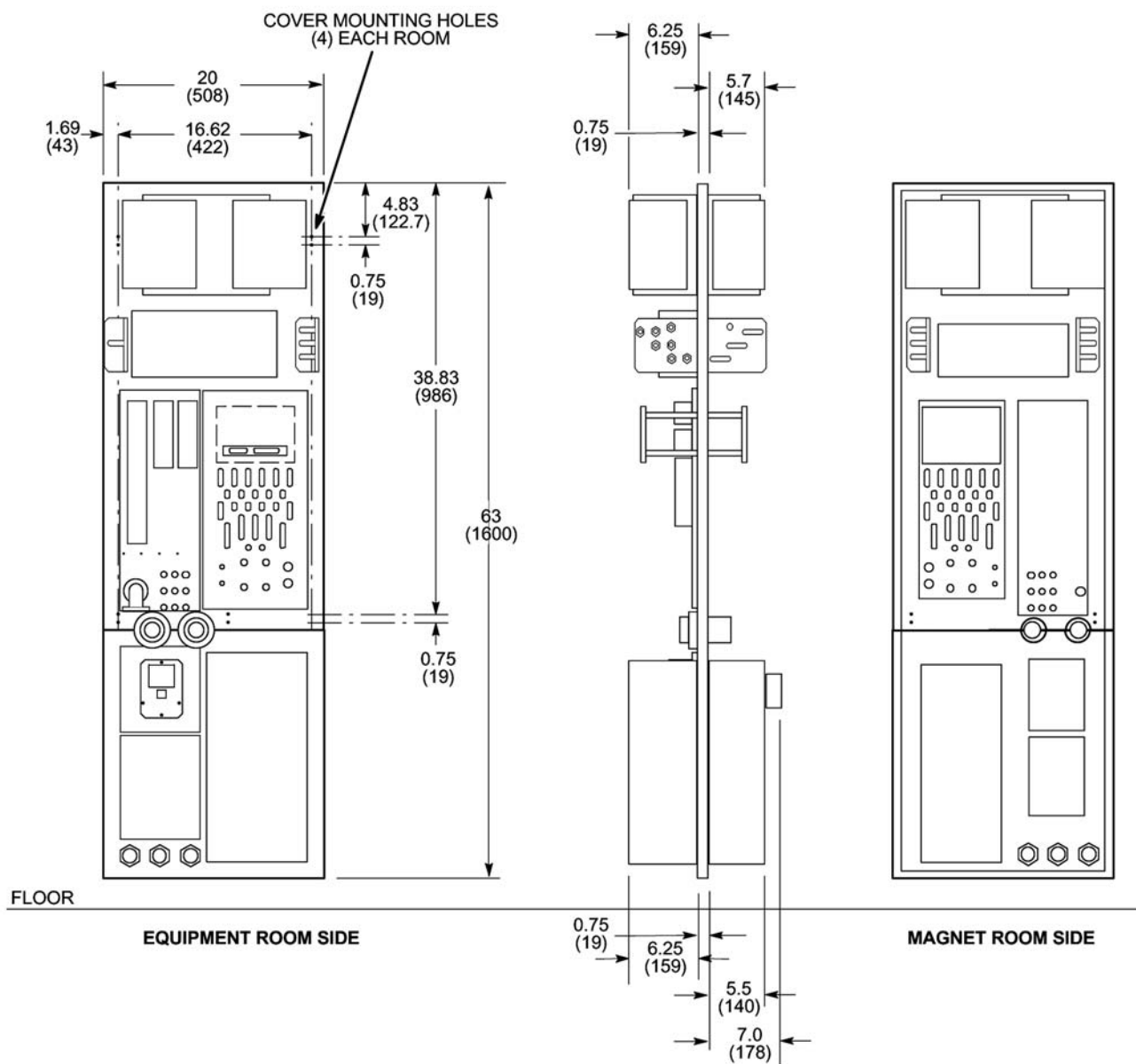
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8.9 Penetration Panel

Illustration 3-19: Penetration Panel (PP1)

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 87 lbs (39.5 kg)



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Chapter 4 Magnetic Field Considerations

1 Upgrades Magnetic Field Considerations

The upgrade catalogs documented in this manual do not change the existing magnet. Magnetic Field Considerations remain the same as specified in the latest revision of the existing system pre-installation document.

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Chapter 5 Site Environment

1 Introduction

The rating and duty cycles of all subsystems are applicable only if the room environment is maintained as specified in the following sections.



CAUTION

The MR system environment must be constantly maintained (i.e. holidays, weekends, etc.) to prevent exceeding the temperature and humidity specifications. Subjecting the equipment to consistent excessive temperatures and humidity above specifications may shorten the life of the internal electrical components.

2 IEC EMC Compliance

Per IEC 60601-1-2 Edition 2 Medical Electrical Equipment needs special precautions regarding Electro Magnetic Compatibility (EMC) and needs to be installed and put into service according to the EMC information provided in the following tables. [Table 5-1](#) , [Table 5-2](#) , and [Table 5-3](#) provide details about the level of testing performed by the manufacturer, and provide information about potential interactions between devices.

Portable and mobile RF communications equipment can affect Medical Electrical Equipment.

Table 5-1: Guidance And Manufacturer's Declaration – Electromagnetic Emissions

The Signa MR System is intended for use in the electromagnetic environment specified below. The customer or the user of the Signa MR System should assure that it is used in such an environment.			
Emissions Test	Type of Test	Compliance Level	Electromagnetic Environment - Guidance
CISPR 11	Conducted Emissions	Class A Group 2	The MR system must emit electromagnetic energy in order to perform its intended function. Near by electronic equipment may be affected.
	Radiated Emissions	Class A Group 2	The MR system must emit electromagnetic energy in order to perform its intended function. Near by electronic equipment may be affected.
IEC 61000-3-2	Emissions: Harmonic disturbance <16A	Not Applicable	System is permanently connected to a supply line with >16A current draw.
IEC 61000-3-3	Emissions: Harmonic disturbance <16A	Not Applicable	System is permanently connected to a supply line with >16A current draw.
IEC 61000-4-3	E-Field RF Immunity	80-2500 MHz, 5 V/m with AM 80% @ 1kHz or 2 Hz	

Table 5-2: Guidance And Manufacturer's Declaration – Electromagnetic Immunity

The Signa MR System is intended for use in the electromagnetic environment specified below. The customer or the user of the Signa MR System should assure that it is used in such an environment.			
Immunity test	IEC 60601 test level	Compliance Level	Electromagnetic environment – guidance
Electrostatic discharge (ESD) IEC 61000-4-2	±8 kV air	Air 2, 4, 6, 8, 10kVolts	Floors should be wood, concrete, or ceramic tile. If floors are covered with a synthetic material, the relative humidity should be at least 30%.
	±6 kV contact	Contact 2, 4, 6, 8kVolts	
		Coupling Plane 2, 4, 6, 8kVolts	
Electrical fast transient / burst IEC 61000-4-4	±2 kV for power supply lines	Power Lines 3kVolts	Power quality should be that of a typical commercial or hospital environment.
	±1 kV for input/output lines	Interconnect Cables 1.5kVolts	
Surge IEC 61000-4-5	±2 kV common mode	Common Mode ±2 kVolts	Power quality should be that of a typical commercial or hospital environment.
	±1 kV differential mode	Differential ±1 kVolts	
Power Frequency (50/60Hz) magnetic field IEC 61000-4-8	3 A/m	4.5A/m at 50 Hz. and 60 Hz.	
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5 % UT (>95 % dip in UT) for 5 sec	5 sec @ >-95%, Each phase individually	Power quality should be that of a typical commercial or hospital environment.

Table 5-2: Guidance And Manufacturer's Declaration – Electromagnetic Immunity (cont'd)

The Signa MR System is intended for use in the electromagnetic environment specified below. The customer or the user of the Signa MR System should assure that it is used in such an environment.			
Immunity test	IEC 60601 test level	Compliance Level	Electromagnetic environment – guidance
Conducted RF IEC 61000-4-6	V1 = 3 Vrms 150 kHz to 80 MHz	0.15-80 MHz @5 VRMS with AM 80% @ 1kHz 1% Frequency steps	Portable and mobile RF communications equipment should be used no closer to any part of the Signa MR System, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance $d = \left[\frac{3,5}{V_1} \right] \sqrt{P} \quad 150 \text{ kHz to } 80 \text{ MHz}$ $d = \left[\frac{3,5}{E_1} \right] \sqrt{P} \quad 80 \text{ MHz to } 800 \text{ MHz}$ $d = \left[\frac{7}{E_1} \right] \sqrt{P} \quad 800 \text{ MHz to } 2,5 \text{ GHz}$ where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey,^a should be less than the compliance level in each frequency range.^b Interference may occur in the vicinity of equipment marked with  the following symbol:
Radiated RF IEC 61000-4-3	E1 = 3 V/m 80 MHz to 2,5 GHz	80-2500 MHz, 5 V/m with AM 80% @ 1kHz or 2 Hz	

NOTE 1 At 80 MHz and 800 MHz, the higher frequency range applies.

NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

^a Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the Signa MR System is used exceeds the applicable RF compliance level above, the Signa MR System should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the Signa MR System.

^b Over the frequency range 150 kHz to 80 MHz, field strengths should be less than [V1] V/m.

Table 5-3: Recommended Separation Distances

The Signa MR System is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the Signa MR System can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the Signa MR System as recommended below, according to the maximum output power of the communications equipment.			
Immunity test	Recommended Separation Distances Between Portable And Mobile RF Communications Equipment And The Signa MR System: Separation distance according to frequency of transmitter m		
Rated maximum output power of transmitter (Watts)	$d = \left[\frac{3,5}{V_1}\right]\sqrt{P}$ 150 kHz to 80 MHz	$d = \left[\frac{3,5}{E_1}\right]\sqrt{P}$ 80 MHz to 800 MHz	$d = \left[\frac{7}{E_1}\right]\sqrt{P}$ 800 MHz to 2,5 GHz
0.01	0.12 m	0.12m	0.23m
0.1	0.36m	0.36m	0.74m
1	1.16m	1.16m	2.33m
10	3.68m	3.68m	7.38m
100	11.66m	11.66m	23.33m
For transmitters rated at a maximum output power not listed above, the recommended separation distance d in metres (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.			
NOTE 1 At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.			
NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			

3 Temperature and Humidity Specifications

3.1 System Suite



NOTICE

If these temperature and humidity specifications are not strictly adhered to, failures of the HFA modules of the HFD/PDU Cabinet may occur.

Use the specifications listed in [Table 5-4](#) for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmental controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

NOTE: To help prevent a patient from feeling uncomfortably warm during a scan, make sure the magnet room temperature does not exceed 69.8°F (21°C) maximum.

Table 5-4: Temperature And Humidity Specifications

Area	Temperature		Humidity		Max. Room Gradient °F (°C)
	Range °F (°C)	Change °F/Hr (°C/Hr)	Range %	Change %/Hr	
Equipment Room at Inlet to Equipment	59-89.6* (15-32)*	5 (3)	30-75*	5	5 (3)**
Magnet Room	59-69.8 (15-21)	5 (3)	30-60*	5	5 (3)
Operator's Control Room	59-89.6* (15-32)*	5 (3)	30-75*	5	5 (3)
Note * Non-condensing humidity with 50% nominal at 65°F (18.3°C). ** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets.					

4 Air Cooling

The Maximum Heat Output For Air Cooling tables contains the heat output of the equipment listed in the typical site location. There may be additional MR system equipment which was not replaced during the upgrade. These values do not include people, lights and non-MR equipment. Actual site average values will vary depending on system use (i.e. protocols used, patient load, etc.). Note any variations of equipment location for your site when calculating your cooling requirements for each room. Refer to [Table 5-5](#) for Signa EXCITE HD with LCC300 Magnet or [Table 5-6](#) Signa EXCITE HD with 3.0T94 Magnet.

Table 5-5: 3T System with LCC300 Magnet Maximum Heat Output For Air Cooling *

MR Component	Magnet Room See Note 1 listed below		Equipment Room See Note 2 listed below		Operator/Control Room	
	BTU/hr	Watt	BTU/hr	Watt	BTU/hr	Watt
RF/Gradient Body Coil Assembly, Magnet Enclosure Equipment	8189	2400				
Patient Blower Box	1366	400				
Penetration Panel			324	95		
GE pre-engineered Main Disconnect Panel			900	264		
NB RF Amp Cabinet			21,336	6253		
3.0T RFS Cabinet			8640	2532		
HFD/PDU Cabinet			34,120	10,000		
Twin Accessory Cabinet			2354	690		
Magnet Monitor			205	60		
Operator Workspace with LCD Color Display (See Note 3)					4950	1450
Shield/Cryo Cooler Compressor			Heat dissipation to air negligible requires water cooling, See Note 4			
Water cooled GWHX** option for providing Gradient Coil water cooling (requires site provided cooling for Shield Cooler Compressor) (See Note 4)			1707	500		
MRCC** option for providing Gradient Coil water cooling (See Note 4)			57,320	16,800		
MRCC** option for providing Shield/Cryo Cooler Compressor water cooling (See Note 4)			52,550	15,400		
Magnet Monitor UPS ** & Modem **			450	132		
3T MNS Cabinet for 4KW MNS option **			5118	1500		
BB RF Amp Cabinet **			24,600	7205		
BrainWave HW Lite Cabinet ** (see Note 5)			2337	685		
DC Lighting Controller Panel **			1024	300		
DC Lighting Controller Autotransformer **			171	50		

Table 5-5: 3T System with LCC300 Magnet Maximum Heat Output For Air Cooling * (cont'd)

MR Component	Magnet Room See Note 1 listed below		Equipment Room See Note 2 listed below		Operator/Control Room	
	BTU/hr	Watt	BTU/hr	Watt	BTU/hr	Watt
Chilled Air Blower (CAB) for IPCM Option **			14,330	4200		
Not used						
Notes * Maximum heat output is defined for temperature and humidity as defined in Section 3 , Temperature and Humidity Specifications. ** Optional equipment 1. Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room specifications as listed in Section 3 , Temperature and Humidity Specifications. It is recommended that cool inlet air be directed towards the Blower Box intake which contain a patient cooling fan. 2. FOR EQUIPMENT ROOM ONLY: Although the air cooling load averaged over a 12 hour working day is approximately 1/2 of the maximum value, the Equipment Room HVAC system must be sized such that Maximum Room Gradient, Temperature Range, Temperature Change per Hour, and Humidity specifications per Section 3 , Temperature and Humidity Specifications are not exceeded at any point during the working day. Actual heat output is site specific and dependent on the specific MR system configuration and customer usage of the MR system and options. 3. Operator Workspace equipment includes the following: LCD Color Monitor, GOC Computer Cabinet, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, and Keyboard. 4. The MR System requires water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor. - Gradient Coil water cooling must be supplied by cooling equipment (MRCC or GWHX) supplied with the MR system to prevent contamination/damage to the coil and for proper image quality. - The Shield/Cryo Cooler Compressor water cooling can be provided by a second MRCC or by customer provided facility water cooling. MRCC can be located outdoor or indoor. 5. Addition of 3rd party provided audio-visual equipment to the BrainWave HW Lite Cabinet may increase cabinet heat maximum output to 2781 BTU/Hr (815 Watts).						

Table 5-6: 3T System with 3.0T94 Magnet System Maximum Heat Output For Air Cooling *

MR Component	Magnet Room See Note 1 listed below		Equipment Room See Note 2 listed below		Operator/Control Room	
	BTU/hr	Watt	BTU/hr	Watt	BTU/hr	Watt
RF/Gradient Body Coil Assembly, Shim Coils, 3T94 Magnet Enclosure Equipment	14,000	4100				
Patient Blower Box	3415	1000				
Penetration Panel			324	95		
GE pre-engineered Main Disconnect Panel			900	264		
NB RF Amp Cabinet			21,336	6253		
RFS Cabinet			8640	2532		
BB RF Amp Cabinet*			24,600	7205		
HFD/PDU Cabinet			34,120	10,000		
Accessory Cabinet			24,600	7205		

Table 5-6: 3T System with 3.0T94 Magnet System Maximum Heat Output For Air Cooling * (cont'd)

MR Component	Magnet Room		Equipment Room		Operator/Control Room	
	See Note 1 listed below		See Note 2 listed below			
	BTU/hr	Watt	BTU/hr	Watt	BTU/hr	Watt
Operator Workspace with LCD Color Display (See Note 3)					4950	1450
Shield/Cryo Cooler Compressor			Heat dissipation to air negligible requires water cooling. See Note 4			
Water Chiller (WC1) for Gradient Coil			14,000	4100		

Notes

* Maximum heat output is defined for temperature and humidity as defined in [Section 3](#), Temperature and Humidity Specifications.

** Optional equipment

1. Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room specifications as listed in [Section 3](#), Temperature and Humidity Specifications. It is recommended that cool inlet air be directed towards the Blower Box intake which contain a patient cooling fan.
2. FOR EQUIPMENT ROOM ONLY: Although the air cooling load averaged over a 12 hour working day is approximately 1/2 of the maximum value, the Equipment Room HVAC system must be sized such that Maximum Room Gradient, Temperature Range, Temperature Change per Hour, and Humidity specifications per [Section 3](#), Temperature and Humidity Specifications are not exceeded at any point during the working day. Actual heat output is site specific and dependent on the specific MR system configuration and customer usage of the MR system and options.
3. Operator Workspace equipment includes the following: LCD Color Monitor, GOC Computer Cabinet, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, and Keyboard.
4. The MR System requires water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor.
 - Gradient Coil water cooling must be supplied by cooling equipment (WC1) supplied with the MR system to prevent contamination/damage to the coil and for proper image quality.
 - The Shield/Cryo Cooler Compressor water cooling requires customer provided facility water cooling.

4.1 Requirements

Physical placement of the air conditioning equipment (compressor, etc.) is an important factor due to the homogeneous field requirements of the magnet. Therefore, it is important this equipment be located outside the 10 gauss line.

The Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in [Section 3](#), Temperature and Humidity Specifications. It is recommended that cool inlet air be directed towards the Enclosure Rear Pedestal and Blower Box air intake for patient cooling.

4.2 Recommendations

A dedicated air conditioner with a dual compressor is preferred to avert shutdowns during repair of the primary air conditioner. Due to the large variation in heat loads, the compressors should be equipped with unloaders or hot gas bypass to prevent moisture stripping of the evaporator coils.

It is recommended that a temperature and humidity recorder be used during preinstallation and during actual installation and placed near the Gradient Cabinet air inlets to establish the true criteria. Refer to cooling table calculator in this section for each room's cooling requirements.

GE recommends the use of a 12 inch high raised flooring system for the equipment room (10 inch minimum clearance from floor slab to underside of access flooring). Care must be taken in locating the air conditioning supply vents in the floor. The air conditioning supply vents should be located directly in front of the cabinet inlets

5 Water Cooling Requirements

The upgrade catalogs documented in this manual do not change the existing MR system water cooling specifications.

6 Altitude

MR System Operating Range: 100 ft (30.5 m) below sea level to 11,808 ft (3600 m) above sea level



NOTICE

If MR System site (Magnet) is located above 8100 ft (2469 m) must obtain approval and direction from Engineering Installed Base Leader at Magnet & Gradient Systems in Florence, SC USA phone number 843-667-9799.

7 Lighting

Magnet Room Lighting requirements are listed in [Table 5-7](#)

Table 5-7: Magnet Room Direct DC Lighting Requirements

Requirements	
	NOTICE Fluorescent lighting is not allowed in the Magnet Room due to the RF noise generated by the fluorescent light tubes. 1. Direct Current (DC) lighting is required in the magnet room to avoid RF broadband noise impacts to image quality. NOTE: Sites previously designed to the GE Pre-Installation Manuals released between 2000 and 2007, and did install AC Lighting, will be allowed to continue to use the AC lighting they have in place. The expectation is that the sites which have installed AC Lighting have MR Systems that are sensitive to AC lighting broadband noise. These sites are expected to have found site specific incandescent bulbs that emit low levels of RF noise; low levels that are not measurable nor do they impact image quality. This subset of sites can continue to use AC Lighting provided their system is RF broadband noise free. In cases where the AC lighting broadband noise surfaces as an image quality concern, the site may then have no choice other than to replace the AC lighting with a DC controlled system. 2. Illumination of 300 lux around the front of the magnet for patient access. 3. Need to provide 300 lux above the magnet service work (non-magnetic, portable lighting is acceptable). 4. The AC ripple from the DC power should be no greater than 5%. 5. Discrete switch or variable lighting level DC Lighting Controller (GE option available refer to Chapter 6, Section 8, DC Lighting Controller (Facility Option)) must be used for selectable light levels. Dimmers (i.e. SCR, rheostats, etc.) are not allowed. 6. Lighting fixtures selection and installation must comply with requirements in Chapter 8, Section 1, RF Shielded Room Requirements to minimize the possibilities of electrical discharge. 7. Light fixtures must have a ground wire from its power source and be grounded to the RF Shielded Room at the RF Common Ground Stud as shown in illustration in Chapter 6, Section 4, Grounding System Ground subsection. 8. Light Emitting Diode (LED) lighting, if used, must meet the following: a. Power source must be located external to the Magnet Room RF Shield. b. All wiring, filters, and ground requirements must be met, refer to Chapter 6, Section 4, Grounding System Ground subsection. 9. Battery chargers (e.g. emergency lighting) are required to be located outside the Magnet Room.
Comments	
• Short filament length is recommended, linear lamps are not recommended because of the filament length and high incidence of filament failure. • Track lighting fixtures do not comply with light fixtures requirement listed above.	

8 Acoustics

8.1 Introduction

The acoustic information is provided for site planning and architectural design activities to address acoustics to meet local regulations and customer requirements. For more information about recommended safety procedures regarding patient exposure to MR-generated acoustic levels, see the MR Safety Guide included with the system Users Manual.

8.2 Acoustics Background

A typical MR suite has two types of acoustic noise issues. The first is the acoustics within the rooms in which the patients and technicians are impacted by the noise of the MR system as the gradients are pulsed. The second is noise transmitted to other spaces via airborne and structureborne paths.

8.2.1 Airborne

The airborne transmission path entails the excitation of air within the magnet room; the resonator module consisting of the magnet, RF coil, and gradient coil generates acoustic noise similar to an intense loud speaker. The airborne noise passes through walls via any openings, i.e. small holes, cracks, HVAC ducts, and waveguides, into surrounding spaces within and possibly beyond the confinements of the building. Acoustic energy can transmit across distances of significant length.

Examples of airborne acoustics issues may include the following (not limited to only these):

- MR Operator exposure at Operator Workstation (i.e. Operator viewing in-line with the patient inside the magnet may require a higher acoustic attenuation window)
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

8.2.2 Structureborne

The structureborne transmission path is the result of mechanical excitation of the floor/building structure causing the building to vibrate. The vibration of the surfaces at surrounding spaces then radiates as acoustic noise. Acoustic energy can transmit across distances of significant length.

NOTE: Less than 5% of installed base sites have experienced structureborne acoustic issues.

Examples of structureborne acoustics issues may include the following (not limited to only these):

- Areas directly above or below the Magnet Room, may not always be an issue
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

8.3 System Acoustic Noise Levels

Any GE factory-installed protocol can be modified by operators, which can increase or decrease acoustic SPL (Sound Pressure Level); or operators may create their own protocol which could produce a higher or lower acoustic SPL as stated under Operating Conditions, Condition 1 below. Typical scans generate acoustic levels as stated under Operating Conditions, Condition 2 below. In addition, the exposure times are completely under operator control. Consequently, hearing protection is required for all people in the Magnet Room during scans to prevent hearing impairment, acoustic levels may exceed 99dBA. Again, for more information about recommended safety procedures regarding patient exposure to MR-generated acoustic noise, see the MR Safety Guide included in the User Manual.

8.3.1 Ambient Conditions

To reduce any background noise due to cabinet blowers, etc., acoustical ceilings, walls, and floors are recommended. The following are typical noise level readings:

- Operator Area 55 dBA
- Equipment Room 75 dBA

8.3.2 Operating Conditions

8.3.2.1 Condition 1

MR scanners under “worst-case” operating conditions, could generate acoustic levels (as measured at the magnet isocenter) as follows:

For Signa EXCITE HD 3.0T System with LCC300 Magnet

- Average SPL 120 dBA SPL= Sound Pressure Level
- Peak 126 dB
- Frequency Range 20 to 20k Hz

For Signa EXCITE HD 3.0T System with 3.0T94 Magnet

- Average SPL 122 dBA SPL= Sound Pressure Level
- Peak 133 dB
- Frequency Range 20 to 20k Hz

8.3.2.2 Condition 2

MR scanners for many typical clinical scanning scenarios though, generate acoustic levels (as measured at the magnet isocenter) somewhat lower as follows:

For Signa EXCITE HD 3.0T System with LCC300 Magnet

- Average SPL 90 to 105 dBA

- Peak 110 to 120 dB
- Frequency Range 20 to 20k Hz

For Signa EXCITE HD 3.0T System with 3.0T94 Magnet

- Average SPL 90 to 105 dBA
- Peak 110 to 120 dB
- Frequency Range 20 to 20k Hz

As recent history has shown an evolution towards more powerful (and hence louder) gradient subsystems, architects should consider the acoustic levels stated in the "worst case" Condition 1, mentioned above. Note that high-field Signa systems have the ability to run scanning protocols which can generate acoustic levels over the entire human perceptible frequency range (20 to 20k Hz), therefore attenuation over this entire range must be considered for site design.

8.4 Acoustic Design Guidelines



NOTICE

The MR system upgrade will have the potential to increase the system generated acoustic levels (refer to [Section 8.3, System Acoustic Noise Levels](#)).

8.4.1 Magnet Room

Noise generated by the MR system is inherent to the operation of the system, refer to [Section 8.3, System Acoustic Noise Levels](#). The sound quality (human perception) within the Magnet Room can be modified by including sound absorbing materials to make the room sound more subdued and less harsh. The measured sound levels via a sound level meter will not change. However, the measured sound levels can be reduced only when the sound level generated by the MR System is reduced.

Sound quality improvements can be achieved by the following:

- Use ceiling tiles with fiberglass panels having a 2 inch (51mm) thickness set into the standard T-bar grid system.
- Adding fiberglass panels to the side walls covering approximately 20% of the side wall surface area. The panels should focus on covering the top half of the side walls. Panels could take many different and decorative shapes to improve the sterile look of the rooms. Typically panels might be on the order of 4ft x 6ft (1.2m x 1.8m) with a thickness of 4 inches (102mm) or equivalent. Panels shape could vary to produce mosaic effects to meet the customer preference. Any decorative materials used to cover the wall panels must be porous so that sound waves can pass through with ease. In principle, a person should be able to breath through the material with ease. Fire retardant cloth should be used. The NRC (Noise Reduction Coefficient) of the panels should be 0.95 or better when mounted against a hard surface such as drywall or concrete.

8.4.2 Inter-Spacial Areas

Acoustic Noise Control to mitigate noise from being transmitted to other spaces often amounts to paying attention to small details while working with ordinary construction materials. The key objectives are to eliminate all cracks and gaps in the wall construction while making sure that the doors, walls, floor, and ceiling have adequate transmission loss via mass or special double wall construction along with good fitting massive doors.

The entire Magnet must be surrounded by walls with substantial mass and/or double wall construction so that noise is contained in the room and not allowed to pass through into nearby spaces. Wall junctions must be sealed with acoustical sealant so that noise waves to do not escape from the room. In principle, if the room were filled with smoke and under a positive pressure, no smoke would leak from the room.

8.4.2.1 Wall Construction

Wall Construction will entail ordinary building materials in a careful configuration.

- The preferred wall would have an ASTM STC 50 construction which entails the use of standard wall construction of steel studs (typically 3-5/8 inch (92 mm)) with 2 layers of Type X drywall (typically 5/8 (16 mm)) on each side totaling 4 layers and fiberglass batt in the stud cavity. All drywall must be overlapped by 6 inches (152 mm) or more. Beads of (USG) Acoustical Caulking (non-hardening) would be used around the entire perimeter of the drywall. Any form of wall penetration should be avoided. Any necessary wall penetrations must be sealed using combination of Acoustical Caulking (non-hardening) and fiberglass batt material. See examples of wall construction shown in [Illustration 5-1](#) and [Illustration 5-2](#).
- The top of the wall must join the ceiling/floor above so that no cracks or gaps occur. If metal pan is used on the ceiling/floor (above), then flute seals would be used to seal the gaps between the drywall and the pan. Alternately drywall can be cut out to fit into the flutes. Acoustical caulking (non-hardening) will be used to seal the remaining cracks and gaps.

Illustration 5-1: Example Of Wall Construction For Airborne Noise Control - Option 1

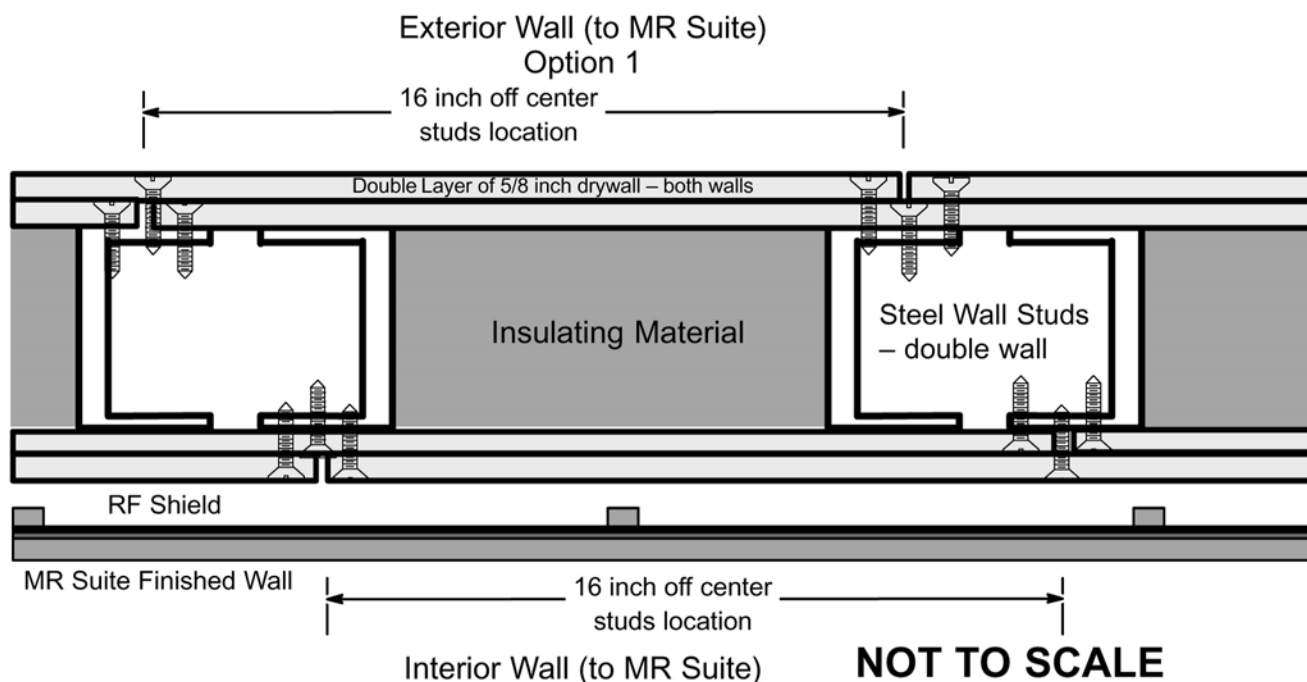
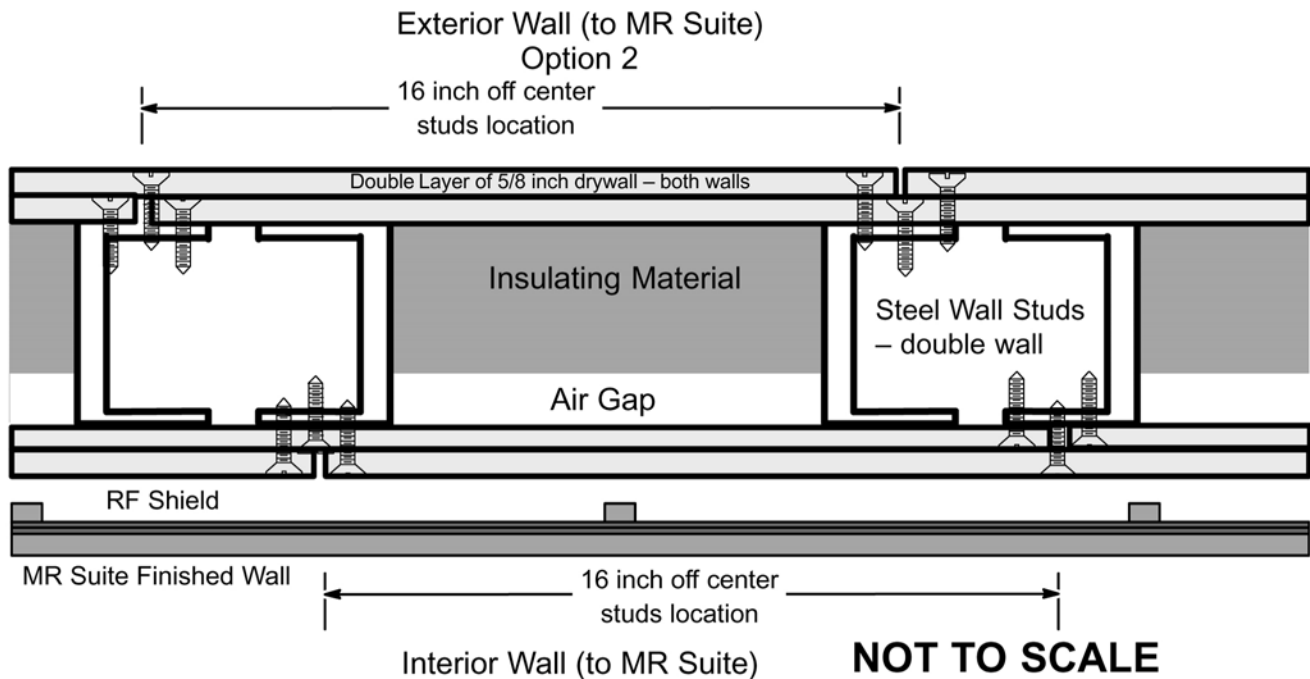


Illustration 5-2: Example Of Wall Construction For Airborne Noise Control - Option 2

8.4.2.2 High Bay RF Room

A high bay RF Room is a self contained RF Room which has open air space between the RF Room ceiling and the building floor above. The air space is an acoustic transmission path. Acoustic energy must be reduced to minimize this transmission of energy through this path.

In cases where the Magnet is to be installed in a high bay, it may be most effective to enclose the RF Room with its own drywall and steel stud room. The key difference being a ceiling assembly that mimics the sidewall construction to contain noise.

- Normal high STC stud walls from above would be used to support a ceiling assembly constructed of structural C channel with two layers of drywall on each side (total of 4 layers) with fiberglass batt in the cavity.
- Penetrations should be avoided via the use of surface mounted lights. HVAC and ducts passing through the ceiling, party wall or side walls would require acoustic noise attenuation in the form of inline silencers. Gaps and cracks would be sealed between the ceiling, party wall or vertical side walls and the cryogen vent plumbing. In essence the Magnet would be enclosed in a drywall "doghouse".

8.4.2.3 Miscellaneous Plumbing, RF Windows and RF Doors

Other construction details are equally important to mitigate noise transmission to meet the intended goal.

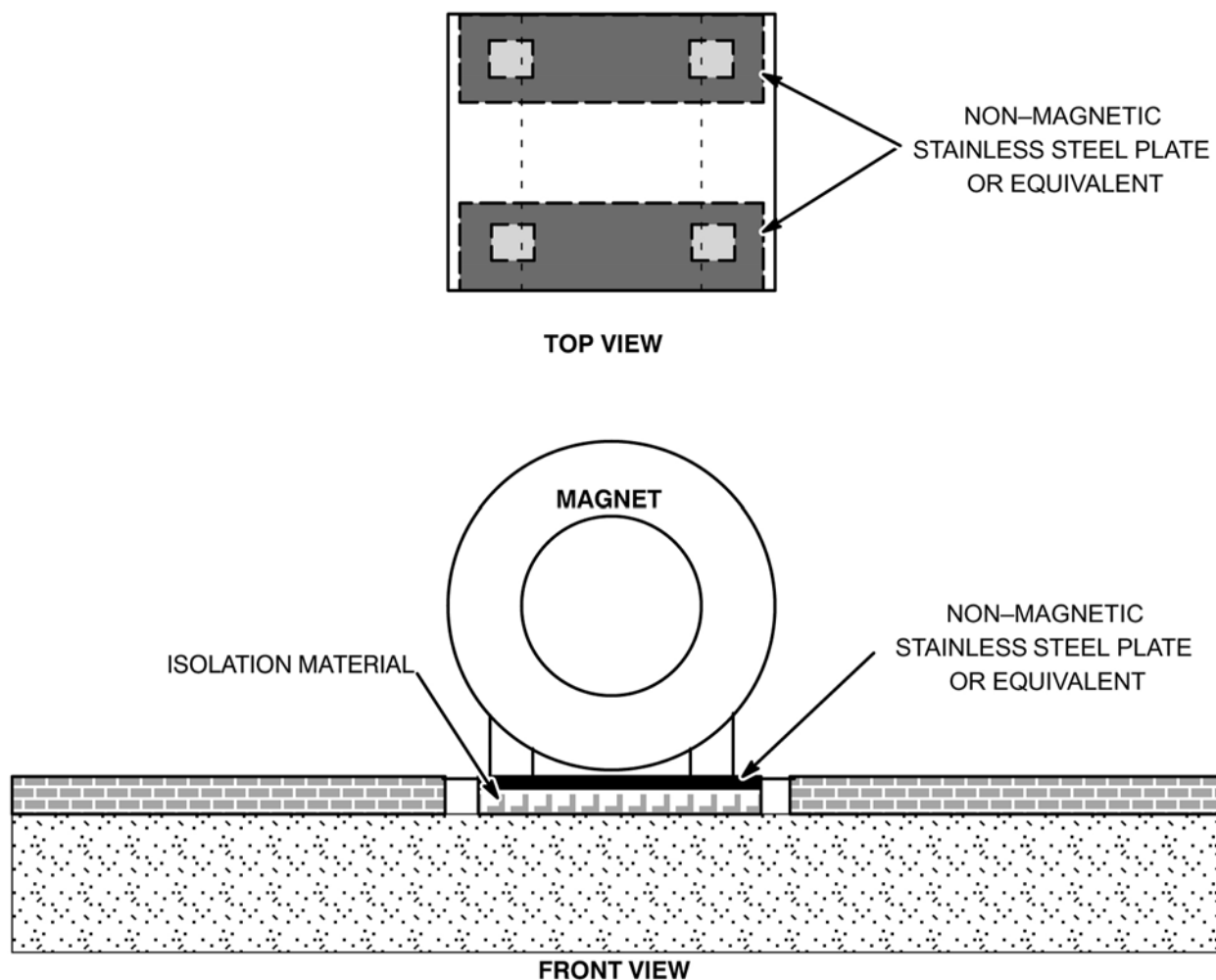
- Pipes (gas or water) and electrical conduit or Magnet Room signal cables must be sealed where they penetrate the walls or ceiling. A heavy mastic material such as Duxseal™ is appropriate.

- RF windows should be purchased as window/frame units with an STC rating obtained from laboratory testing per ASTM standards. STC 50 to 60 windows are needed. The installation must include proper sealing to avoid sound leaks.
- RF doors should be selected to provide an STC 50 to 60 to quell the noise. Contact RF Shield Room supplier for selection of RF doors that meet the local acoustic codes and site acoustic requirements. RF door seals must be selected to prevent small gaps around the door perimeter and at the door threshold. RF door seals would either require periodic replacement or a door seal that would last the life of the Magnet Room.

8.4.3 Structureborne Vibration Control

Upper floor MR installations represent the largest population of sites which may experience structureborne acoustic issues.

The GE VibroAcoustic Damping Option provides a method to reduce the probability of acoustic structureborne transmission throughout the customer facility. See [Illustration 5-3](#) for an overview of the VibroAcoustic damping mats concept. The Magnet Room floor requirements are defined in [Chapter 8, Section 6.4](#), Magnet Room Floors.

Illustration 5-3: Overview Structureborne Vibration Control Solution Concept

8.4.4 VibroAcoustic Damping Option

GE customers may purchase the VibroAcoustic Damping Option to address site vibration which may contribute to potential structureborne acoustic issues. Amount of vibration attenuation provided by the VibroAcoustic Damping Option will be site dependent.

When this option is planned for the Magnet Room floor area then directly beneath the Magnet needs to be properly prepared to ensure the Magnet isocenter to top of finished floor requirement is met, refer to Direction 5133303 Signa EXCITE HD 3.0T Pre-Installation, RF Shielded Room chapter, Magnet Room Floors section.

9 Room Ventilation

Refer to [Table 5-8](#) for ventilation specifications for the magnet. Refer to [Section 12](#), Pollution for air quality specifications.

Table 5-8: Magnet Room Ventilation Requirements/Recommendations

Requirements	
1.	Sufficient air ventilation in the Magnet Room must be maintained, not only for patient comfort during scans but also to maintain proper oxygen level during cryogen replenishment.
2.	An exhaust fan to be placed above RF shielding with appropriate wave guide filtering for quick removal of helium gas if large amounts of helium disperse into magnet room. Inert gas containers, such as dewars, are not air tight.
3.	The Magnet Room exhaust fan intake vent must be located at the highest ceiling plane near the magnet cryogen vent.
4.	The Magnet Room exhaust fan to exhaust to safe outside area and be independent of cryogenic venting.
5.	The Magnet Room exhaust fan and air inlet must be sized for a minimum of 1200 CFM (34 m ³ /minute) and minimum of room 12 air exchanges per hour.
6.	Two manual exhaust fan controls connected parallel, one to be located near the Operator Workspace and second control located in the Magnet Room. - The Magnet Room ventilation switch should be mounted near the Magnet Room door and is the responsibility of the architect and mechanical contractor. - Refer to Table 5-9 for exhaust fan recommended set-up or Table 5-10 for recommended set-up with optional Oxygen Monitor.
7.	Exhaust fan (customer supplied) to be installed and operating before magnet is moved into room.
8.	Annual customer inspection and cleaning / maintenance of the exhaust fan system (fan, inlet grill/filter, ducts, etc.) is needed to meet the minimum airflow requirement to an outside area.
9.	Provide minimum 2 ft x 2 ft (0.61 m x 0.61 m) pressure equalizing waveguide vent in the magnet room ceiling or in the wall (with waveguide top edge located at ceiling) to prevent positive or negative pressures from interfering with opening of the magnet room door per IEC 60601-2-33 6.8.3 cc.
Comments	
•	Minimum 5-7% of outside makeup air to be vented into the Magnet Room. For example, with an air input rate of 1200 cubic feet per minute (CFM) (34 cubic meters per minute), there must be a minimum of 60 CFM (1.7 cubic meters per minute) (5%) of outside makeup air.

Table 5-9: Exhaust Fan Set-Up

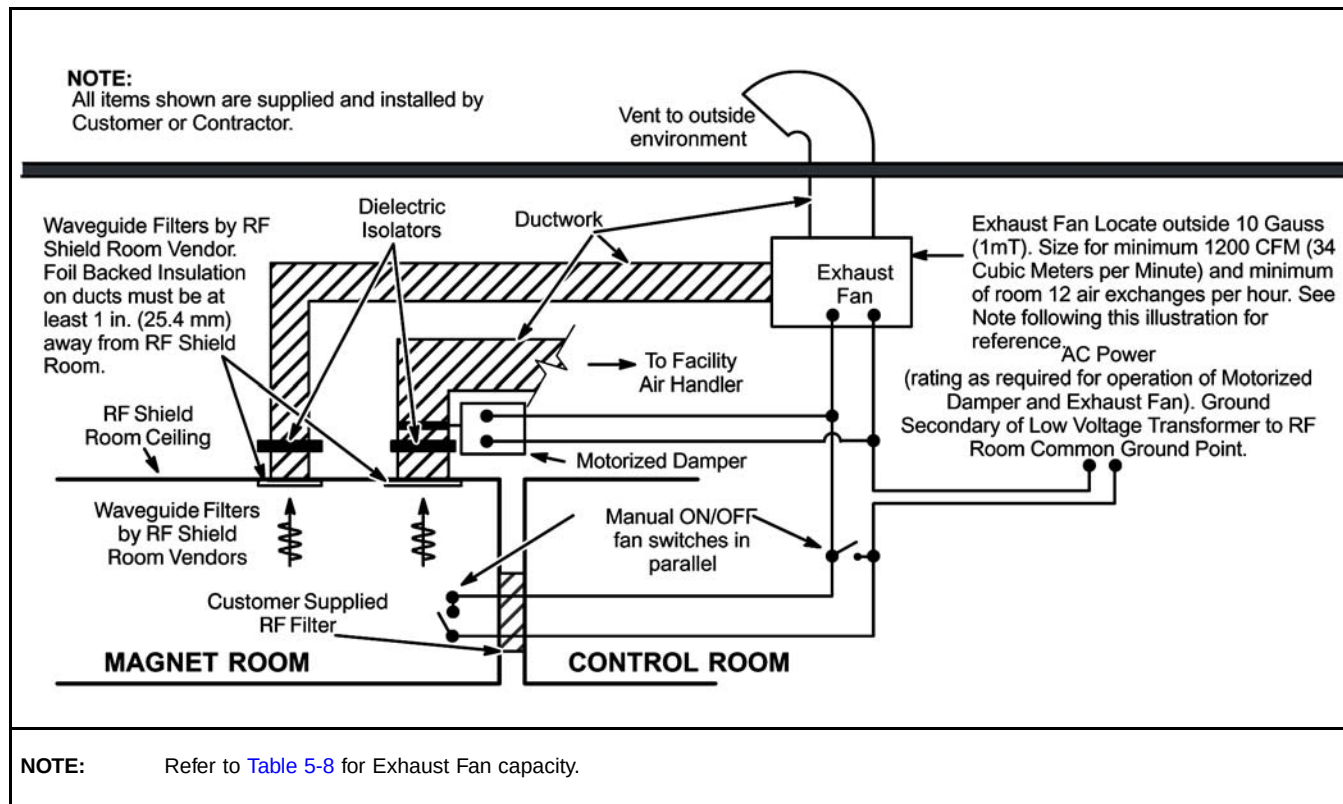
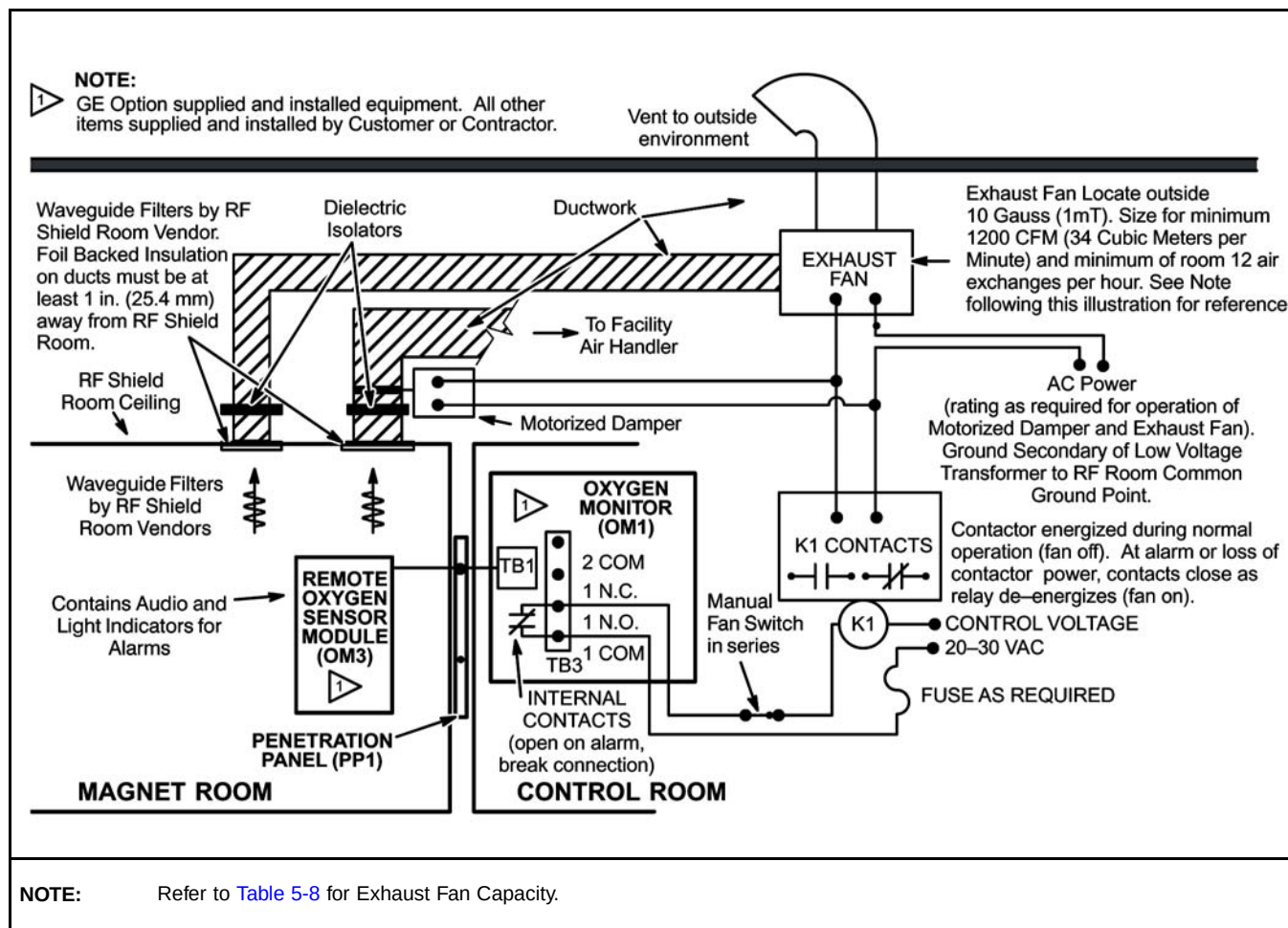


Table 5-10: Exhaust Fan Set-Up With Optional Oxygen Monitor



10 Alarm Devices, Water Sensors and Thermostats

10.1 System Cabinet

EXCITE (Release 11.x) systems upgrading to Release 12.x retain their EXCITE System Cabinet.

The EXCITE System Cabinet has 104°F (40°C) and 122°F (50°C) temperature sensors. When the temperature in the System Cabinet reaches 104°F (40°C) a Warning Message will be displayed on the Operator Workspace. When the temperature in the System Cabinet reaches 122°F (50°C) or above a Warning Message will be displayed on the Operator Workspace and the system scanning operation will be inhibited. Normal scanning operation will be allowed when the System Cabinet temperature returns to normal. Any external alarm device for room temperature monitoring must be supplied by the customer.

10.2 RFS Cabinet

Pre-EXCITE systems upgrading to Release 12.x will install an RFS Cabinet during the upgrade.

The RFS Cabinet has 104°F (40°C) and 122°F (50°C) temperature sensors. When the temperature in the RFS Cabinet reaches 104°F (40°C) a Warning Message will be displayed on the Operator Workspace. When the temperature in the RFS Cabinet reaches 122°F (50°C) or above a Warning Message will be displayed on the Operator Workspace and the system scanning operation will be inhibited. Normal scanning operation will be allowed when the RFS Cabinet temperature returns to normal. Any external alarm device for room temperature monitoring must be supplied by the customer.

10.3 Water Sensor Alarm and Floor Drain

It is recommended that customer supplied water sensor alarms and floor drain be located on floors where water cooled cabinets are positioned, especially under raised flooring.

10.4 Pneumatic Patient Alert

The Pneumatic Patient Alert Control Box provides an audible and visual alarm near the operator when the patient depresses the hand held squeeze bulb. The control box is to be mounted with consideration for ease of use by operator, remaining in sight of operator, and remaining within 5 ft (1.5 m) of an electrical outlet. Note, an outlet on the Operator Workspace may be used. Options for control box location include mounting box vertically (on a wall or other vertical surface), horizontally (place box on a counter top, desk top, or other horizontal surface), or under a shelf within sight of operator.

11 Ambient Radio Frequency Interference (RFI)

The MR System utilizes spatially encoded radio frequency information to create the MR image. Therefore, it is sensitive to ambient RFI. To protect the MR from ambient RFI (as well as the local environment from Magnetic Resonance RF), all sites require a 100 dB RF Shield, refer to [Chapter 8, Section 1](#), RF Shielded Room Requirements for exact requirements. It is very unlikely that local signals will affect an MR System with a properly designed and installed RF Shield. During the site evaluation visit, GE notes the location of nearby sources of RFI and will advise if further information or on-site testing is required. Most sites do not require on-site testing. Listed in [Table 5-11](#) are the recommended centerband and bandwidth frequencies to be used when measuring radio frequency interference. This table includes those frequency bands which are important for both proton imaging and spectroscopy.

Table 5-11: 3.0T System Radio Frequency Survey Specifications

Isotope	Bandcenter MHz	Bandwidth Hz
¹ H	127.72	681,183
¹⁹ F	120.23	641,229
³¹ P	51.75	276,010
²³ Na	33.80	180,291
¹³ C	32.13	171,335

When required, RFI site surveys are to be performed by cycling through the preceding frequency bands and a broad band range of 150MHz $\pm$ 10MHz. Special emphasis, however, should be placed on the 1H band since this is used in proton imaging. The RFI site survey should be performed for a length of time necessary to determine, within a reasonable degree of certainty, that the RFI noise at the site will not exceed the 100 db attenuation provided by the RF shielded room. Note that any RFI site survey no matter how thorough, will not preclude the possibility of future or unmeasured RFI caused by new or intermittent sources.

The ambient RF noise measured should be less than 100 millivolt per meter (100 dB microvolt per meter). When a RFI site survey is required, it must be completed before the purchase and installation of the RF shielded room.

To ensure that 100 millivolt (or greater) RF noise peaks outside the bandwidths specified above do not actually extend into these bandwidths and exceed the 100 millivolt limit, adjust the resolution of the test equipment (spectrum analyzer) according to the equation:

$$BW \text{ (resolution)} = f_0 / 50$$

where: BW = Bandwidth (resolution)

f_0 = Center frequency (for 1H: at 3.0 Tesla 127.72 MHz)

12 Pollution

The site must be clean prior to delivery of the equipment. Although individual components have filters for optimum air filtration, care should be taken to keep air pollution to a minimum.

When cleaning tile floors, do not use steel wool which could enter cabinet enclosures and cause internal shorts.

The computer/equipment area requires that the air be filtered to remove 90 percent of all particles down to 10 microns and 80 percent of all particles from 10 to 5 microns in size.

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Chapter 6 Power Requirements

1 System Power Introduction

The MR system includes a Power Distribution module (PD1) in the lower portion of the HFD/PDU Cabinet (MR3) which distributes power to most MR system components. Refer to [Section 2](#), Critical Power Requirements for required customer power specifics. For specifications of required facility input to the Signa EXCITE HD 3.0T with LCC300 Magnet upgraded system refer to [Table 6-1](#) and to the Signa EXCITE HD 3.0T with LCC300 Magnet upgraded system refer to [Table 6-2](#).



WARNING

THE FACILITY TRANSFORMER AND FEEDER WIRES NEED TO BE CORRECTLY SIZED FOR THE SIGNA SYSTEM POWER DEMANDS.



WARNING

CUSTOMER SUPPLIED MAIN DISCONNECT PANEL DESIGN NEEDS TO HAVE CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE MR SYSTEM POWER REQUIREMENTS.



WARNING

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM POWER AND REGULATION DEMANDS.

Customers should carefully consider the advantages and disadvantages of raised flooring, conduits, floor ducts, and surface raceways for running cables in accordance with local codes. If used, conduits should be large enough to pass any cable and its connector through with all other cables in the conduit.

To reduce voltage regulation problems and wiring costs, minimize the cable length between the primary power source and the Power Distribution Unit. When routing cables, keep all phase conductors and ground for a circuit in the same trough. Whenever possible, keep power cables away from signal and data cables. Use separate trough or dividers in duct.

Table 6-1: Required Customer Power

Signa EXCITE HD 3.0T System with LCC300 Magnet					
MR Component	Voltage (VAC)	Frequency	Phase	Max. Amps	Comments
GE pre-engineered Main Disconnect Panel (MDP) See Notes 1, 2, 3, & 4 See Note 5 if customer MDP	480Y/277 VAC ± 10% or 400Y/230 VAC ± 10%	60 Hz 50 Hz	(3+GND) See Comments	See Note 6	Recommend input configuration: 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to PDU or inside the Main Disconnect Panel and not brought to the PDU or its cabinet. (See Note 7) Optional input configuration: 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.
Magnet Rundown Unit	100-120 or 200-240	50/60 Hz	1	1.0	Hard wired in unit.
Service Receptacle in Magnet Room	110-120 See Comments	50/60 Hz	1	2.0	Receptacle required for small power tools. Local voltage and portable transformers for voltages values.
* Remote Magnet Rundown Unit	100-120 or 200-240	50/60 Hz	1	1.0	Hard wired in unit
* O ² Monitor	110-120 or 200-240	50/60 Hz	1	3.0	Hard wired in monitor

Table 6-1: Required Customer Power (cont'd)

Signa EXCITE HD 3.0T System with LCC300 Magnet					
MR Component	Voltage (VAC)	Frequency	Phase	Max. Amps	Comments
* Integrated Patient Comfort Module (IPCM) for LCC300 Magnet ONLY	200-240	50/60	1	30.0	Refer to Section 9 , IPCM Option Power Requirements
	200-277	60	1	30.0	
Notes					
* Optional equipment.					
1. Power phase conductors, neutral (if present), and ground conductor must be routed inside the same raceway, cable tray, trench cable. or cord per National Electric Code (NEC) 2005 or 2002 Articles 250.134, 300.3, 517.13. 2. Signa TwinSpeed MDP controls power to the following system equipment: - Power Distribution Unit - Coldhead MRCC (option to provide water cooling for Shield/Cryo Cooler Compressor) - Gradient MRCC - Shield/Cryo Cooler Compressor - Magnet Monitor equipment including the Magnet Monitor, Modem, Uninterruptible Power Supply (UPS*) (optional) for Magnet Monitor, Multiplexer Box (optional). 3. MDP power circuits for MRCC if providing Shield/Cryo Cooler Compressor water cooling, Magnet Monitor, and Shield/Cryo Cooler Compressor Cabinet, along with cooling for these units, are required immediately upon magnet arrival to minimize cryogen consumption. If permanent site power is not ready, temporary power drop line and cooling must be made available. If site voltage is not any of the voltages listed above, customer must provide transformer and secondary circuit breaker to provide correct voltage and/or configuration. MDP power circuits for MRCC if providing Shield/Cryo Cooler Compressor water cooling, Magnet Monitor, and Shield/Cryo Cooler Compressor Cabinet, along with cooling for these units, are required immediately upon magnet arrival to minimize cryogen consumption. If permanent site power is not ready, temporary power drop line and cooling must be made available. If site voltage is not any of the voltages listed above, customer must provide transformer and secondary circuit breaker to provide correct voltage and/or configuration. Refer to Facility Options for listing of step up transformers options. 4. The complete MR System Digital Energy SG Series UPS 100 KVA (E4502FB) option requires 480 VAC, 3-phase, 4 wire + ground, 60 Hz input power. 5. If customer provided MDP has been selected then Customer provided MDP MUST meet all MDP requirements, refer to Section 3.1, Main Disconnect Panel (MDP) Requirements. 6. Maximum amps dependent on voltage selected. Refer to Section 2, Critical Power Requirements for configuration. 7. PDU Module is located in the lower portion of the HFD/PDU Cabinet (MR3) or HFD-S/PDU Cabinet (MR3).					

Table 6-2: Required Customer Power

Signa EXCITE HD with 3.0T94 Magnet					
MR Component	Voltage (VAC)	Frequency	Phase	Max. Amps	Comments
Power Distribution Unit (PD1) Module located in the lower portion of HFD/PDU Cabinet (MR3) See Note 1	480Y/277 VAC ± 10% or 400Y/230 VAC ± 10%	60 Hz 50 Hz	(3+GND) See Comments	See Note 2	Recommend input configuration: 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to PDU or inside the Main Disconnect Panel and not brought to the HFD/PDU Cabinet. Optional input configuration: 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.
Other system Components	Systems with 3.0T94 : other system components which require power are not impacted by the upgrade, refer to Direction 2221775 Signa 3.0T/94 & VH3 Pre-installation for power requirements.				
Notes					
* Optional equipment.					
1. Power phase conductors, neutral (if present), and ground conductor must be routed inside the same raceway, cable tray, trench cable. or cord per National Electric Code (NEC) 2005 or 2002 Articles 250.134, 300.3, 517.13.					
2. Maximum amps dependent on voltage selected. Refer to Section 2 , Critical Power Requirements for configuration.					

2 Critical Power Requirements

The power requirements for the upgraded system is dependent on the system configuration: magnet type, MR system cooling equipment configuration present at site, and system Power Distribution Unit (PDU) present. After the EXCITE HD upgrade the system will have the PDU located either a HFD/PDU Cabinet or a HFD-S/PDU Cabinet (an upgraded ACGD/PDU Cabinet). Below information includes the PDU requirements.



NOTICE

Signa EXCITE HD 3.0T With 3.0T94 Magnet: After the EXCITE HD upgrade the system will have the PDU located in the lower portion of either a HFD/PDU Cabinet or a HFD-S/PDU Cabinet (an upgraded ACGD/PDU Cabinet). System components beyond the PDU in the HFD/PDU Cabinet which require power are not impacted by the upgrade, refer to Direction 2221775 Signa 3.0T/94 & VH3 Pre-installation for other system component power requirements.



NOTICE

If customer provided MDP has been selected then customer provided MDP MUST meet all MDP requirements.

Refer to [Section 3.1](#), Main Disconnect Panel (MDP) Requirements for Main Disconnect Panel capability and set up.

All work is to be done in accordance with national and local electrical codes.



WARNING

THE FACILITY TRANSFORMER AND FEEDER WIRES NEED TO BE CORRECTLY SIZED FOR THE SIGNA SYSTEM POWER DEMANDS.



WARNING

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM POWER AND REGULATION DEMANDS.

Table 6-3: Critical Power Requirements

Parameter	Requirements
Configuration	- Recommend input configuration 3 phase solidly Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to or inside the Main Disconnect Panel and not brought to the HFD/PDU Cabinet or HFD-S Cabinet. - Optional input configuration 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.
Frequency	50 ± 3 Hz or 60 ± 3 Hz

Table 6-3: Critical Power Requirements (cont'd)

Parameter	Requirements
Regulation	4% maximum at system maximum power demand (averaged over 5 seconds) from source to PDU (i.e. includes all feeders and transformer to utility)
Phase Balance	Difference between the highest phase line-to-line voltage and the lowest phase line-to-line voltage must not exceed 2%
Daily Voltage Variation	10% from nominal under worst case line and load regulation
PDU Voltage	200/208/380/400/415/480 VAC 10%
Shield/Cryo Cooler Compressor Voltage	380/400/415 VAC 50 Hz or 460/480 VAC 60Hz
Coldhead MRCC Voltage Gradient MRCC Voltage for system with LCC300 Magnet	380/400 VAC 10% 50 Hz or 460/480 VAC 10% 60Hz
Magnet Monitor equipment Voltage	100/120 or 200/220 VAC
Voltage Transients	Phase-to-phase voltages must be within 2% of the lowest phase-to-phase voltage. Maximum allowable transient voltage above or below nominal waveshape not to exceed 200 V at a maximum duration of 1 cycle and frequency of 10 times per hour.
Facility Zero Voltage Reference Ground	• Main facility ground conductor to Main Disconnect Panel (MDP) shall be copper and the minimum as required by the local coding regulations, such as the National Electric Code (NEC) 2005 or 2002 Article 250.122. • Main facility ground wire to be insulated. • Ground impedance to earth at power source to be 2 ohms or less. • Main facility ground wire to be bonded at every distribution box in an approved grounding block.

Table 6-3: Critical Power Requirements (cont'd)

Parameter	Requirements
Maximum Momentary Demand	For Signa EXCITE HD 3.0T With LCC300 Magnet & MRCC or GWHX System Cooling Equipment: The power demands specified as a function of the duration of the power demand. Table 6-4 lists points on the curve. For Signa EXCITE HD 3.0T With LCC300 Magnet & TSCC or TGWC System Cooling Equipment: The power demands specified as a function of the duration of the power demand. Table 6-5 lists points on the curve. For all systems: The power system feeding the Signa system must be designed to meet the specifications of less than 4% regulation when loaded at the 5.0 second allowable consumption. For short intervals the Signa system power demands can exceed the 5.0 second value and the line voltage delivered to the system will sag below the 4% regulation. The Signa system is designed to tolerate these short voltage sags.
Average (while scanning) Power Demand	For Signa EXCITE HD 3.0T With LCC300 Magnet & MRCC or GWHX System Cooling Equipment: Refer to Table 6-6 For Signa EXCITE HD 3.0T With LCC300 Magnet & TSCC or TGWC System Cooling Equipment: Refer to Table 6-7
Standby (no scan) Power Demand	For Signa EXCITE HD 3.0T With LCC300 Magnet & MRCC or GWHX System Cooling Equipment: 27.7 KVA at 0.9 lagging Power Factor including 4.4 KVA for PDU, 9.8 KVA for 2 MRCC units for system water cooling, 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet, and 1.5 KVA 1 phase for Magnet Monitor equipment (4.5 KVA 3 phase equivalent). For Signa EXCITE HD 3.0T With LCC300 Magnet & TSCC or TGWC System Cooling Equipment: 26.9 KVA at 0.9 lagging Power Factor including 4.4 KVA for PDU, 9 KVA for for system cooling equipment, 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet, and 1.5 KVA 1 phase for Magnet Monitor equipment (4.5 KVA 3 phase equivalent).

Table 6-4: System With TRM Coil Peak Power Demand

Cooling Equipment	MR System Cooling Equipment For Both Gradient Coil & Shield/Cryo Cooler Compressor Water Cooling See Note 1	MR System Cooling Equipment For Gradient Coil ONLY See Note 2	
System Equipment ∨	2 MRCC Units	1 MRCC Unit	1 GWHX Unit
PDU draw for 5.0 sec	~61.2 KVA	~61.2 KVA	~61.2 KVA
PDU draw for 1.0 sec or less See Note 3	~65 KVA	~65 KVA	~65 KVA
Magnet Monitor See Note 4	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	15.8 KVA	7.9 KVA	0 KVA See Note 5
TOTAL for 5.0 sec	~90.5 KVA	~82.6 KVA	~74.7 KVA
TOTAL for 1.0 sec or less	~94.3 KVA	~86.4 KVA	~78.5 KVA
Notes - Two MRCC units are used to provide water cooling for the Gradient Coil and for the Shield/Cryo Cooler Compressor. - Customer provided water cooling for the Shield/Cryo Cooler Compressor is required when either 1 MRCC or the GWHX is used to provide Gradient Coil water cooling. Customer provided water cooling equipment power demands are not included in the values in this table. - The PDU draw on the line will not exceed list values. The ACGD Power Supply may provide up to 170 KVA for 0.003 seconds from supply internal capacitance but the supply will recharge capacitors at a power level less than 65 KVA. The PDU draw on the line will not exceed list values. The ACGD Power Supply may provide up to 170 KVA for 0.003 seconds from supply internal capacitance but the supply will recharge capacitors at a power level less than 65 KVA. - The Magnet Monitor equipment power is 1.5 KVA 1 phase on an unbalanced leg of 3 phase input (4.5 KVA 3 phase equivalent). - The GWHX is powered from the PDU and therefore included in the PDU draw value.			

Table 6-5: System With TRM Coil Peak Power Demand

Cooling Equipment	TSCC See Note 1			TGWC See Note 2		
System Equipment ∨	Outdoor TSCC	Indoor Water Cooled TSCC	Indoor Air Cooled TSCC	Outdoor TGWC	Indoor Water Cooled TGWC	Indoor Air Cooled TGWC
PDU draw for 5.0 sec	~61.2 KVA	~61.2 KVA	~61.2 KVA	~61.2 KVA	~61.2 KVA	~61.2 KVA
PDU draw for 1.0 sec or less See Note 3	~70 KVA	~70 KVA	~70 KVA	~70 KVA	~70 KVA	~70 KVA
Magnet Monitor See Note 4	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA

Table 6-5: System With TRM Coil Peak Power Demand (cont'd)

Cooling Equipment	TSCC See Note 1			TGWC See Note 2		
System Equipment ∨	Outdoor TSCC	Indoor Water Cooled TSCC	Indoor Air Cooled TSCC	Outdoor TGWC	Indoor Water Cooled TGWC	Indoor Air Cooled TGWC
System Cooling equipment (configuration indicated in column heading)	23 KVA	20 KVA	22.7 KVA	16 KVA	0 KVA See Note 5	11.6 KVA
TOTAL for 5.0 sec	~98.7 KVA	~948.7 KVA	~97.4 KVA	~90.7 KVA	~74.7 KVA	~86.3 KVA
TOTAL for 1.0 sec or less	~106.5 KVA	~103.5 KVA	~106.2 KVA	~99.5 KVA	~83.5 KVA	~95.1 KVA

Notes

1. TSCC provides water cooling for the Shield/Cryo Cooler Compressor and Gradient Coil.
2. TGWC provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor. Customer provided water cooling equipment power demands are not included in the values in this table.
3. The PDU draw on the line will not exceed list values. The HFD Power Supply may provide up to 170 KVA for 0.003 seconds from supply internal capacitance but the supply will recharge capacitors at a power level less than 65 KVA.
4. The Magnet Monitor equipment power is 1.5 KVA 1 phase on an unbalanced leg of 3 phase input (4.5 KVA 3 phase equivalent).
5. The GWHX is powered from the PDU and therefore included in the PDU draw value.

Table 6-6: System With TRM Coil Average (Continuous) Scanning Power Demand

Cooling Equipment	MR System Cooling Equipment For Both Gradient Coil & Shield/Cryo Cooler Compressor Water Cooling See Note 1	MR System Cooling Equipment For Gradient Coil ONLY See Note 2	
System Equipment ∨	2 MRCC Units	1 MRCC Unit	1 GWHX Unit
PDU draw See Note 3	48.1 KVA	48.1 KVA	49.5 KVA
Magnet Monitor	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	12.4 KVA	6.2 KVA	0 KVA See Note 4
TOTAL See Note 5	74.0 KVA	67.8 KVA	63.0 KVA

Notes

1. Two MRCC units are used to provide water cooling for the Gradient Coil and for the Shield/Cryo Cooler Compressor.
2. Customer provided water cooling for the Shield/Cryo Cooler Compressor is required when either 1 MRCC or the GWHX is used to provide Gradient Coil water cooling. Customer provided water cooling equipment power demands are not included in the values in this table.
3. The PDU is rated for 50 KVA continuous power.
4. The GWHX is powered from the PDU and therefore included in the PDU draw value.
5. GE pre-engineered Main Disconnect Panel (MDP) is rated continuous power draw is 77 KVA but MDP continuous draw does not exceed listed demands.

Table 6-7: System With TRM Coil Average (Continuous) Scanning Power Demand

Cooling Equipment	TSCC See Note 1			TGWC See Note 2		
System Equipment ∨	Outdoor TSCC	Indoor Water Cooled TSCC	Indoor Air Cooled TSCC	Outdoor TGWC	Indoor Water Cooled TGWC	Indoor Air Cooled TGWC
PDU draw See Note 3	48.1 KVA	48.1 KVA	48.1 KVA	48.1 KVA	49.5 KVA	48.1 KVA
Magnet Monitor	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	18.3 KVA	14.0 KVA	13.0 KVA	15.0 KVA	0 KVA See Note 4	8.8 KVA
TOTAL See Note 5	79.9 KVA	75.6 KVA	74.6 KVA	76.6 KVA	63.0 KVA	70.4 KVA
Notes 1. TSCC provides water cooling for the Shield/Cryo Cooler Compressor and Gradient Coil. 2. TGWC provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor. Customer provided water cooling equipment power demands are not included in the values in this table. 3. The PDU is rated for 50 KVA continuous power. 4. The Water Cooled TGWC is powered from the PDU and therefore included in the PDU draw value. 5. GE pre-engineered Main Disconnect Panel (MDP) is rated continuous power draw is 77 KVA but MDP continuous draw does not exceed listed demands.						

3 Power Distribution System

3.1 Main Disconnect Panel (MDP) Requirements



NOTICE

If customer provided MDP has been selected then customer provided MDP MUST meet all MDP requirements.



WARNING

CUSTOMER SUPPLIED MAIN DISCONNECT PANEL DESIGN NEEDS TO HAVE CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE MR SYSTEM POWER REQUIREMENTS.



WARNING

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM POWER AND REGULATION DEMANDS.

Upgraded MR System With LCC300 Magnet

After the MR system upgrade is installed the PDU will be located in the lower portion of the HFD-S/PDU Cabinet (an upgraded ACGD/PDU Cabinet). The Main Disconnect must meet the PDU requirements stated in [Section 2](#), Critical Power Requirements.

Upgraded MR System With 3.0T94 Magnet

After the MR system upgrade is installed the PDU will be located in the lower portion of either a HFD/PDU Cabinet or a HFD-S/PDU Cabinet (an upgraded ACGD/PDU Cabinet). The Main Disconnect must meet the PDU requirements stated in [Section 2](#), Critical Power Requirements.

3.2 System Power Distribution Unit

The PDU Module in the lower portion of HFD/PDU Cabinet has an integrated filter for a level of power conditioning. The largest allowable phase conductor the PDU will accept is 3/0 AWG (83 mm²). Larger feeder wires can be connected to the MDP with 3/0 AWG (83 mm²) between the MDP and PDU.

- NOTE:** The ground conductor between the MDP and PDU shall be minimum size of 1/0 AWG copper or the same size as the feeder wire, which ever is larger. Lug connector for the ground wire is to be provided by the contractor, recommended Amp Inc. number 36919 lug.
- NOTE:** The resistance between any two grounded devices in the MR system must not exceed 0.1 ohm (ie. PDU and MDP).
- NOTE:** Neutral, if present, must be terminated prior to or inside the Main Disconnect Panel and not brought to the PDU Module in the lower portion of HFD/PDU Cabinet (MR3).
- NOTE:** When the full MR system UPS option [Signature 5000 Series 3 UPS 100KVA (E4502FB)] is installed the feeder wiring from the UPS to the PDU Module must be sized to maintain voltage regulation of <5% at 100KVA.

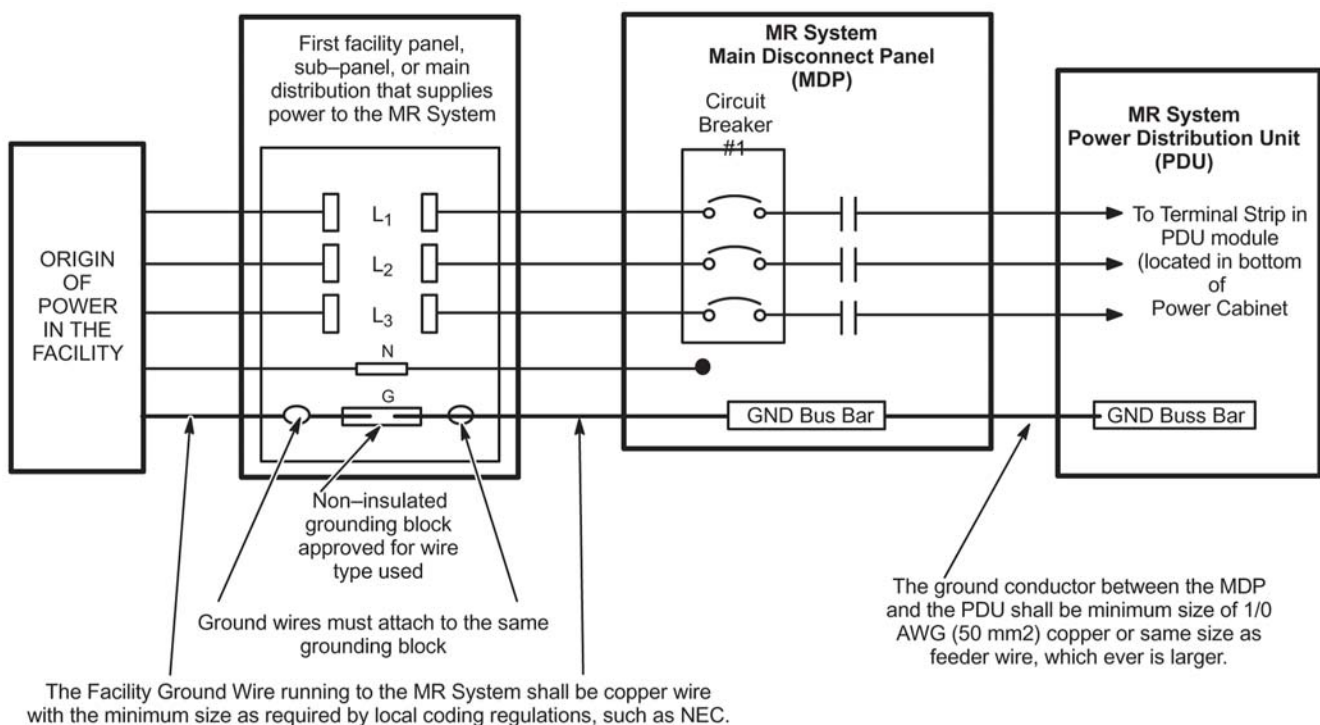
4 Grounding

4.1 Facility Ground

The ground for the MR system shall originate at the system power source, ie. transformer or first access point of power into the facility, and be continuous to the MR system Main Disconnect Panel (MDP) in the room. This ground can be spliced with "High Compression Fittings" and should be terminated at each distribution panel it passes through. When it is broken for a connection to a panel, it shall be connected into an approved non-insulated grounding block with the incoming and outgoing ground in this same grounding block, which is then connected to the steel panel, never using the steel or other material of the panel as the block. See [Illustration 6-1](#).

The connection at the power source shall be at the grounding point of the "Neutral - Ground" if a "Wye" transformer is used, or typical grounding points of separately derived system. In the case of an external facility, it shall be bonded to the facility ground point at the service entrance.

Illustration 6-1: Ground Wire To MR System & Ground Connection At Distribution Panel



Ground Wire

The main facility ground conductor to the MDP shall be copper wire and the minimum size as required by the local coding regulations, such as the NEC. A dedicated copper ground wire the same size as the feed wires or 1/0 AWG (which ever is larger) must be run from the MR system MDP to the PDU. See [Illustration 6-1](#). The ground wire impedance from the MR system disconnect, including the ground rod, shall not have an impedance greater than 2 ohms to earth as measured by one of the applicable techniques described in Section 4 of ANSI/IEEE Standard 142 - 1982 which can be accomplished using 3-point Fall Of Potential (3

point measurement) method or Clamp-On Ground Resistance measurement which requires a ground measurement device such as AEMC 3730.

4.2 System Ground

The MR system is designed with minimum ground loops to prevent noise currents and natural disturbances from flowing through the low-level signal reference path.

The three major grounding points in the MR system are: the system ground point (bus) in the System PDU (PD1), the enclosure ground points (ground studs located in each cabinet or enclosure), and the RF shielded room common ground point. This RF shielded room common ground point is to be located within 6 in. (152 mm) of the GE supplied Penetration Panel. Refer to [Chapter 8, Section 4](#), Electrical for a further description of the RF shielded room common ground point.

To ensure patient safety and system performance, the conditions defined in [Illustration 6-2](#) must be met when running power lines into the Magnet Room.

Any modifications or non-MR equipment grounds added to the MR ground system must be approved by your GE Service Representative in order to ensure safety and performance.

Illustration 6-2: MR Magnet Room Grounding Requirements And Typical Diagram

5 Ground Fault Protection

MR suites and radiology departments are considered health care facilities pursuant to National Electric Code (NEC) 2005 or 2002 Article 517.2 definitions and as such must be powered from sources that comply with the ground fault requirements of NEC Article 517.17. NEC 2005 Article 517.17 (B) or NEC 2002 Article 517.17 (A) states "Where ground fault is required for the operation of the service disconnecting means or feeder disconnecting means as specified in NEC 2002 Article 230.95 or 215.10, an additional step of ground fault protection shall be provided in the next level of feeder disconnecting means downstream towards the load."

NEC 2005 or 2002 Article 230.95 or 215.10 requires ground fault protection on service disconnecting means rated 1000 Amps or more on solidly grounded WYE services over 150 volts to ground but not over 600 volts phase to phase.

The two or more levels of ground fault shall be coordinated to provide selectivity between each level of ground fault such that a ground fault on the load side of the feeder would cause the feeder and not the service disconnect to open on a ground fault. Six cycles of separation between the different levels of ground fault tripping is required for the system to be considered selective in accordance with NEC 2005 Article 517.17 (B) or NEC 2002 Article 517.17(B).

Check national and local electrical codes.

6 Power Source Monitoring

The facility input power for the proposed system should be checked using a power line disturbance monitor for average line voltage, surges-sags, impulses, and frequency. Some of the recommended line analyzers which are designed for unattended monitoring are the Dranetz Models 656A or 658 and RPM Models 1651, 1656, or 1658.

Analysis should span a period to include two weekends so as to cover several days of normal use. The possibility of "brown-out" conditions which may be experienced in summer must be considered. Any existing power problems with large power consuming systems (x-ray units, CT scanners, etc.) or other computer installations at the proposed site should be reviewed as they may affect the MR system. Results of this analysis should be reviewed with your GE representative to determine if line conditioning is needed.

7 Emergency Power

Primary power should be distributed from the customer's emergency life-safety power branch to an emergency lighting source in the Magnet Room. All input power lines must be filtered upon entrance into the RF shielded room (Magnet Room) and grounded according to the requirements listed under System Grounding heading in [Section 4](#), Grounding. Always check national and local codes for other emergency power requirements.

8 DC Lighting Controller (Facility Option)

Direct current (DC) powered lighting is required in the Magnet Room per [Chapter 5, Section 7, Lighting](#). A constant lighting level DC Light Controller is available from GE as well as a variable DC lighting controller system. The wiring diagrams for these units are shown in [Illustration 6-3](#) and [Illustration 6-4](#). The input power, interconnect cabling, RF shielded room filters, lighting fixtures, and conduit are customer furnished.

The DC lighting systems output is rated nominally 115 VDC. Determining whether the 20 or 28 Amp system is required can be calculated by: $I = \text{Total Lamp Wattage} \div 115V$. If $I \leq 20$ then a 20 Amp system can be used. If $I \geq 20$ but < 28 then use the 28 Amp system.

Illustration 6-3: DC Lighting Controller (Facility Option) Wiring Diagram

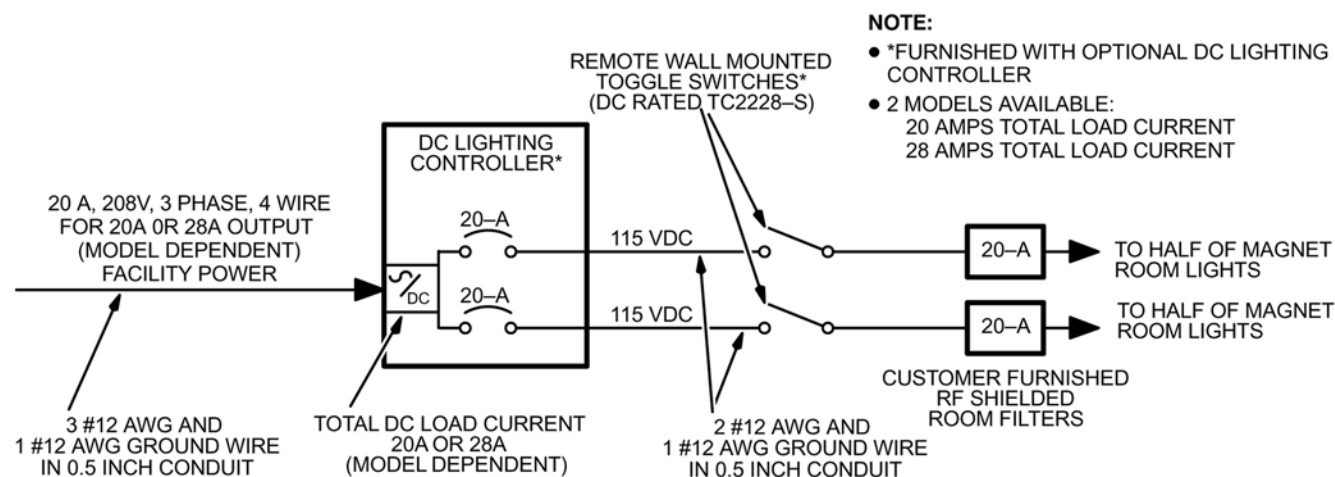
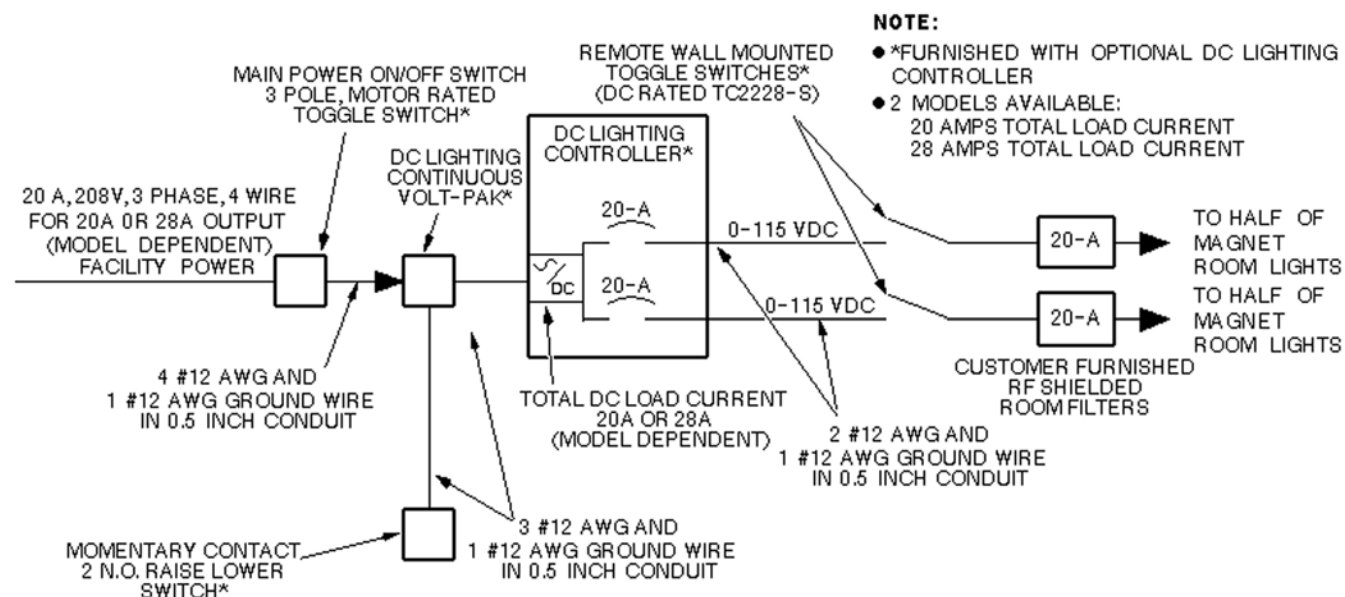


Illustration 6-4: Variable DC Lighting Controller (Facility Option) Wiring Diagram



9 IPCM Option Power Requirements

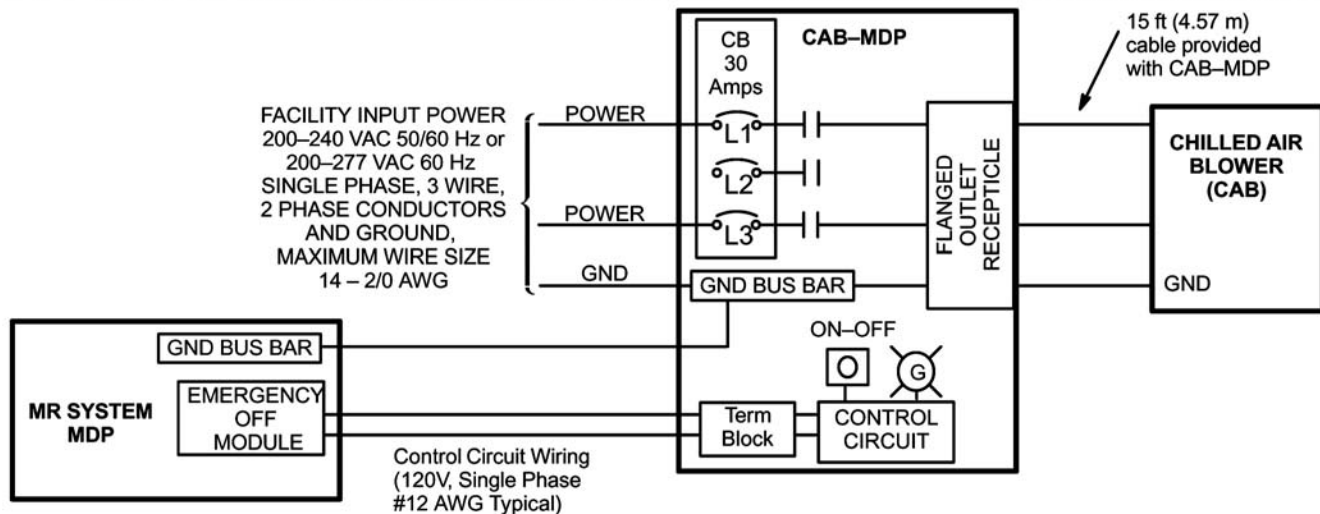
The IPCM Option hardware consists of the Chilled Air Blower (CAB), CAB Main Disconnect Panel (CAB-MDP), Bore Liner hardware, modified Penetration Panel waveguide, Bridge with notched sides, Bore Liner hardware including hoses and connectors, and IPCM Option Key.

The Chilled Air Blower (CAB) provides chilled air to the 2 piece Bore Liner via an air hose routed through the Penetration Panel (provided waveguide) to the back end of the Rear Pedestal. At the Rear Pedestal additional provided hardware, hoses and connectors are used to connect the air hose to both pieces of the Bore Liner.

The CAB-MDP is not powered from the MR system MDP or PDU and therefore requires facility provided power. The CAB-MDP must also be connected to the MR system MDP Emergency Power Off control circuit via a customer supplied cable. Refer to [Illustration 6-5](#) for IPCM Main Disconnect Panel set-up.

Illustration 6-5: IPCM Main Disconnect Panel Set-up

NOTE:• THE CAB-MDP TO CAB POWER CORD IS GE SUPPLIED IN THE IPCM OPTION. ALL OTHER WIRING IS CUSTOMER SUPPLIED.



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Chapter 7 Interconnect Data

1 Definition of Terms

The definition of terms used in the interconnects details tables throughout this chapter are:

Usable Length

Amount of cable/wire/hose/etc. available for site routing point to point of the FROM and TO equipment. The interconnect cable/wire/hose total length MINUS any required takeup at or within both the FROM and TO equipment determines the usable length.

Group Number

Area

Cross-sectional area of the combined cables in a group.

NOTE: The group area was found by adding up the circular cross-sectional areas of all individual cables within a group. It does not take any fill factors or space between cables into account. Adhere to applicable electrical codes for fill factors.

Between Units (From/To)

Component Designators as found in interconnects list tables throughout this chapter, for designators descriptions refer to Component Designators.

Run Number

Unique number assigned to each GE-supplied cable.

NOTE: The Run Number must be used when making special cable order inquiries.

Cable Diameter

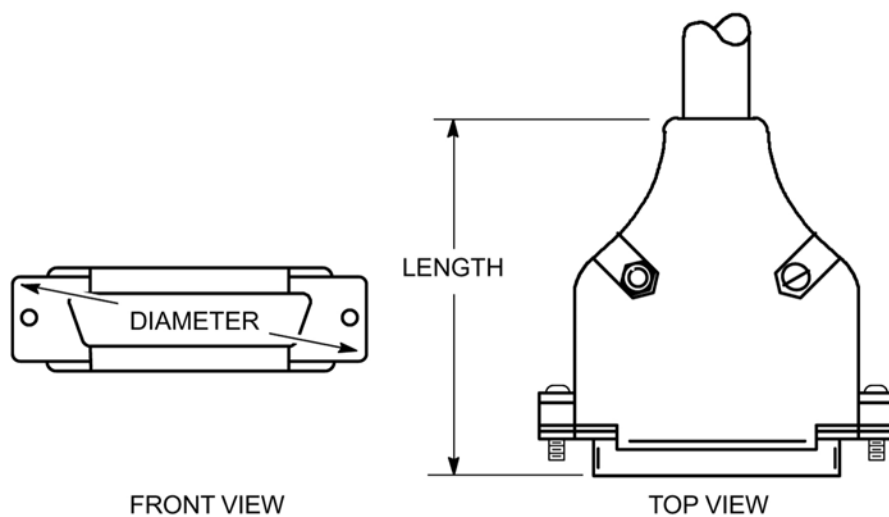
Diameter of an individual cable

Plug Pulling Diameter x Length

Cable plug dimensions as shown in [Illustration 7-1](#)

NOTE: In some cases, a cable has more than one connector on an end. These cables will have the number of connectors following the english dimensions of the plug pulling diameter times length (e.g. '2.0x3.25 -x2' means there are 2 connectors with dimensions of 2.0 in. diameter and 3.25 in. length). The same number of connectors apply to the metric dimensions as well.

Illustration 7-1: Subminiature-D Connector Plug Pulling Dimensions



2 Power Interconnects

Signa EXCITE HD 3.0T upgrades do not change the quantity of power interconnects to the MDP.

Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. Power cables may be pulled by the lug terminal ends so the connector pulling dimensions on the plug ends will not be a factor for routing power cables.

NOTE: Unless otherwise specified, cables and components listed in the following subsections are supplied by GE.

3 Emergency Off Wiring

3.1 Introduction

This section addresses wiring for the Emergency Off circuit (also known as protective disconnect circuit). Refer to [Chapter 6, Section 3.1](#), Main Disconnect Panel (MDP) for information on the recommended protective disconnect device and emergency off button locations and mounting.

The emergency off wiring for the MR system is unique because the wiring into the Magnet Room must be RF tight.

3.2 Main Disconnect Panel Connections

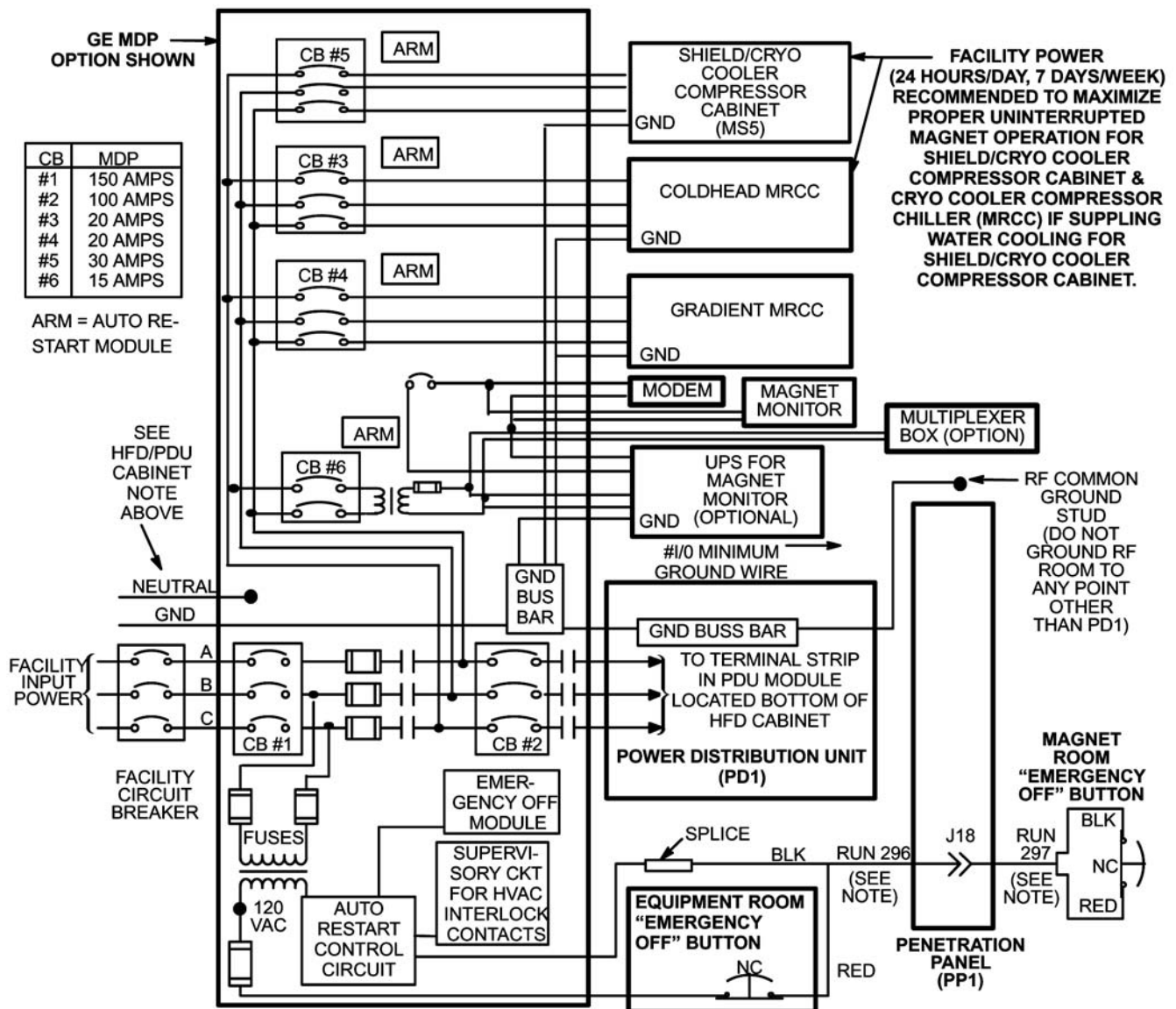
NOTE: Existing systems retain their TSCC or TGWC system cooling equipment. These existing units are not documented in this section.

The emergency off circuit is shown in [Illustration 7-2](#) . The circuit utilizes the normally closed series loop shown. The GE pre-engineered MDP Option provides 2 emergency off buttons.

Illustration 7-2: Protective Disconnect Set-Up

NOTE: • RUNS 296 AND 297, & POWER CORDS FOR SHIELD/CRYO COOLER COMPRESSOR CABINET, MRCC, & MAGNET MONITOR EQUIPMENT (MAGNET MONITOR, UPS INPUT & OUTPUT, MODEM, OPTIONAL MULTIPLEXER) ARE GE SUPPLIED CABLES. **ALL OTHER WIRING IS CUSTOMER SUPPLIED.**

- TWO REMOTE EMERGENCY "OFF" BUTTONS ARE SUPPLIED WITH GE MDP OPTION, **EMERGENCY OFF BUTTONS ARE CUSTOMER SUPPLIED IF GE MDP OPTION NOT USED.**
- CIRCUIT BREAKERS ARE PROVIDED FOR PDU, CRYO COOLER COMPRESSOR CHILLER, GRADIENT CHILLER, SHIELD/CRYO COOLER COMPRESSOR CABINET, MAGNET MONITOR EQUIPMENT CIRCUITS.
- ALL BRANCH CIRCUITS DROP OUT ON LOSS OF POWER. COLDHEAD MRCC, GRADIENT MRCC, SHIELD/CRYO COOLER COMPRESSOR CABINET, & MAGNET MONITOR EQUIPMENT AUTOMATICALLY RESTART AFTER 3 SEC TIME DELAY UPON RESTORATION OF POWER. EMERGENCY OFF LOCKS OUT ALL CONTACTORS.
- IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT USED THEN NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT PANEL AND NOT BROUGHT TO THE HFD/PDUCABINET
- SUPERVISORY CIRCUIT FOR HVAC INTERLOCK CONTACTS OPEN ON LOSS OF DC POWER OR EMERGENCY OFF OPERATION.



3.3 Magnet Room Wiring

GE provides two cables for routing the emergency off circuit through the Penetration Panel and into the magnet room (Runs 296 and 297). Alternate wiring may be used by the customer; however, the use of these cables ensures that the emergency off wiring will be RF tight.

[Section 3.2](#) illustration black and red wires are used for connections on the ends of Runs 296 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. (Runs 296 and 297 are actually nine wire cables.)

4 EXCITE HD 3.0T Upgrades Group Interconnects

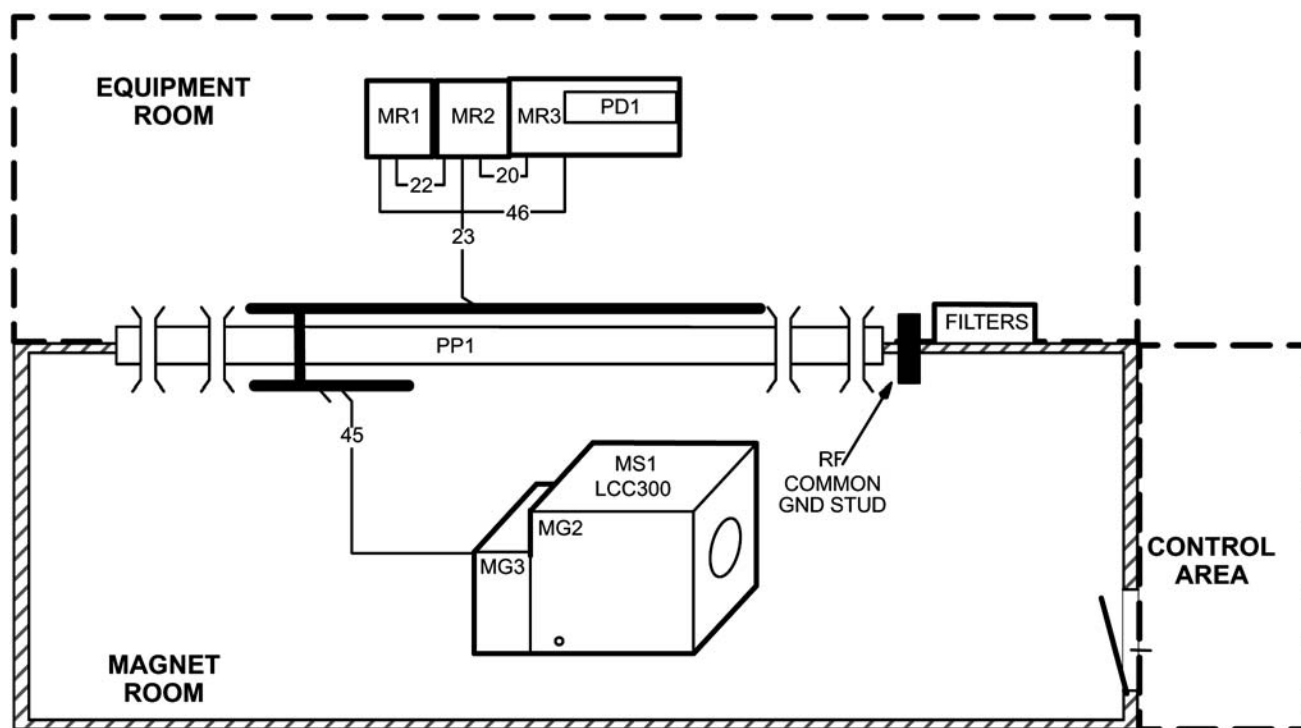
4.1 Release 11.x To Release 12.x Upgrades

4.1.1 System With LCC300 Magnet

NOTE: Existing systems with LCC300 Magnet retain their system cooling equipment (i.e. TSCC, TGWC, MRCC unit(s), GWHX). These existing units are not documented in Interconnect Data sections.

Illustration 7-3 shows the Group Interconnect Diagram for a Release 11.x upgraded to Release 12.x for a system with a LCC300 Magnet. This diagram should be referred to for when using the tables in this chapter for Release 11.x upgrade to Release 12.x Signa EXCITE HD 3.0T system with a LCC300 Magnet.

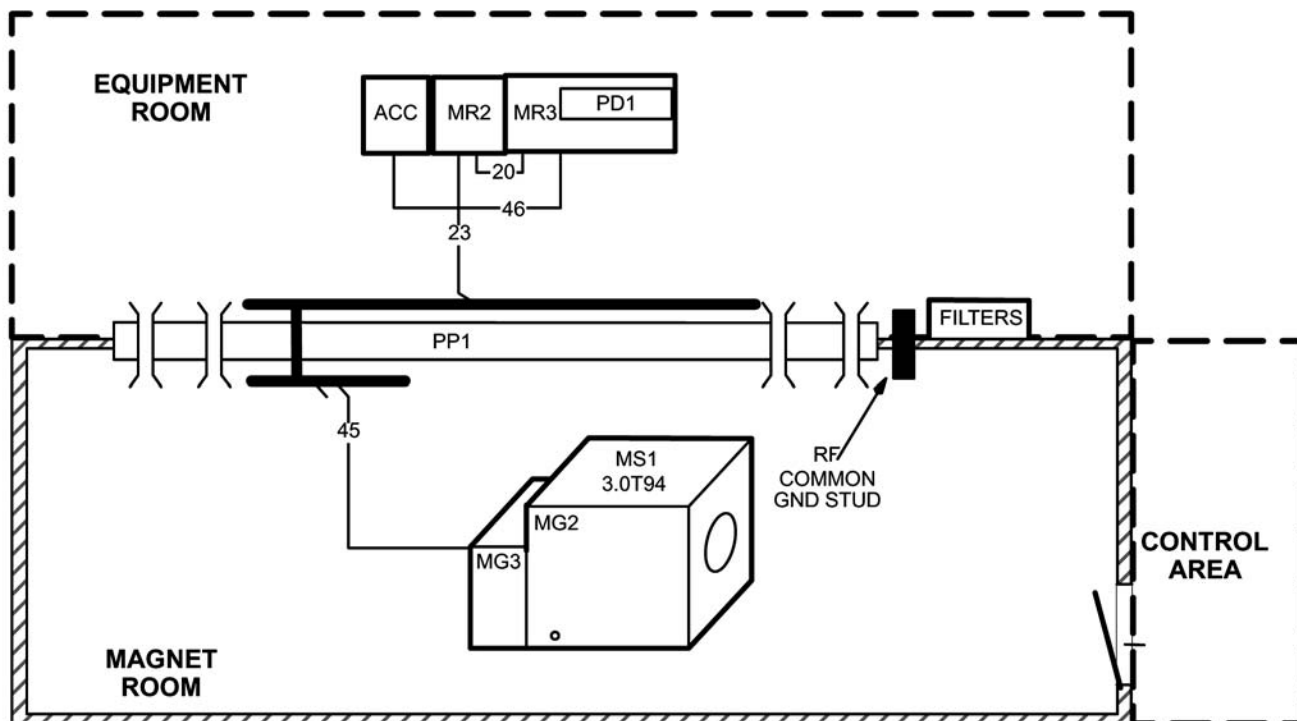
Illustration 7-3: Release 11.x To Release 12.x With LCC300 Magnet Group Interconnect Diagram



4.1.2 System With 3T94 Magnet

Illustration 7-4 shows the Group Interconnect Diagram for a Release 11.x upgraded to Release 12.x for a system with a 3.0T94 Magnet. This diagram should be referred to for when using the tables in this chapter for Release 11.x upgrade to Release 12.x Signa EXCITE HD 3.0T system with a 3.0T94 Magnet.

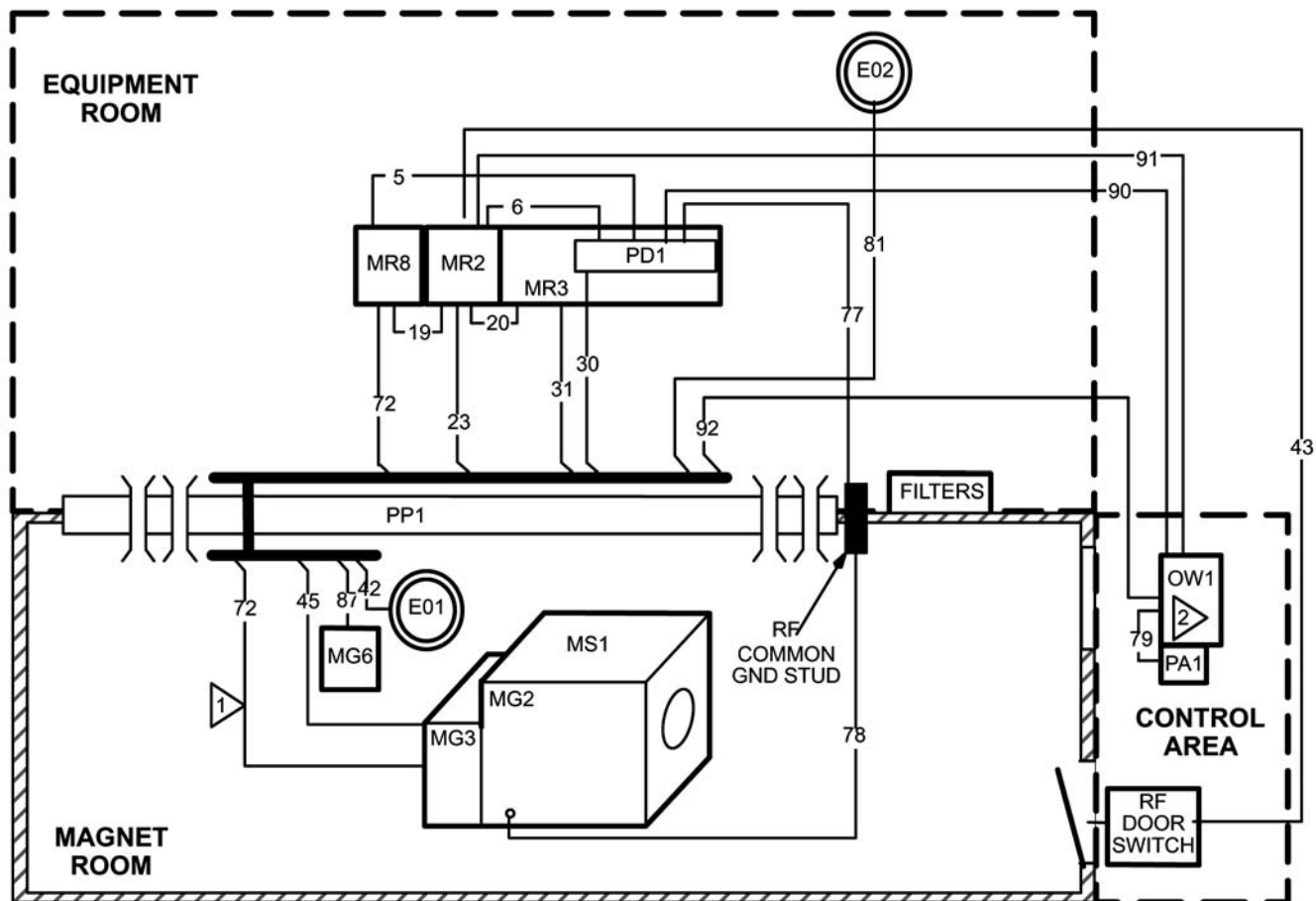
Illustration 7-4: Release 11.x To Release 12.x With 3.0T94 Magnet Group Interconnect Diagram



4.2 Pre-EXCITE Signa With 3.0T94 To Release 12.x Upgrades

[Illustration 7-5](#) shows the Group Interconnect Diagram for a Pre-EXCITE Signa With 3.0T94 upgraded to Release 12.x. This diagram should be referred to for when using the tables in this chapter for Pre-EXCITE Signa With 3.0T94 upgraded to Release 12.x.

Illustration 7-5: Pre-EXCITE To Release 12.x With 3.0T94 Magnet Group Interconnect Diagram



5 Upgrade Catalogs And Interconnects Information Index

NOTE: Systems retain their existing system cooling equipment (TSCC, TGWC, MRCC units, GWHX unit, etc.). These existing units are not documented in this section.

lists the upgrade catalogs which provide system interconnects and the location of interconnects details information.

Table 7-1: Upgrade Catalogs And Interconnects Information Index

Catalog #	Name	Interconnects Information
M3000PT	HFD Gradient Driver Upgrade	Section 6 , M3000PT Interconnects
M3033LH	Integrated Patient Comfort Module (IPCM) Option for LCC300 Magnet	Section 7 , IPCM Option Interconnects
M3334WA	LCC300 Magnet EXCITE III Fixed Site Hardware Upgrade	Section 8 , M3334WA And M3334WC Interconnects
M3334WB	LCC300 Magnet EXCITE III Relocatable Hardware Upgrade	Provides interconnect lengths compatible with GE certified Van configurations.
M3334WC	3T94 Magnet EXCITE III Fixed Site Hardware Upgrade	Section 8 , M3334WA And M3334WC Interconnects
M3334WH	8KW MNS Option for system with EXCITE System Cabinet	Section 9 , M3334WH Interconnects
M3334WK	3T/94 Magnet Pre-EXCITE to E-III Fixed Site Upgrade 8 ch, R400, ES	Section 10 , M3334WK And M3334WL Interconnects
M3334WL	3T/94 Magnet Pre-EXCITE to E-III Fixed Site Upgrade 16 ch, R800, ES	Section 10 , M3334WK And M3334WL Interconnects

6 M3000PT Interconnects

M3000PT HFD Gradient Driver Upgrade provides the interconnects listed in [Table 7-2](#) .

Table 7-2: M3000PT Equipment Room Interconnects Details

Group	Between Units		Usable Length ft (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
20	MR2	MR3	16 (4.88)	1180	0.305 (7.75)	1.30x2.00 (33.0x50.8)	1.30x2.00 (33.0x50.8)	
46	MR1 / ACC	MR3	16 (4.88)	1263	0.305 (7.75)	1.30x2.00 (33.0x50.8)	1.30x2.00 (33.0x50.8)	MR1 used for system with LCC300 Magnet. ACC used for system with 3T94 Magnet.

7 IPCM Option Interconnects

The IPCM Option for LCC300 Magnet (M3033LH) consists of the following items:

- Chilled Air Blower (CAB)
- CAB Main Disconnect Panel (CAB-MDP)
- Bore Liner (2 pieces) installed in Magnet
- Interconnect cables.

Illustration 7-6 shows the system interconnect groups for IPCM Option for LCC300 Magnet.

For cable lengths refer to Table 7-3 and for cable details refer to Table 7-4 and Table 7-5 .

Illustration 7-6: IPCM Option Group Interconnects Diagram

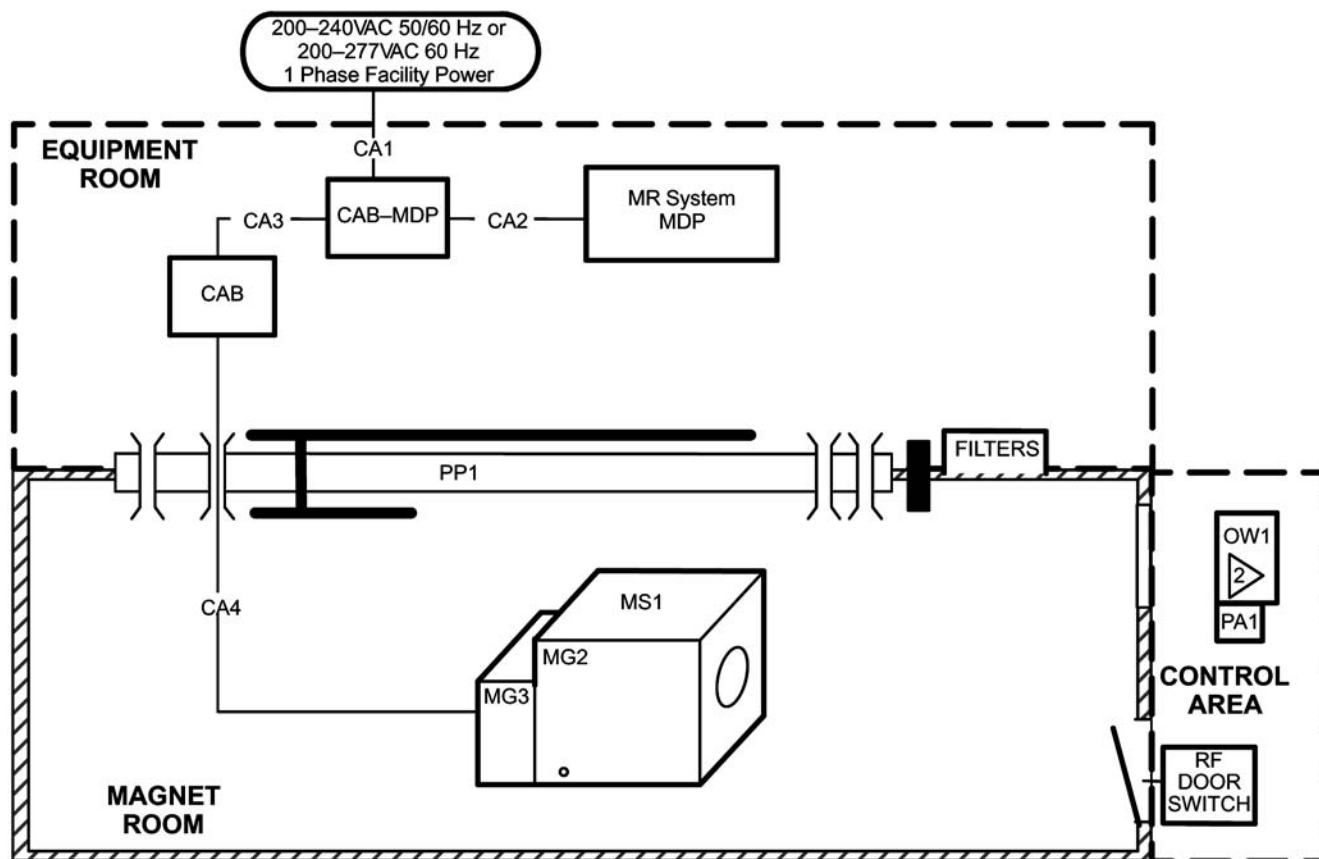


Table 7-3: IPCM Option Interconnects Length Provided

Location	Group	Between		Usable Length ft (m)
		From	To	
L1/L2	CA4	CAB	MG3	72 (22)
L3	CA1	Facility Power	CAB-MDP	Customer supplied
	CA2	CAB-MDP	MDP	Customer supplied
	CA3	CAB-MDP	CAB	10 (3.05)

Table 7-4: IPCM Option Interconnects - L1/L2

Group	Group Area in. ² (mm ²)	Between Units		Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
		From	To			From	To	
CA4	3.98 (2570)	CAB	MG3	--	2.25 (57.2)	air hose	air hose	Air hose routed to MG3. Outside MG3 connects to Y junction and two 1.75 in. (44.5mm) hoses routed to Bore Liner 2 pieces.

Table 7-5: IPCM Option Interconnects - L3

Group	Group Area in. ² (mm ²)	Between Units		Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
		From	To			From	To	
CA1	--	Facility Power	CAB-MDP	--	--	--	--	Customer supplied, refer to Chapter 6, Section 9 , IPCM Option Power Requirements
CA2	--	CAB-MDP	MDP	--	--	--	--	Customer supplied 120 control circuit wire, refer to Chapter 6, Section 9 , IPCM Option Power Requirements
CA3		CAB-MDP	CAB	--	0.5 (12.7)	2.00x3.25 (50.8x82.6)	2.00x3.25 (50.8x82.6)	Power cable

8 M3334WA And M3334WC Interconnects

M3334WA LCC300 Magnet EXCITE III Fixed Site Hardware Upgrade and M3334WC 3T94 Magnet EXCITE III Fixed Site Hardware Upgrade provide the interconnects listed in [Table 7-6](#) and [Table 7-7](#).

Table 7-6: M3334WA & M3334WC Magnet Room Interconnects Details

Group	Between Units		Usable Length ft. (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
45	PP1	MG2/3	40 (12.19)	1197	0.305 (7.75)	1.30x2.00 (33.0x50.8)	1.30x2.00 (33.0x50.8)	RF Transmit Cable
				1198	0.59 (15.0)	1.50x2.50 (38.1x63.5)	1.50x2.50 (38.1x63.5)	RF Transmit Cable Cut to length at site
				1199	0.59 (15.0)	1.50x2.50 (38.1x63.5)	1.50x2.50 (38.1x63.5)	RF Transmit Cable Cut to length at site
				1200	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1201	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1203	0.525 (13.3)	2.80x2.00 (71.1x50.8)	2.80x2.00 (71.1x50.8)	
				1216	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				1219	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable

Table 7-7: M3334WA & M3334WC Equipment Room Interconnects Details

Group	Between Units		Usable Length ft (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
23	MR2	PP1	55 (16.76)	1185	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1186	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1218	0.45 (11.4)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1251	0.525 (13.3)	2.80x2.00 (71.1x50.8)	2.80x2.00 (71.1x50.8)	
22	MR2	MR1	14 (4.27)	1252	0.234 (5.9)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	These Runs are used for systems with LCC300 Magnet
				1253	0.234 (5.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	

9 M3334WH Interconnects

Illustration 7-7 shows the system interconnect groups which have additional system interconnects M3334WH. Table 7-8 contains the details for the additional interconnects.

Illustration 7-7: M3334WH Additional Cable Groups Interconnects

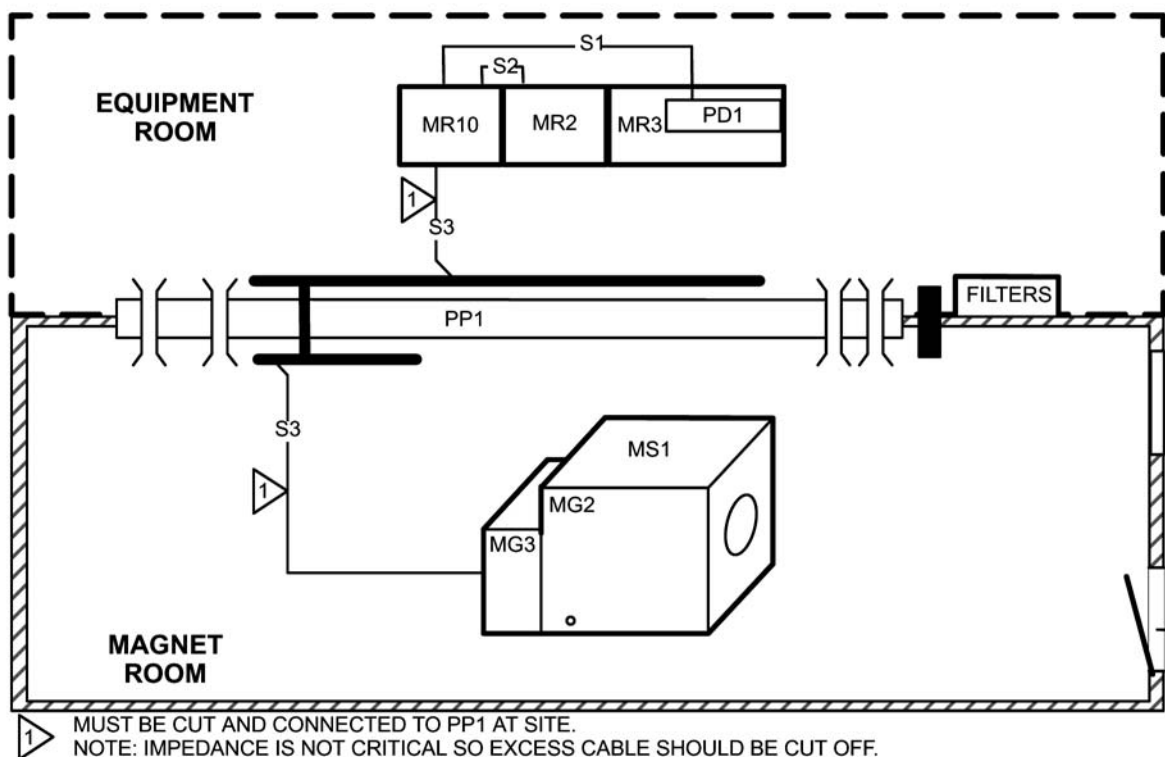


Table 7-8: M3334WH Additional Interconnects Details

Group	Between Units		Usable Length ft (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
S1	PD1 See Note 1	MR9	16 (4.88)	1138	0.59 (11.8)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable

Table 7-8: M3334WH Additional Interconnects Details (cont'd)

Group	Between Units		Usable Length ft (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
S2	MR2	MR9	21 (6.40)	1139	0.24 (6.10)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1140	0.44 (11.18)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				1141	0.24 (6.10)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1142	0.24 (6.10)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1143	0.24 (6.10)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1144	0.24 (6.10)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1145	0.23 (5.94)	0.57x1.125 (14.5x25.58)	0.57x1.125 (14.5x25.58)	
S3	MR10	MG2/3	88 (26.8)	1146/1147	0.59 (14.99)	1.44x0.87 (36.6x22.1)	1.44x0.87 (36.6x22.1)	RF Transmit cable , must be cut to length and connected via PP1 at site. Usable length provided for routing in Equipment Room and Magnet Room.

10 M3334WK And M3334WL Interconnects

M3334WK 3T/94 Magnet Pre-EXCITE to E-III Fixed Site Upgrade 8 ch, R400, ES and M3334WL 3T/94 Magnet Pre-EXCITE to E-III Fixed Site Upgrade 16 ch, R800, ES provide the interconnects listed in [Table 7-9](#)

Table 7-9: M3334WK & M3334WL Interconnects Details

Group	Between Units		Usable Length ft. (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
5	PD1 See Note 1	MR8	0.73 (462)	1113	0.2 (4.95)	Hard Wired	4.5x9.5 (114.3x241.3)	#8 AWG / 4 wire Power cable
				1150	0.94 (23.75)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
6	PD1 See Note 1	MR2	0.803 (518)	037	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				968	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
				971	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	
19	MR2	MR8	0.436 (281)	1231	0.44 (11.18)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				1264	0.45 (11.4)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	
				1239	0.23 (5.84)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1240	0.23 (5.84)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1242	0.23 (5.84)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
20	MR2	MR3	0.923 (595)	1179	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1179 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
				1180	0.305 (7.75)	1.30x2.00 (33.0x50.8)	1.30x2.00 (33.0x50.8)	

Table 7-9: M3334WK & M3334WL Interconnects Details (cont'd)

Group	Between Units		Usable Length ft. (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
23	MR2	PP1	55 (16.8)	1045	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
				1184	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				1185	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1186	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1187	0.45 (11.4)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1188	0.34 (8.6)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				1189	0.525 (13.3)	2.80x2.00 (71.1x50.8)	2.80x2.00 (71.1x50.8)	
				1190	0.525 (13.3)	2.80x2.00 (71.1x50.8)	2.80x2.00 (71.1x50.8)	
				1191	0.19 (4.8)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1192	0.19 (4.8)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1193	0.19 (4.8)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1194	0.19 (4.8)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1195	0.19 (4.8)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1238	0.59 (150)	1.50x2.50 (38.1x63.5)	1.50x2.50 (38.1x63.5)	RF Receive Cable Cut to length at site
30	PD1 See Note 1	PP1	59 (17.98)	1182	0.433 (11.25)	Ring Terminals	1.78x2.59 (45.2x65.8)	
				1183	0.433 (11.25)	Ring Terminals	1.78x2.59 (45.2x65.8)	
31	MR3	PP1	61 (18.59)	1181	0.305 (7.75)	1.30x2.00 (33.0x50.8)	1.30x2.00 (33.0x50.8)	
42	PP1	EO1	68 (20.7) allows EO1 takeup of 15 ft (4.57 m)	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	Refer to Section 3 , Emergency Off Wiring
43	MR2	RF Door Switch	83 (25.3)	701	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	RF Door Switch provided by RF Screen Room vendor. Usable length allows RF Door Switch takeup of 15 ft (4.57 m).

Table 7-9: M3334WK & M3334WL Interconnects Details (cont'd)

Group	Between Units		Usable Length ft. (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
45	PP1	MG2/3	40 (12.2)	282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1197	0.305 (7.75)	1.30x2.00 (33.0x50.8)	1.30x2.00 (33.0x50.8)	RF Transmit Cable
				1198	0.59 (15.0)	1.50x2.50 (38.1x63.5)	1.50x2.50 (38.1x63.5)	RF Transmit Cable Cut to length at site
				1199	0.59 (15.0)	1.50x2.50 (38.1x63.5)	1.50x2.50 (38.1x63.5)	RF Transmit Cable Cut to length at site
				1200	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1201	0.71 (18.0)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1202	0.45 (11.4)	2.75x2.25 (69.9x57.2)	2.75x2.25 (69.9x57.2)	RF Receive Cable
				1203	0.525 (13.3)	2.80x2.00 (71.1x50.8)	2.80x2.00 (71.1x50.8)	
				1216	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
72	MR2	MG2/3	88 (26.8)	1125 & 1127	0.87 (22.1)	1.5x2.50 (38.1x63.5)	1.5x2.50 (38.1x63.5)	
								Note: These runs are RF Transmit Cables. Runs 1125/1127 must be cut to length and connected via PP1 at site. Usable length provided is for routing in Equipment Room and Magnet Room.
77	PD1 See Note 1	RF Common Ground Stud	83 (25.3)	1254	0.464 (11.79)	Ring Terminal	Ring Terminal	1 ground wire
78	RF Common Ground Stud	MS1 Ground Stud	40 (12.2) minus takeup at RF Common Ground Stud	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire
81	EO2	PP1	50 (15.24)	296	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	Refer to Section 3 , Emergency Off Wiring

Table 7-9: M3334WK & M3334WL Interconnects Details (cont'd)

Group	Between Units		Usable Length ft. (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
87	PP1	MG6	37 (11.3)	045	0.195 (4.95)	Ring Terminal	Ring Terminal	The usable length is based on MG6 being mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1 See Note 1	WC1		970	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	#10 AWG / 3 wire Power cable
90	PD1 See Note 1	OW1	80 (24.4)	1081	0.70 (17.78)	Hard Wired	Hard Wired	#10 AWG / 4 wire power cable
91	MR2	OW1	80 (24.4)	1082	0.25 (6.35)	0.5x0.75 (12.7x19.1)	0.5x0.75 (12.7x19.1)	
				1083	0.33 (8.30)	1.75x2.00 (44.5x50.8)	1.75x2.00 (44.5x50.8)	
				1255	0.31 (7.75)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
92	PA1	OW1	80 (24.4)	1085	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
Note 1 The PDU is a module (PD1) in the lower portion of the HFD/PDU Cabinet (MR3).								

11 Remote MRU Option Interconnects

The Remote MRU Option consists of the following items:

- Secondary MRU
- Remote MRU Connector Box (to be mounted inside the Magnet Enclosure)
- Interconnect cables

Illustration 7-8 shows the Interconnect Diagram , Table 7-10 contains the cable data.

Illustration 7-8: Remote MRU Interconnect Diagram

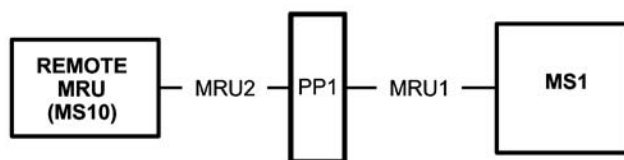


Table 7-10: Remote MRU Interconnects Details

Group	Between Units		Usable Length ft. (m)	Run #	Cable Diameter in. (mm)	Plug Pulling Diameter x Length in. (mm)		Notes
	From	To				From	To	
MRU1	PP1	MS1	81.0 (24.69)	1265	0.30 (7.6)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
MRU2	MS10	PP1	88.0 (26.82)	1266	0.30 (7.6)	0.787x2.33 (20.0x59.2)	1.30x2.00 (33.5x50.8)	

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Chapter 8 RF Shielded Room

1 RF Shielded Room Requirements

NOTE: The following information applies to areas of the site impacted by the MR system upgrade.

Improvements in MR imaging technology have increased imaging capabilities. MR procedures require a stable RF environment to achieve high resolution image quality. RF sources both inside and outside the Magnet Room have the potential to adversely affect image quality. Therefore the Magnet Room must be properly RF shielded to prevent external RF sources from entering the room. In addition, the selection of materials and construction methods of the RF Shield must be designed and installed to minimize the generation of adverse signals within the RF Shield and Magnet Room.

1.1 RF Shielding Background

RF sources which can adversely affect image quality may be generated by discrete frequency or broadband noise (RF) sources.

1.1.1 Discrete Frequency

Discrete RF interferences are narrowband and are fixed frequency in nature. The Magnet Room must be RF shielded from RF sources so external RF energy does not degrade the MR system RF receivers at the system imaging frequencies, refer to [Chapter 5, Section 11](#), Ambient Radio Frequency Interference (RFI). Some potential sources for discrete frequency signals are radio station transmitters, mobile or hand-held RF transmitting devices, etc.

1.1.2 Broadband RF Noise

Broadband RF noise is a single transient or continuous series of transient disturbances caused by an electrical discharge, for system imaging frequencies refer to [Chapter 5, Section 11](#), Ambient Radio Frequency Interference (RFI). Low humidity environmental conditions will have higher probability of electrical discharge, refer to [Chapter 5, Section 3](#), Temperature and Humidity Specifications. The electrical discharge can occur due to electrical arcing (micro arcing) or merely a static discharge. Some potential sources capable of producing electrical discharge include:

- loose hardware/fasteners vibration or movement (electrical continuity must always be maintained)
- flooring material including raised access flooring (panels & support hardware) and carpeting
- electrical fixtures
 - lighting fixtures
 - track lighting
 - emergency lighting
 - battery chargers
 - outlets
- ducting for HVAC and cable routing
- RF Shield seals (walls, doors, windows, etc.).

1.2 RF Shielded Room Requirements

The Magnet Room RF Shield must meet the requirements defined in [Table 8-1](#) for the system to produce high quality MR images.

NOTE: The RF Attenuation and Ground Isolation are intended to control discrete RF signals and RF interference sources. The remaining parameters contained in [Table 8-1](#) are intended to minimize potential broadband noise sources.

Table 8-1: RF Shielded Room Requirements

Parameter (See Note *)	Requirements (See Note *)
RF Attenuation	100dB (150MHz $\pm$ 10MHz) planewave, refer to Chapter 12, Section 1 , RF Shielded Enclosure Test Guideline Frequency Range subsection.
Ground	1. Ground Isolation: 1,000 ohms or greater 2. Primary Ground: All RF Shield components (walls, floor, ceiling, etc) must be electrically bonded together to form one common ground plane which is connected to the Facility Grounding Conductor. The RF Shield must be grounded back to the facility ground via the RF Common Ground Stud connection to the MR system PDU. NOTE: Introduction of facility power into the RF Shielded Room must not compromise the RF Shield Primary Ground. 3. Secondary Ground (Other grounds that connect the outside of the RF Shield Room to earth grounds are called secondary grounds): Secondary grounds must not compromise Ground Isolation of 1,000 ohms or greater.

Table 8-1: RF Shielded Room Requirements (cont'd)

Parameter (See Note *)	Requirements (See Note *)
Materials	1. The choice of RF Shield material including fasteners is the responsibility of the customer's architect and RF vendor. 2. The choice of material must not affect magnet homogeneity (i.e. copper, brass or treated aluminum are non-magnetic and will not affect Magnet homogeneity). NOTE: Any steel RF Shield will affect the magnet's homogeneity and must be reviewed by GE Healthcare MR Siting and Shielding Group. 3. The floor under the Magnet in a 10 ft x 12 ft (3.048 m x 3.658 m) area must not be fabricated from magnetic materials, refer to Magnet Room Floors Magnetic Properties in Construction Materials. 4. The Magnet Room floor materials must meet the requirements in Section 6.4, Magnet Room Floors. 5. The door or any other moving or non-rigid parts must not be fabricated from magnetic materials. 6. The RF Shield integrity (attenuation) must not be compromised by corrosion for the anticipated duration of usage for MR imaging. The following items are critical: a. Avoid direct contact of materials of different solution potential (e.g. dissimilar metals galvanic corrosion) when selecting fastens to secure the RF screen material to the RF supporting structure. b. Ensure the RF shield seams/joints have overlaps that are properly dressed selecting proper material to avoid galvanic corrosion. c. Introduce sacrificial anodes to prevent corrosion of critical RF shield components. NOTE: Typically the RF Shield surrounds the Magnet Room finished walls, ceiling, and floor. Therefore the RF Shield may not be in a temperature and humidify controlled environment. d. It is highly recommended the RF Shield vendor use a water seal / thin coat of lacquer to protect any exposed RF shield material from condensation produced when servicing the Magnet cryogens or from cleaning liquids. e. RF Screen Room including all openings (i.e. windows, doors, vents, etc.) need acoustic properties to meet local regulations and customer requirements. NOTE: RF Screen Room doors with <55 db acoustic attenuation have caused customer acoustics issues. Refer to Chapter 5, Section 8.4, Acoustic Design Guidelines for additional information.

Table 8-1: RF Shielded Room Requirements (cont'd)

Parameter (See Note *)	Requirements (See Note *)
Construction	- The design of the shield support system is the responsibility of the customer's architect and RF vendor. - RF Shield construction methods must not compromise RF Shield integrity (attenuation) for the anticipated duration of usage for MR imaging. - When RF shield seams/joints are sealed with solder: all solder drips, cold solder joints, and cracked solder joints must be cleaned and repaired. - RF Shield movement can cause micro electrical arcs (static discharge or broadband RF noise) that will cause MR image artifact known as White Pixel. - Physical fluctuation of the RF Shield material can result in RF leaks due to seam openings, cracks, enlarged holes at fasteners. Physical fluctuation of the RF Shield can be generated by cyclic air pressure changes and door closures/openings. - All electrical and mechanical connections and fasteners including screws, nails, nuts, bolts, clips clamps, concrete anchors, seismic anchors, etc. must be tightened and secured to supplier specifications so as not to become a potential broadband noise source. - All fasteners must be solid locking devices such as t-nuts, PEM nuts or welded nuts; no self tapping screws allowed. - The Magnet Room floor design and construction must meet the requirements in Section 6.4, Floors.   CAUTION For safety reasons, magnetic materials must be secured to ensure magnetic components do not become projectiles. Methods of securing must not loosen due to repeated use, some options include (but not limited to) redundant restrains straps or anchors/bolts with locking nuts.
Testing	- The customer's architect and RF vendor are responsible for conducting testing to verify compliance with the requirements for RF attenuation and ground isolation. - The RF shielded room verification test is to be performed in the presence of a GE representative. - The FINAL RF Shielded room acceptance test shall be performed in accordance with Chapter 12, Section 1, RF Shielded Enclosure Test Guideline.
Maintenance	- The customer is responsible for maintaining the RF Shield service life integrity for the anticipated duration of usage for MR imaging per the RF vendor's recommended maintenance. - The customer is responsible to notify the GE Service Representative of any RF shielded Room maintenance issues since there may be system performance impacts.
NOTE:	* The RF Shielded Room design, materials, construction, and installation shall be such to meet the requirements for the anticipated duration of usage for MR imaging.

2 Vents

2.1 Cryogenic Vent

The upgrade catalogs documented in this manual do not change the existing magnet. RF Shielded Room specifications for the Cryogenic Vent remain the same as specified in the existing system pre-installation document.

2.2 HVAC

RF shielded room contractor is to install HVAC waveguides (open pipes or honeycomb-type) which penetrate room and to ensure waveguides are non-magnetic and electrically isolated. HVAC contractor is to determine size and number of vents, consistent with local codes.

Honeycomb-type waveguides must be accessible for annual customer inspection and cleaning / maintenance.

An exhaust fan placed outside the RF shielding with appropriate wave guide filtering is required for quick removal of helium gas in the event large amounts of helium disperse into the Magnet Room. The exhaust fan can be connected to the output relay of the optional oxygen monitor. The fan will then be activated in the event the room oxygen level is less than 18%. Refer to [Chapter 5, Section 9](#), Room Ventilation for other exhaust fan requirements.

3 Plumbing

All metallic pipes entering the RF Room, excluding cryogenic vent and floor drains, must be located within 30 inches (762 mm) of the RF common ground.



NOTICE

Mounting and support of all metallic pipes must comply with requirements in [Section 1](#), RF Shielded Room Requirements to minimize the possibilities of electrical discharge which can cause RF broadband noise.

NOTE: When welding in an MR room with system equipment installed, the return path for the welding must be in very close proximity to the welding. The close proximity is needed to make sure the welding currents do not cause damage to the system. Never use the building structure as a return path for welding.

3.1 Water

All pipe waveguides must comply with requirements in [Section 1](#), RF Shielded Room Requirements to minimize the possibilities of electrical discharge which can cause RF broadband noise. All plumbing must be in accordance with local and national codes.

3.2 Medical Gases

The customer should consider if medical gases are to be piped into the Magnet Room along with suction service for patient life support. Remember, all non-electrical entries into the Magnet Room must use appropriate waveguide. Special precaution must be taken to ensure that ferromagnetic medical gas cylinders are not brought into the Magnet Room.

3.3 Sprinklers

If using sprinklers in the Magnet Room, dry pipe systems have the advantage of reducing ground problems. However, all decisions regarding fire protection systems are the customer's responsibility. If wet-type sprinkler system is used, pipe penetration should be limited to one location.

4 Electrical

4.1 Electrical Lines and Filters

The entry of any electrical lines into the RF Shielded Room must be filtered to ensure compliance with the RF Shielded Room attenuation requirements. The RF Shielded Room vendor must supply filters for all penetrations of the RF shielding excluding the lines entering through the GE supplied RF penetration panel. All filters (for electrical lines) must be located outside the 200 gauss line.

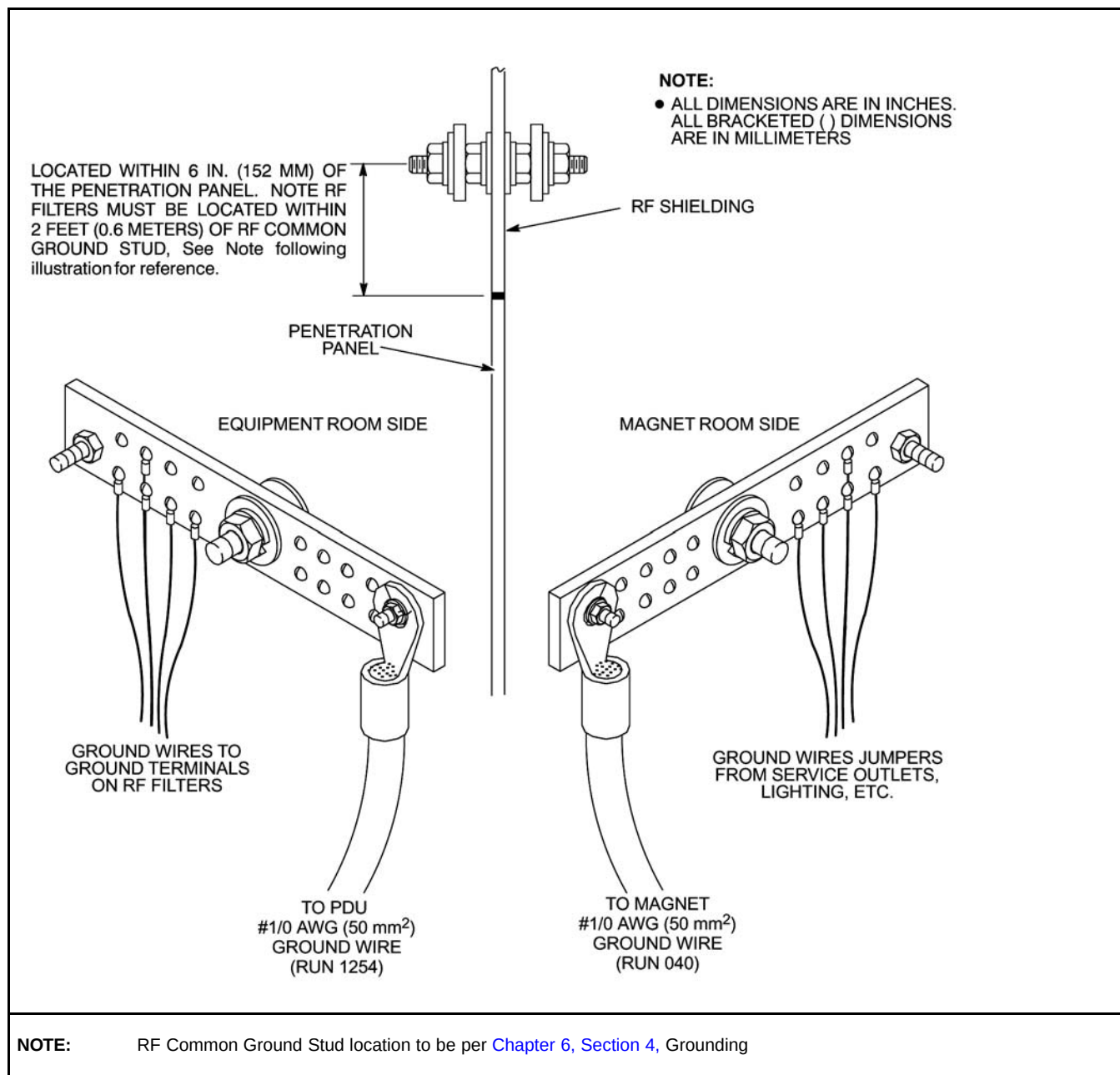
NOTE: All electricals lines and filters must comply with requirements in [Section 1](#), RF Shielded Room Requirements to minimize the possibilities of electrical discharge can cause RF broadband noise.

RF Shielded Room vendor should review with the electrical contractor the number of incoming power lines to the Magnet Room to determine the number of filters needed for electrical requirements.

Grounding from customer supplied power filters to the RF Shield Common Ground Stud, see [Chapter 6, Section 2](#), Critical Power Requirements and [Chapter 6, Section 4](#), Grounding for power and grounding requirements of all incoming power lines to the RF shielded room.

Common ground connection for shielded room must be located within 6 in. (152 mm) of the RF shielded room Penetration Panel with RF filters located within 2 feet (0.6 meters) of the RF Common Ground Stud. RF shielded room vendor to provide this common ground connection on both sides of shielded room by means of a stud extending through the shielded room (see [Table 8-2](#)). The RF Common Ground Stud and terminal bars must be accessible for servicing purposes on both sides of shield room. It is recommended that the RF Common Ground Stud be positioned above the Penetration Panel so it is concealed behind the Penetration Panel Covers (see [Section 5](#),) RF Penetration Panel.

Table 8-2: RF Common Ground Penetration Stud



4.2 Room Lighting



NOTICE

Installation and selection of lighting fixtures must comply with requirements in [Section 1](#), RF Shielded Room Specifications to minimize the possibilities of electrical discharge which can cause RF broadband noise.

Fluorescent lighting is not allowed in the Magnet Room due to the RF noise generated by the fluorescent light tubes. Dimmer switches must not be used; however, a selectable switch may be used to change the light intensity.

For additional Magnet Room lighting information refer to [Chapter 5, Section 7](#), Lighting, [Chapter 6, Section 4](#), Grounding, [Chapter 6, Section 7](#), Emergency Power, [Chapter 6, Section 8](#), DC Lighting Controller (Facility Option).

5 RF Penetration Panel

The RF Shielded Room Vendor must provide the opening in the RF shielding and appropriate mounting hardware for the GE Penetration Panel.



NOTICE

Penetration Panel electrical and mechanical connections, mounting hardware, and installation must comply with requirements in [Section 1](#), RF Shielded Room Requirements to minimize the possibilities of electrical discharge can cause RF broadband noise.



NOTICE

The Penetration Panel mounting hardware must not loosen over time to maintain RF attenuation requirement in [Section 1](#), RF Shielded Room Requirements for the anticipated duration of usage for MR imaging. Some of the design parameter that can contribute to loosen of the Penetration Panel mounting hardware are: wall material compression over time, insufficient fasteners quantity or spacing, over or under tightness of mounting fasteners, insufficient locking mechanism (i.e. Locktight, double/locking nuts), etc.

The RF shielded room acceptance test must be performed after the opening is cut in the RF shielding for the GE Penetration Panel. This acceptance test must be conducted with vendor supplied blank panel and the same mounting hardware to be used with the GE Penetration Panel. It is the facility's responsibility to ensure that the RF Shielded Room Vendor testing meets the attenuation specifications listed in [Section 1](#), RF Shielded Room Specifications.

The Penetration Panel must be covered on both sides for safety. If GE supplied adjustable covers are not used, customer must furnish covers or enclosures with key or tool required for opening to limit access to the panel. The mounting and clearance dimensions for the Penetration Panel GE covering are shown in Penetration Panel/Covering Mounting Requirements illustration in [Table 8-3](#).

[Illustration 8-1](#) and [Illustration 8-2](#) show two possible methods for mounting the GE MR Penetration Panel. Either method may be used depending on RF shielded room wall thickness. Make sure if the mounting method in [Illustration 8-1](#) is used then the RF wall thickness must be 0.75 in. (19 mm) 0.0625 in. (1.6 mm). Refer to the two preceding Notices. Check with RF Shielded Room Vendor to determine appropriate mounting method.

The Penetration Panel is to be mounted above the finished floor on the Equipment Room side of the RF shielded room. GE supplies only the Penetration Panel as shown in [Illustration 8-1](#) and [Illustration 8-2](#).

Illustration 8-1: Penetration Panel Cut Out For 0.75 Inch (19 MM) Thick RF Wall

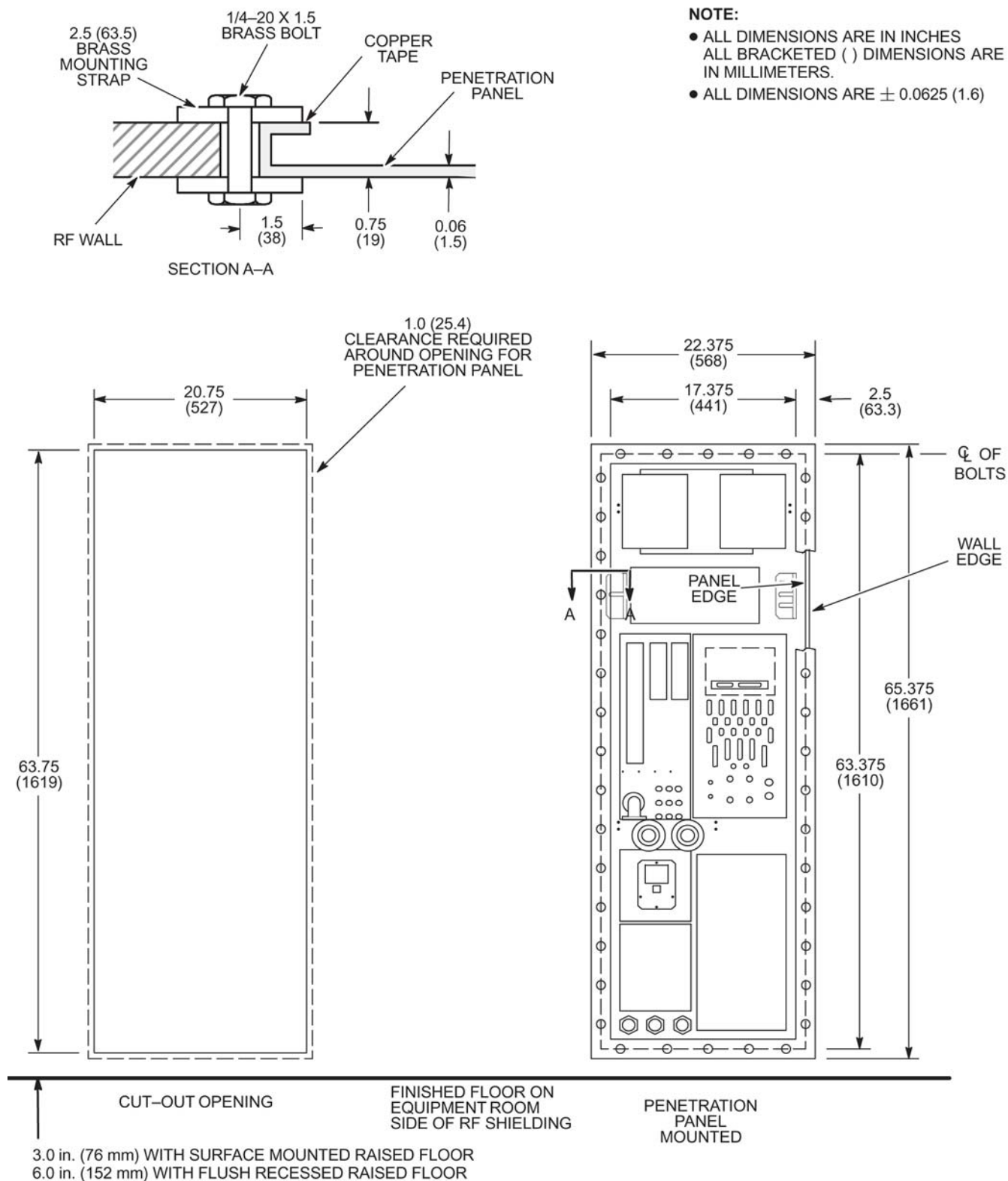


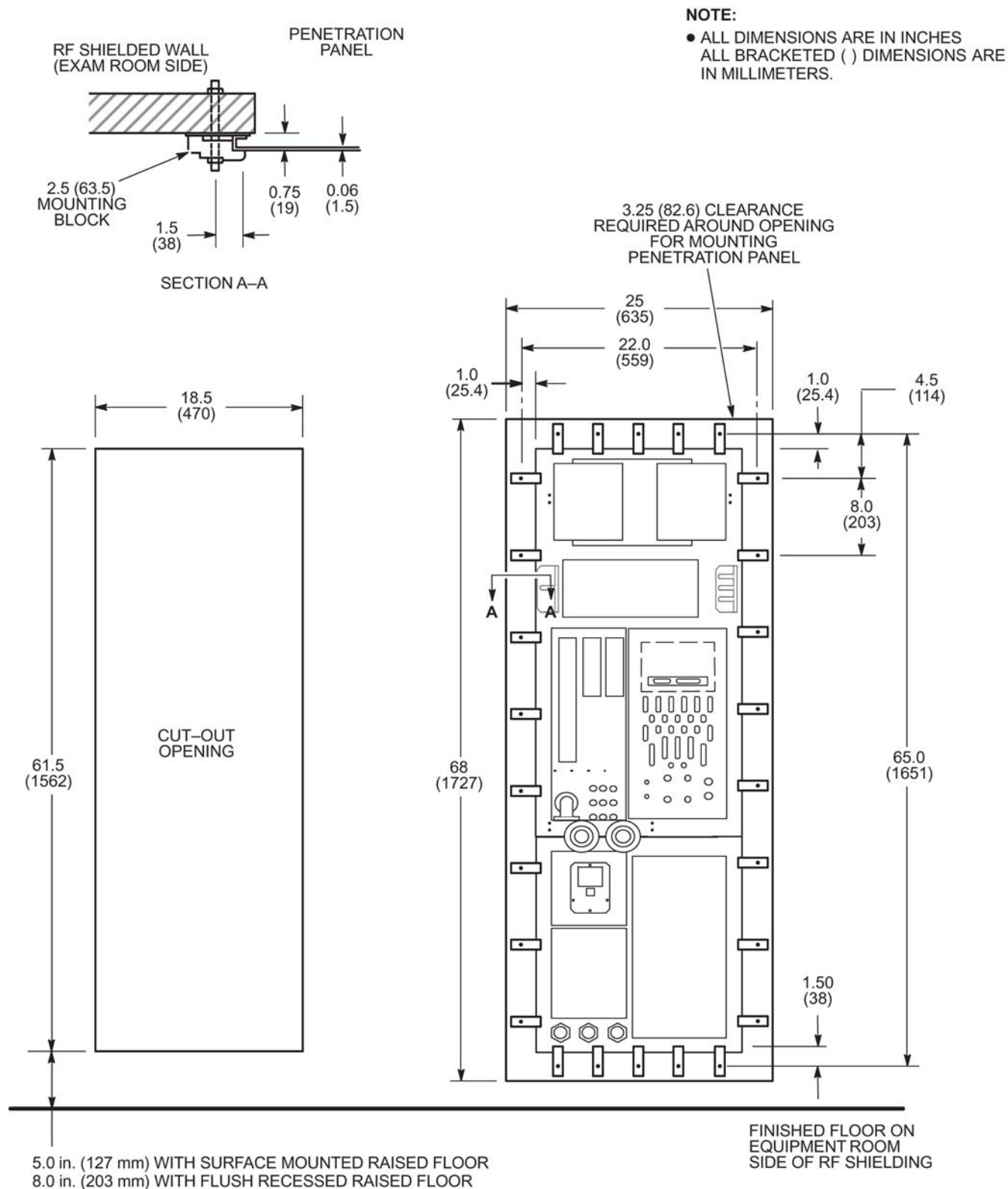
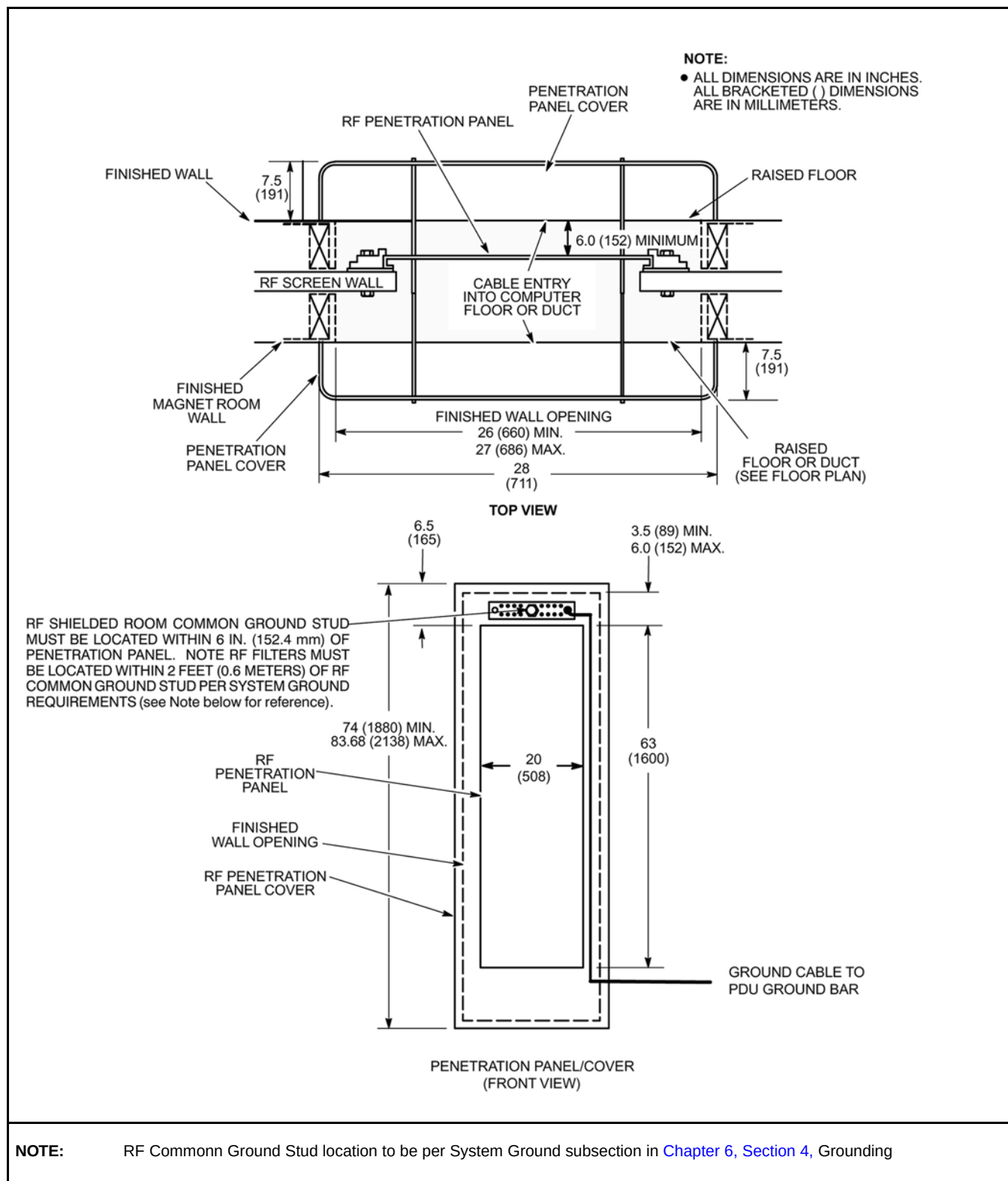
Illustration 8-2: Penetration Panel Cut Out For RF Wall Thickness Varying From Small To Large

Table 8-3: Penetration Panel/Covering Mounting Requirements



6 Physical Considerations

6.1 Introduction

The RF shielded room can be either a free standing shielded structure or a shielded room within an existing room. All styles of RF Shielded Rooms must be electrically isolated from earth ground complying with requirements in [Section 1](#), RF Shielded Room Requirements.

NOTE: All physical considerations must comply with requirements in [Section 1](#), RF Shielded Room Requirements to minimize the possibilities of electrical discharge that can cause RF broadband noise.

6.2 Doors and Other Openings

Shielded room doors are a major source of RFI leaks and must comply with requirements in [Section 1](#), RF Shielded Room Requirements.

The main door requires a minimum finished opening of 43 in. (1092 mm) to allow for helium dewars and patient tables to pass through the opening. However, a 48 in. (1219 mm) wide door is recommended for easy maneuvering of the Patient Table. Maximum door sill height is 1 in. (25 mm) with a 10 degree maximum threshold inclination.

6.3 Walls

It is recommended that walls be covered to protect RF material and to add to the aesthetics of the room for patient comfort. Fire retarding material must be used per building codes. Consult RF shield room vendor for RF shielding service requirements prior to covering RF walls. Removable wall covering may be needed if periodic RF shield servicing is required to maintain RF integrity. For wall designs impacts to sound quality improvements within the Magnet Room and to address airborne acoustics to areas outside the Magnet Room, refer to [Chapter 5, Section 8.4](#), Acoustic Design Guidelines.

NOTE: Walls materials and installation including all electrical and mechanical connections, mounting hardware, and installation must comply with the requirements in [Section 1](#), RF Shielded Room Requirements to minimize the possibilities of electrical discharge can cause RF broadband noise.

The recommended patient viewing window dimensions are 48 in. wide by 42 in. high (1219 mm x 1067 mm). The location of the window is dependent on the position of Operator Workspace position. For potential acoustic impact of window location refer to [Chapter 5, Section 8.2](#), Acoustics Background, [Chapter 5, Section 8.3](#), System Acoustic Noise Levels, [Chapter 5, Section 8.4](#), Acoustic Design Guidelines.

NOTE: The operator at the Operator Workspace must be able to view the patient during a scan.

6.4 Magnet Room Floors

NOTE: The following information applies to areas of the site impacted by the MR System upgrade.
Refer to [Table 8-4](#) for Magnet Room floor requirements and comments.



NOTICE

Magnet Mounting recessed area is required for all installations. The recessed area enables future system upgrades.

Table 8-4: Magnet Room Floor Requirements And Comments

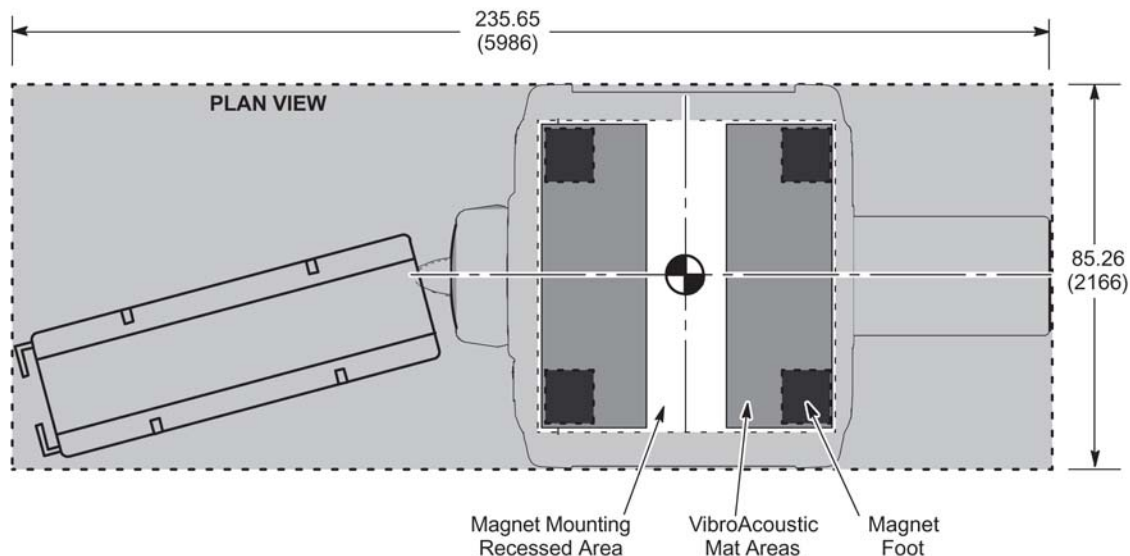
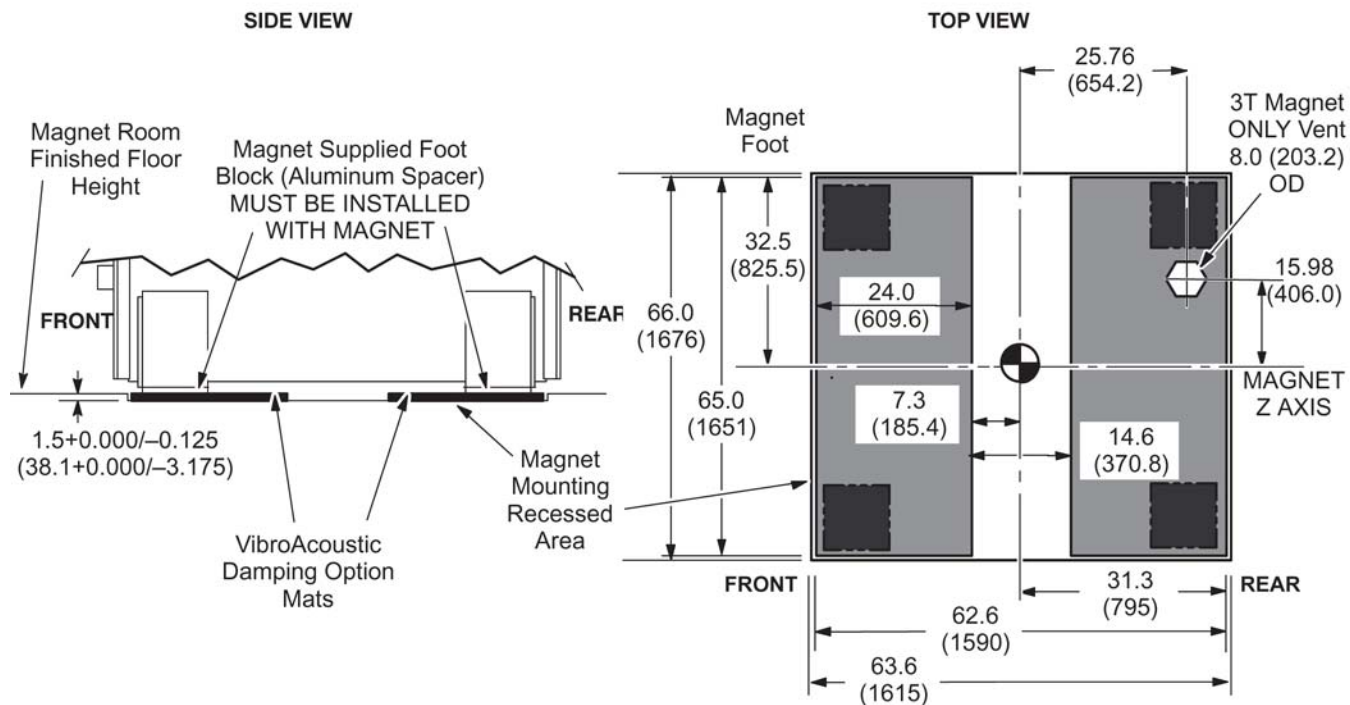
Requirements
- The floor/flooring (i.e. structural, sub-floor and finished flooring) must meet local and national codes. - The floor under the Magnet in a 10 ft x 12 ft (3.048 m x 3.658 m) area must not be fabricated from magnetic materials per Magnet Room Floors Magnetic Properties in Construction Materials. - Floor/flooring must support the weight of equipment during system installation and throughout the operation and Service life. - Magnet, Enclosure, and Patient Table areas (shown shaded in Illustration 8-3) must be level and flat defined as less than 0.125 in. (3.175 mm) between high and low spots in the area. - Magnet Mounting Recess Area shown in Illustration 8-3 : - Area depth must be 1.500 + 0.000 / - 0.125 inch (38.1 + 0.000 / - 3.175 mm) from the top of the finished floor material to all points of the bottom of the recess floor. - Area must be level and flat defined as not greater than 0.125 inches (3.175 mm) between high and low points in the entire projected area of each of the 2 VibroAcoustic Mats. RF shield seams, joints or overlaps MUST NOT be located in the projected areas of the 2 VibroAcoustic Mats. - The material in the bottom of the Magnet Mounting Recess must not be compressible. No plywood or filler board type materials are allowed within this recess. - For VibroAcoustic Damping Option sites: self-leveling grout, self-leveling epoxy, or self-leveling mortar are acceptable non-compressible materials - For non-VibroAcoustic Damping Option sites: RF Shield Vendor to provide Aluminum, Brass or Stainless Steel plates support the Magnet feet (non-compressible material to maintain the depth and level requirements stated in preceding bullets). - Metal access floor tiles are NOT allowed anywhere in the Magnet Room. - Rear Pedestal floor support: - The floor under the Rear Pedestal must support 550 lbs (250 kg) distributed across the 4 leveling feet (1.2 meter). - The Rear Pedestal leveling feet must be positioned on one solid floor member, no joints or seams are allowed. - The Magnet Room finished floor surface resistance must measure between 10E6 ohms and 10E9 ohms. - The finished floor grounding device must be fastened/secured to the RF Shield Common Ground Stud. - No carpets, antistatic wax or spray on treatments are allowed for use on the magnet room finished flooring. - Electrically conductive materials utilized must comply with requirements in Section 1, RF Shielded Room Requirements to minimize the possibilities of electrical discharge which can cause RF broadband noise. - The finished floor needs to be water resistant and protect the RF/Magnetic Shield and subfloor from possible water damage.

Table 8-4: Magnet Room Floor Requirements And Comments (cont'd)

Comments
• MR System Interconnection cables are FT4 or equivalent rated; not plenum rated. • The VibroAcoustic Damping Option mats must be continuously supported to function per design. Failure to meet all levelness requirements will result in excessive settlement of the Magnet and mats contributing to mis-alignment of the Magnet Enclosure, Patient Table and Table Docking Assembly. • The flooring grounding device and fastener/securing equipment are supplied by the customer or customer contractor and must be installed as the manufacturer defines. Local and national codes must be met when grounding the floor system to the RF Ground Stud.

Illustration 8-3: Magnet Room Floor Levelness Area**NOTE:**

- All dimensions are in inches. All bracketed () dimensions are in millimeters.

**MAGNET MOUNTING RECESSED AREA DETAILS**

Chapter 9 Shipping and Delivery Data

1 Shipment

Domestic transportation for the MR system, excluding the magnet, will be via an air-ride moving van. The magnet will be shipped on an air-ride flat-bed truck. Export transportation for the MR system overseas will be via air shipment in a pressurized cargo hold. Refer to [Table 9-2](#) for transportation environmental conditions.

for the shipping weights and dimensions of the major Signa EXCITE HD 3.0T system upgrade components. Actual shipping may vary, international shipment may require equipment to be crated.

Table 9-1: Transportation & Storage Environmental Conditions

System Equipment	Temperature Range °F (°C)	Temperature Change °F/Hr (°C/Hr)	Relative Humidity %	Humidity Change %/Hr	Atmospheric Pressure hPa
Electronics Cabinets & equipment	-30 to 140 (-34 to 60)	176 (80)	0-90 non-condensing	30	1012 to 525
Magnet	-31 to 122 (-35 to 50)	176 (80)	0-90 non-condensing	30	1012 to 525

Table 9-2: Transportation & Storage Environmental Conditions

System Equipment	Temperature Range °F (°C)	Temperature Change °F/Hr (°C/Hr)	Relative Humidity %	Humidity Change %/Hr	Atmospheric Pressure hPa
Electronics Cabinets & equipment	-30 to 140 (-34 to 60)	176 (80)	0-90 non-condensing	30	1012 to 525

If the system is stored before installation, it must be stored in a warehouse protected from weather and meet the environmental conditions in [Table 9-1](#).

Table 9-3: Signa EXCITE HD 3.0T Upgrade Shipping Data

MR Component	Approximate WxDxH		Approximate Weight		Method Of Shipment
	in.	mm	lbs	kg	
3T94 Magnet Enclosure Bridge	24 x 161 x 10	610 x 4090 x 254	75	34	box
HFD/PDU Cabinet without Fan Module installed	24 x 37 x 75	610 x 940 x 1905	1810	823	on cabinet casters, wrapped with plastic
Fan Module for HFD/PDU Cabinet	28 x 38 x 15	711 x 965 x 381	160	73	on pallet
RFS Cabinet	24 x 36 x 77	610 x 914 x 1956	585	265	on cabinet casters, wrapped with plastic
SPT Phantom Set	34 x 32.5 x 60	864 x 826 x 1524	350	159	on cart casters with box cover

Table 9-3: Signa EXCITE HD 3.0T Upgrade Shipping Data (cont'd)

MR Component	Approximate WxDxH		Approximate Weight		Method Of Shipment
	in.	mm	lbs	kg	
Operator Workspace Cabinet	24 x 35 x 31	600 x 900 x 780	243	110	wood pallet with cardboard cover
Operator Workspace LCD Panel	27 x 33 x 27	686 x 838 x 686	125	57	skid
Operator Workspace equipment	32 x 32 x 23	813 x 813 x 584	100	45	box
Chilled Air Blower for IPCM Option *	24 x 42 x 45	610 x 1060 x 1140	375	170	special crate
Notes * Optional Equipment					

Chapter 10 Pre-Installation Checklist

1 Introduction

"Preinstallation" refers to work necessary to plan and prepare a site for delivery and installation of equipment. Delay, confusion, and waste of manpower can be avoided by completing preinstallation work. It is recommended to have GE Service Representative make on-site inspections during construction.

The purpose of this section is to outline key areas of concern in the preparation of a customer's site for magnetic resonance. It is intended as a guide for GE's Project Manager, Installations and/or Installation Leaders when making on-site inspections during the construction phase of an MR project. Note that these inspections by GE are intended to aid the overall site preparation process. They do not relieve the customer and project architect from responsibility for the design, engineering, and coordination efforts necessary to ensure a successful MR project.

During the course of the site preparation process, GE's Project Manager, Installations and/or Installation Leaders may observe that the Requested On-Site Date (ROSD) for the MR equipment is not realistic. If this is the case, appropriate actions must be taken by the GE Field Sales/Service Team to adjust the ROSD accordingly.

All work must be in compliance with national and local codes.

2 Upgrade Pre-Installation Checklist

NOTE: CTIQ numbers listed in following items are defined in [Chapter 1, Section 2](#), Critical To Image Quality.

2.1 General Pre-Installation Reminders

1. The customer's architect has fully reviewed final site construction drawings using the guidelines provided by GE Healthcare Design Center.
2. **CTIQ#2 Magnetic Field:** The customer has confirmed that the site design contains the magnetic field and site design complies with the moving metal requirements. Refer to:
 - [Chapter 3, Section 1](#), Proximity Limits
 - for magnetic field containment refer to Magnetic Field Consideration information in pre-installation manual appropriate for the magnet system:
 - LCC300 refer to Direction 5133303 Signa EXCITE HD 3.0T Pre-Installation or Direction 2355548 Signa EXCITE 3.0T Pre-Installation and for 3T94 Magnet refer to 2221775 Signa 3.0T/94 & VH3 Pre-Installation.
3. **CTIQ#5 Acoustics:** The site has been designed and constructed to contain the MR System acoustic levels to meet local regulations and customer requirements. Refer to [Chapter 5, Section 8.3](#), System Acoustic Noise Levels and [Chapter 5, Section 8.4](#), Acoustic Design Guidelines.
4. **CTIQ#6 Power Quality:** The site power plan meets the power quality requirements. Refer to [Chapter 6, Section 1](#), System Power Introduction and [Chapter 6, Section 2](#), Critical Power Requirements.
5. **CTIQ#7 Lighting:** The Magnet Room lighting design and installation complies with the requirements so as to not to generate RF noise, which would be detrimental to the ability of the MR System to produce images of good quality. Refer to [Chapter 5, Section 7](#), Lighting.
6. **CTIQ#8 Multiple MR System Site:** The site design complies with the requirements for sites with more than one magnet in the magnetic field area or the Equipment Room is shared by more than one MR system of the same field strength. Refer to [Chapter 3, Section 2.1](#), Two Magnet Site Layout and [Chapter 3, Section 2.2](#), Equipment Room Shared By Multiple MR Systems.

2.2 Safety

1. **CTIQ#10 Broadband Network Connection & Telephone:** Functioning Broadband (network) connections and telephone are available at the site for the duration of the installation and system operation. Broadband is needed for Magnet Monitor and MR System computer access. Telephone is needed to dial out in case of an emergency. Refer to [Chapter 3, Section 6.1](#), System Monitoring and Support Connectivity.
2. **CTIQ#11 Room Ventilation & Air Conditioning:** All of the required Magnet Room ventilation items including exhaust fan and controls must be operational to make sure sufficient ventilation is available. Need to re-verify exhaust fan system operation meets requirements especially minimum CFM value. The customer needs to maintain a copy of the Exhaust Fan System test reports with the site documentation. Refer to [Chapter 5, Section 9](#), Room Ventilation.

2.3 Required Before System Delivery and Install

The following items must be completed prior to upgrade system delivery. A site inspection by GE Service Representative must be completed prior to upgrade system delivery to ensure site readiness.

1. **CTIQ#11 Room Ventilation & Air Conditioning:** Heating, Ventilating & Air Conditioning (HVAC) is installed, functional and available 24 hour/day, 7 days/week for Magnet Room and Equipment Room. Refer to [Chapter 5, Section 3](#), Temperature and Humidity Specifications and [Chapter 5, Section 4](#), Air Cooling.
2. **CTIQ#15 Main Disconnect Panel (MDP):** The site power meets the power quality requirements. The Main Disconnect Panel (MDP) has been installed, electrician wiring complete, and the MDP is operational with power available 24 hour/day, 7 days/week for powering the Shield/Cryo Cooler Compressor cabinet, Magnet Monitor equipment, and Coldhead Compressor Chiller (MRCC) if providing water cooling for the Shield/Cryo Cooler Compressor cabinet. Refer to [Chapter 6, Section 1](#), System Power Introduction and [Chapter 6, Section 2](#), Critical Power Requirements.
3. **CTIQ#16 Dust-Free Environment:** All of the areas in the MR System Magnet Room, Equipment Room, and Control Room are complete, so a dust-free environment is available for the installation of the equipment. Refer to [Chapter 5, Section 12](#), Pollution.
4. **CTIQ#17 Environmental Controls:** Environmental systems and controls are functioning to provide the required site environment for all MR System equipment. This includes Heating, Ventilating & Air Conditioning (HVAC) and Water Cooling. Refer to
 - [Chapter 5, Section 3](#), Temperature and Humidity Specifications
 - [Chapter 5, Section 4](#), Air Cooling
 - [Chapter 5, Section 5](#), Water Cooling Requirements.
5. The MR System interconnects routing design/method equipment has been installed and ready for interconnects installation. Refer to [Chapter 3, Section 4](#), MR System Interconnects Routing.
6. Delivery route has been defined for equipment including plans for protecting flooring for heavy equipment and carts. Refer to [Chapter 3, Section 3](#), Minimum Delivery Route Sizes and Capacity.
7. Power Distribution Unit power is available for connection to the Main Disconnect Panel. Refer to [Chapter 6, Section 3.1](#), Main Disconnect Panel (MDP) Requirements and [Chapter 6, Section 3.2](#), System Power Distribution Unit.
8. **CTIQ#18 RF Shielding:** The Magnet Room has been tested to ensure that the RF shielding meets the attenuation and electrical isolation requirements. Refer to [Chapter 8, Section 1](#), RF Shielded Room Specifications. The customer needs to maintain a copy of the RF shielded room vendor test reports with the site documentation.
9. Power has been connected from the Main Disconnect Panel to Power Distribution Unit. Refer to [Chapter 6, Section 3.1](#), Main Disconnect Panel (MDP) Requirements and [Chapter 6, Section 3.2](#), System Power Distribution Unit.

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Chapter 11 Tools and Test Equipment

1 Moving Metal Measurement Equipment

Table 11-1: Moving Metal Measurement Equipment

Item	GE Part Number	Description
1	2175019 or equivalent	Multiwave II Kit with Barrington 3 axis probe

2 Rigger/Customer Supplied Equipment

Table 11-2: Rigger/Customer Supplied Equipment

Supplied By	Item	Description
Rigger	1	Refer to <i>Direction 2340869 GE LCC300 Active Shield Magnet Delivery and Installation</i> for equipment and requirements.
Customer	2	Equipment for off loading electronics and other miscellaneous components. (e.g. fork lift, hand trucks, straps, etc.)
	3	Panel lifters for computer flooring: Smooth floor Indicon Industries PL2DC Standard carpeted floor Indicon Industries PL30P Level loop carpet floor Indicon Industries SCLV1.
Customer/Contractor	4	Resistance meter: • Megger Insulation Tester - preferred test unit • Analog d'Arsonval Meter (meter must have test source >9 VDC per specification in <i>Appendix Chapter 12, Section 1</i>, RF Shielded Enclosure Test Guideline titled <i>Enclosure Power Reference Isolation</i>)

3 Cryogenic Equipment

Table 11-3: Cryogenic Equipment

Item	GE Part Number	Description
1	5111049	Gaswatch personal Oxygen monitor
2	46-306734G1	Low Pressure Regulator Kit (Non-magnetic gas cylinder regulator / hose assembly consisting of regulator, hose, and case) (See Note 2)
3	---	Liquid helium in non-magnetic dewars is needed for refilling the magnet (See Note 2). Refer to appropriate magnet manual for helium volume.
4	46-306717G1	Non-magnetic gaseous helium cylinder cart (See Note 2)
5	46-294705G1	Universal Fill Line Kit (See Note 2)
6	46-294511P1	250 Liter Dewar Stinger (See Note 2)
7	46-294511P2	500 Liter Dewar Stinger (See Note 2)
8	46-294512P1 (12 ft) 46-294512P2 (8 ft)	Transfer Line (See Note 2)
9	46-282336P2	Dewar stinger assembly (High Efficiency) (See Note 2)
10	46-306812G1	Dewar tube "Thumper" tool (See Note 2)
11	2253802	Burst Disk and Gasket Assembly for LCC Magnet Helium Vessel (See Note 1)
12	46-271136G1	Dewar Adapting Kit including O-ring Kit for Dewar Adapting kit (46-271135P9) (See Note 2)
13	46-294804G1	Non-magnetic Aeroquip wrench Set (See Note 2) includes the following: • 46-294800G1 case for bronze wrench kit • 46-294805P1 bronze 1 5/8 in. open end wrench • 46-294805P2 bronze 1 3/8 in. open end wrench • 46-294805P3 bronze 1 3/16 in. open end wrench • 46-294805P4 bronze 1 1/8 in. open end wrench • 46-294805P5 bronze 1 in. open end wrench
14	46-281088G3	Shield Cooler Installation / Maintenance Kit (See Note 2)
15	46-265273G1	Liquid Helium Level Meter Kit (See Note 2)
16	2362622	Remote Helium Meter
17	46-306781G1	Helium Mechanical Gas Flowmeter (See Note 2)
18	46-301477G1	Helium Mechanical Gas Flowmeter (See Note 2)
19	46-252210P1	3 Inch Valve Operator (vacuum break tool) (See Note 1)
20	46-294872G2	SAV-CON / Instrumentation Lead Service Kit (See Note 2)
21	46-318784G2	Shield Cooler Test Kit (See Note 2)
22	46-318696G1	Water Tee for Shield Cooler Water Samples access (See Note 2)
23	46-294052G1	Water Flow Meter Kit (for checking flow of water to Shield Cooler Compressor) (See Note 2)
24	2171219	RUO Temperature Monitor for LCC 1.5T , 1.0T, & 3.0T Magnet (See Note 2)
25	---	Torch for de-icing
26	---	Fuel for torch

Table 11-3: Cryogenic Equipment (cont'd)

Item	GE Part Number	Description
27	46-252805P2 (10 ft) 46-252805P3 (15 ft)	Nitrogen Transfill line (only needed if magnet is warmer than nitrogen temperature)
28	46-260201P1	N2 Precool Syphon (only needed if magnet is warmer than nitrogen temperature)
29	5160887	Standard mid arm cryo gloves medium size
30	5160896	Standard mid arm cryo gloves large size
31	5160897	Standard mid arm cryo gloves X-L size
32	TBD	safety galses
33	TBD	face shield
Notes - Supplied as part of Signa. - Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		

4 Installation Equipment

Table 11-4: Installation Equipment

Item	GE Part Number	Description
1	---	Ramp for removing cabinets from pallets for International shipments (See Note 1)
2	---	Wrecking bar
3	---	Claw hammer, 3/4 lb
4	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5	---	Magnet Log Book
6	---	Installation log book
7	---	4 foot or equivalent carpenter level
8	2319156	Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
9	2134776	Gradient Cable Crimper/Stripper Kit (Note 2) consisting of: • 2134586 Cable stripping tool • 2134586-2 Stripping tool replacement blade • 2134587 Cable slicer • 46-282853P1 Ratcheting crimper • 2135839 1/2 inch terminals • 2135839-2 3/8 inch terminals
10	46-260776G3	Magnet Service Tool 750 Amp Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 375 lbs (170 kg).
11	46-260776G4	Magnet Service Tool 1000 Amp Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 375 lbs (170 kg).
12	46-260777G3	Shim Service Tool Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 290 lbs (132 kg).
13	2135558	Shim case / cable kit
14	2386028	LCC300 Shim Camera Kit
15	2386029	LCC300 Passive Shimming Kit
16	46-306763G1	Subminiature-D Connector Removal/Re-termination Repair Kit for Robinson-Nugent Sub-D connectors
17	46-251865G5 or 46-251865G4 + 2295525-5	3T Field Plotting Kit - Metro Lab Teslameter to include #6 Probe with Pre-Amp Metro Lab Teslameter Kit + Probe
18	46-260703G5	Magnet Ramping Equipment Kit
19	2135435	750 Amp Ramp Cart / Cable Kit
20	46-318833G1	Ramp Cable Holder
21	46-294998G1	Ramping Supply and Equipment kit includes the following: • 46-260703G3 Magnet Ramping Equipment Kit (see Item 15) • 46-260776G3 Magnet Service Tool Power Supply Cabinet (see Item 10)

Table 11-4: Installation Equipment (cont'd)

Item	GE Part Number	Description
22	M1060SR	Magnet Low Ceiling Height (2.5 Meter) Siting Option NOTE: This option reduces the minimum required ceiling height for the Magnet Room to 2.5 meters from 2.67 meters. Check with GE Project Manager, Installations to determine if this is a required option.
23	5112581	Large (Installation) Titanium Non-Magnetic Tool Kit
24	5113258	Small (Maintenance) Titanium Non-Magnetic Tool Kit
Notes 1. Supplied as part of Signa. 2. Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		

5 Test Equipment

Table 11-5: Test Equipment

Item	GE Part Number	Description
1	2284754-2	Textronic TDS3012B digital scope 100 MHz Bandwidth
2	2284754	Textronic TDS3012 digital scope 100 MHz Bandwidth
3	2284763	Fluke FLK196M scopemeter for medical applications
4	46-194427P226	Dual trace oscilloscope, 100 MHz bandwidth, 2 channel, digital storage, Textronic 2232 (End of Product Life)
5	46-194427P222	Dual trace oscilloscope, 350 MHz bandwidth, 4 channel, Textronic 2465 (End of Product Life)
6	46-194427P284	Battery operated digital multi-meter, Fluke 87, 4.5 digits with frequency counter and capacitance
7	46-208572P9	Clamp-on Ammeter, 1-200A, AC/DC
8	2372868	3.0T RF Power Measurement Kit with 2500 Watt load
9	2218826	RPM 1650 Power Analyzer Kit
10	46-328143G3	Dranetz 626 analyzer kit with 1 and 3 phase modules
11	Catalog E6320DA or 46-194427P144	Densitometer
12	46-306801G1	Bell Gauss Meter Kit
13	2293050	Portable Gauss Meter with probe and case
14	46-194427P248	Microguard or ECOS leakage tester
15	46-306797G1	Fogg (ECG) Simulator and Memory Module
16	46-320433P1	Infrared Scanner 0 - 100°C
17	46-317830G1	Fiber Optic Light Meter Kit
18	46-294047G1	Shield Cooler Vacuum Pump Kit
19	46-251867G1	Main Vacuum Pump Down Kit
20	46-1944278P448	AEMC Clamp-On Ground Rod Impedance Meter

6 Calibration Tools and Fixtures

Table 11-6: Calibration Tools and Fixtures

Item	GE Part Number	Description
1	2371509	System Performance Test (SPT) Phantom Set for Systems with LCC300 (Active Shield) systems (See Note 1) consisting of the following: 2125247-13 Nesting Plate 2360054 Large Volume Shim Phantom Assembly which includes sphere (2360051), small half (236049), and large half (2360050) 2371511 Short Loader Assembly 2371512 Daily Quality Assurance (DQA) Phantom 2170481 EPI Foam Positioner 2291292 EPI Head Loader Positioner 5110241 Head Loader Positioner
2	2373141	Head TLT Sphere (2359877) and Loader (2360031) See Note 1
3	46-265434G1	Magnet Rundown Unit Test Box
4	2386042	3.0T ECMT/GRAFIDY Kit
5	46-328021G1	Enmet Oxygen Monitor Calibration consisting of the following: 1800 PSI cylinder of 20.9% Oxygen in Nitrogen 1800 PSI cylinder of 17% Oxygen in Nitrogen - Regulators with calibration adapter For use with portable or permanent Oxygen Monitor.
6	2106236	Portable Oxygen Monitor (Connecticut Analytical)
7	2107184	Permanent Oxygen Monitor (Enmet)
8	46-306712G1	Torque driver kit
9	46-306864G1	Magnet Helium Resistance Box Kit
10	2101360	Power Supply Calibration Kit
11	2371477	100 mm Silicon Oil sphere (2360034) and Universal Phantom Holder (46-328383P1) (See Note 1)
12	2372868	RF Power Measurement Kit
13	Not used	
Notes - Supplied as part of Signa. - Supplied by GE until turnover of system to customer, then available as part of a GE Service Contract. - Customer may purchase these items.		

7 Tool Kit

Table 11-7: Tool Kit

Item	GE Part Number	Description
1	---	Extension cords, with ground conductor
2	---	Power strip, grounded type, with minimum of five outlets
3	---	Soldering iron, pencil type with solder
4	---	Solder sucker
5	---	Assortment of Brady Quick labels
6	---	Micro clip leads
7	---	14 pin and 16 pin DIP clips
8	46-258218P3	Vinyl electrical tape
9	46-258218P4	Copper tape, 3 in. wide
10	46-258218P5	Copper tape, 2 in. wide
11	46-258218P6	Copper tape, 1 in. wide
12	---	Alcohol cleaning solution
13	---	Plastic or aluminum flashlight
14	---	Plastic or aluminum pen light
15	AMP No. 458994-1	Pin extractor, Universal Mate'n'Lock
16	AMP No. 305183-R	Pin extractor, M-series
17	46-237072P1	Pin extractor, Sub-D
18	46-307307G1	Crimping tool for coax cable and BNC connectors Inserts for • RG8, 9, 11, 214 46-255841P103 • RG58, 223 46-255841P100 • RG59 46-255841P101 • RG174 46-255841P102 • Die Removal Tool 46-255841P201
19	---	Assorted crimp tools.
20	---	Non-magnetic level
21	---	Non-magnetic tape rule, 12 ft
22	---	Assorted drill bits
23	---	Inspection mirror
24	---	Tap set, standard, and tap handle, T-type
25	---	rule, standard and metric markings
26	---	Alignment tool (tweaker)
27	---	Hex/alignment tool
28	---	Hemostat, 5 in., curved
29	---	Wrist grounding strap
30	Xcelite 110CG	Diagonal Cutting Pliers, 4-1/2 in.

Table 11-7: Tool Kit (cont'd)

Item	GE Part Number	Description
31	---	Screw Starter, aluminum or plastic shaft
32	---	Hobby and utility knives
33	---	Spring scales, 0-10 lbs and 0-50 lbs
34	46-313413P1	Extractor for 20 - 100 pin PLCC Chips
35	5112581	Large set Non-Magnetic Titanium Tool kit with aluminum case (generally used for installations)
36	5113258	Small Non-Magnetic Titanium Tool kit with soft case (generally used for routine maintenance)

Chapter 12 Appendixes

1 RF Shielded Enclosure Test Guideline

1.1 Introduction

This document describes the procedure and methodology of performing an RF shielding effectiveness verification test on enclosures which will house GE Healthcare Magnetic Resonance Imaging (MRI) equipment. MRI equipment is sensitive to RF energy from sources outside of the shielded enclosure. To ensure proper operation of the MRI equipment, the shielded enclosure must attenuate local RF signals to levels that do not cause interference.

NOTE: RF Shielding Performance is based on planewave measurements. "H" field and "E" field tests are not required, but are allowed as needed for diagnostic purposes.

1.1.1 Purpose of Test Plan

The purpose of this test plan is to describe a series of RF shielding effectiveness tests to demonstrate compliance of an MRI shielded enclosure to the requirements of GE Healthcare.

The test procedure described in this guideline is a modification of MIL-STD-285 and IEEE Std 299-1991. This procedure provides a thorough evaluation of the shield integrity at the upper end of the frequency range of interest showing any RF leakage which may cause imaging problems. These test guidelines ensure that the electromagnetic environment inside of the enclosure will meet the requirements of GE Healthcare.

1.1.2 Not used

1.2 Applicable Documents

MIL-STD-285	MILITARY STANDARD ATTENUATION MEASUREMENTS FOR ENCLOSURES, ELECTROMAGNETIC SHIELDING, FOR ELECTRONIC TEST PURPOSES, METHOD OF; 25 June 1956
IEEE Std 299-1991	IEEE STANDARD FOR MEASURING THE EFFECTIVENESS OF ELECTROMAGNETIC SHIELDING ENCLOSURES; 2 July 1991

1.3 Test Sample Setup

The shielded enclosure under test shall be set-up in a normal configuration, which consists of the following:

1. Magnet installed including all floor mounting bolts
2. RF shielded door(s)
3. Waveguide penetrations, HVAC, vents, medical gas lines, etc.
4. AC power supplied through low-pass filters
5. Patient view window, skylights, windows, hatches, etc.
6. Blank penetration panel installed, dimensionally equivalent to the GE panel and the same mounting hardware to be used with the GE penetration panel.

For safety reasons, the enclosure shall be electrically grounded during the shielding effectiveness test. Any variances from the normal configuration shall be noted in the certification report.

1.4 Shielding Effectiveness

This test procedure determines the worst case shielding effectiveness based on the lowest test point reading obtained. The lowest reading obtained will be the reading of the room.

1.5 Measurement Procedure

To simulate the effects of external RF sources, the transmit antenna shall be located outside the enclosure on a plane parallel to the face of the enclosure wall, at a distance of 6 feet (1.8 meters) unless physically constrained to a lesser separation. The areas of least effectiveness are located by searching the inside of enclosure with the antenna connected to the spectrum analyzer.

1.5.1 Test Position

The transmitting antenna will be positioned in front of all critical areas (doors, windows, filters, penetration areas, etc.) and at a minimum of every 20 feet (6.1 meters) of wall. The receiving antenna is scanned over all panel section joints (where accessible), at floor, wall, and ceiling for a minimum of 10 feet (3.05 meters) in all directions from the location of transmitting antenna. The receiving antenna shall be at a minimum of 1 foot (0.3 meters) from the shield. For areas that are inaccessible for direct location of the transmitting antenna, the inside of that area will still be scanned using the receiving antenna with the transmitting antenna positioned in front of the adjacent wall or adjacent test position.

1.5.2 Frequency Range

The standard frequency for shielding measurements shall be 100 MHz $\pm$ 10 MHz (150 MHz $\pm$ 10 MHz for 3T). This allows the frequency to be adjusted slightly to avoid interference from local active transmitters and/or RF noise from other sources. Test frequency utilized shall be noted in the certification report.

1.5.3 Free Field Calibration

The incident field, i.e. "free field", is measured by the following procedure:

Position the transmit antenna parallel to the exterior wall of the enclosure at a distance of 6 feet (1.8 meters) (unless physically constrained to a lesser separation, in which case a separate reference will be established and documented at the new test distance), using horizontal polarization. The receive antenna shall be placed between the transmit antenna and 1 foot (0.3 meters) from the exterior wall of the enclosure. The receive antenna will be moved vertically and horizontally to achieve maximum signal strength. The receive antenna shall be placed no closer than 2 inches (51 mm) from the exterior wall of the enclosure and in line with the transmit antenna. The maximum received voltage at the test frequency will be recorded.

1.6 Enclosure Power Reference Isolation

To prevent personnel hazard, it is necessary for the enclosure to be properly grounded.

To minimize common mode currents, the ungrounded enclosure should be isolated from ground with a minimum of 1000 ohms of DC resistance. The isolation measurement is performed by the following procedure:

All power to the enclosure is removed. For safety reasons, an AC voltage measurement will be made to verify that no power is connected. With electrical power and intentional ground disconnected, connect the test instrument between the shielded enclosure and AC power ground. Take a reading and record the value. This test shall be made using either an isolated, current limited high voltage (>150 VDC) DC source and DMM to read drop across the limiting resistor or a Megger instrument capable of reading values less than 1000 ohms. Conventional resistance meters employing test sources of 9 VDC or less shall not be used.

1.7 Test Equipment

Test equipment shall be selected to provide measuring capabilities as described in this test guideline. The signal source, amplifier, antennas, and receiver or spectrum analyzer shall be such that the difference between the induced reference voltage and the receiver sensitivity shall be at least 6 dB greater than the required attenuation specification.

The signal source and power amplifier shall output a CW signal for a nominal test frequency of 100 MHz (150 MHz for 3T). Receiver or spectrum analyzer and preamplifier (if required), shall provide adequate sensitivity to permit attenuation measurements to be made to the specified limits. Dipole antennas and other miscellaneous equipment required to transmit and receive the proper RF fields shall be used.

The absolute performance calibration, of the equipment requiring calibration, shall be performed on an as needed basis in accordance with MIL-STD-45662. The calibration period shall not exceed one year. The test equipment tolerances of at least $\pm 2\%$ frequency and ± 2 dB amplitude shall be met. Equipment certifications shall be traceable to the National Institute of Standards and Technology (NIST). All equipment will be verified for proper operation between and after each series of tests by repeating the reference readings at the specified frequency(s).

1.8 Data Recording and Verification

Measurements shall be performed by qualified responsible EMC test personnel. The test must be performed in the presence of a GE representative unless other arrangements have been made by GE Healthcare. All data collected during the course of the tests will be recorded on standardized data sheets. The data sheets will include the test location, frequency, reference level, measured enclosure level, and attenuation level.

1.9 Test Report

A final certification report will be provided after the test is performed. This report will include all recorded data necessary for the evaluation of the shielded enclosure test results and will list any changes pertinent to the test set-up or the shielding effectiveness. The certification report will also include the test procedures and a list of the actual equipment used during the test.

Along with the data sheet, there will be a presentable drawing showing the shape of the enclosure, all test point locations, doors, filters, windows, and existing building walls.

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